

Population AI: Modeling Societies of Interacting Agents

by
Ayush Chopra

Submitted to the Program in Media Arts and Sciences, School of Architecture and Planning,
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ABSTRACT

Many of the most consequential challenges facing modern society emerge from how millions of decisions interact. Pandemics spread through contact networks. Supply chains fail through cascading dependencies. Labor markets evolve as workers and firms continuously respond to one another. These are population-scale phenomena. AI has built agents that can observe, act, and reason about the world individually. This thesis builds populations of these agents that interact to model entire societies. The premise: intelligence in agentic populations emerges not just from how smart each agent is, but from how they interact.

Large Population Models (LPMs) are the computational foundation of this approach. LPMs represent societies as populations of autonomous agents, each carrying a parameterized behavioral model that captures its decision process and local context, and that can be learned from population data. The framework models how agent behavior propagates through networks of interaction, allowing population-level outcomes to emerge from many local decisions. To make this practical at scale, we built AgentTorch, an open-source framework that runs LPMs from a single machine simulating a digital city of millions to national supercomputing infrastructure modeling populations as large as the entire US workforce. AgentTorch has been adopted by governments, national labs, and organizations globally.

We validate the framework on two fronts. On human populations, LPMs have been deployed to monitor health security, food security, and labor market disruption across million-scale populations, surfacing structural patterns of risk that aggregate statistics do not resolve. On AI populations, modeling millions of interacting agents has enabled the discovery of coordination protocols that outperform state-of-the-art AI systems and expert teams on resource management tasks that have resisted coordination by human organizations for over thirty years. In both cases, the improvements arise from how agents interact, not from the capability of any individual agent.

This research lays the foundations for digital populations at the scale of nations: systems with millions of interacting agents that expand human capacity to understand and shape the world.

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In memory of my grandparents

Swaran Chopra

Raj Kumar Khera

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Part I

Foundations

Chapter 1

Introduction

Many of the grand challenges facing human societies emerge from the collective consequences of millions of interacting decisions. A pandemic spreads because contact decisions compound through social networks into waves that overwhelm entire health systems. A supply chain collapses because rational decisions to protect against uncertainty amplify through sequential dependencies into economy-wide disruption. Labor markets shift because new capabilities propagate through networks of skills and industries faster than institutions can adapt. In each case, the outcome emerged from the structure of the interactions.

These systems exhibit population-level behavior that depends on how agents interact over time and how those interactions accumulate across the population. Understanding and predicting these phenomena requires models that represent populations of autonomous agents and the interaction structures through which individual behavior becomes collective outcome.

This thesis develops the computational tools, mathematical frameworks, and algorithms to do exactly that.

1.1 Grand Challenges Emerge from Collectives

Consider two classes of disruption that motivate this thesis.

Biological disruptions, including pandemics, zoonotic spillovers, and outbreak cascades, arise from how populations move, contact, and respond to one another over time. The COVID-19 pandemic made this visible at global scale. Decisions about mobility, gathering, and compliance accumulated through contact networks into waves that overwhelmed health systems across the world. Behavioral response, economic disruption, and policy intervention interacted continuously and simultaneously across health, behavior, and infrastructure. Models focused on individual agents cannot capture these dynamics. The behavior emerges at the level of the population, through the structure of interactions.

Emerging threats share this structure. A pathogen crossing species boundaries, a financial contagion spreading through interbank networks, or a social behavior propagating through online platforms all follow the same mechanism. Local decisions propagate through interaction networks and produce collective outcomes that no individual agent controls or anticipates.

Labor market disruptions driven by artificial intelligence follow the same structure at a different scale. When a new capability automates a task in one part of the economy, the

effects propagate through networks of skills, occupations, and regional economies. These effects accumulate through interdependencies across firms, supply chains, and geographic regions. The resulting exposure is distributed across the system and is not visible in aggregate statistics. It becomes measurable only when the population of workers, firms, and institutions is modeled as interacting agents rather than isolated units [1].

Both classes of disruption emerge from populations of agents interacting over time. Understanding them requires models that represent populations directly and capture how local decisions propagate through interaction networks.

1.2 Intelligence is in the Interaction

Collective outcomes emerge from interaction structure. This principle appears across biological and social systems at every scale.

Consider the liver. A single cell does not contain a blueprint for the organ and has no global awareness of the structure it is part of. Cells exchange local chemical signals with nearby neighbors: morphogen gradients diffusing across short distances, contact-dependent inhibition between adjacent cells, and mechanical forces transmitted through surrounding tissue. Through these local interactions, billions of cells self-organize into a structured organ [2]. Cells do not coordinate by announcing plans. When stressed, they emit local signals that neighboring cells detect and respond to automatically. Local conditions become collective behavior because adaptive signals propagate across the tissue. The structure emerges from the interaction between cells.

The same principle operates at the scale of thousands. Army ants construct living bridges across gaps in their foraging paths. Each ant follows a simple local rule based on the presence or absence of nearby ants. Through these local interactions, the colony forms stable bridge structures without centralized control [3]. Fish schools coordinate motion in response to predators through local alignment rules based on the behavior of nearby neighbors [4]. In each case, coordinated collective behavior arises from interaction structure rather than centralized reasoning or individual sophistication.

The same structure appears in human systems. Consider Fishbanks [5,6], a game that MIT’s John Sterman ran with human teams for over thirty years. Six fishing companies share the same ocean. Each round, every company decides how many boats to deploy. More boats means more immediate profit, but fish regenerate slowly. If companies collectively fish too aggressively, the population collapses and everyone loses. Across experiments, companies overfished early, stocks collapsed within five to six rounds, and populations did not recover. Participants understood the system dynamics and the long-term consequences of their actions. The outcome arose from the inability to coordinate timing across agents making simultaneous decisions. The behavior emerged from the structure of interaction.

These examples motivate a common abstraction. To study any such system, whether a tissue, a colony, a market, or a population at risk, we model each individual entity as an **agent**: an autonomous unit that observes its local environment, makes decisions according to some behavioral rule, and acts in ways that affect its neighbors. The collective behavior of a population of interacting agents is what produces the phenomena we care about. Intelligence, understood as the capacity to produce adaptive, coordinated, functional outcomes, is expressed

through the structure of interaction.

1.3 Personal Intelligence versus Population Intelligence

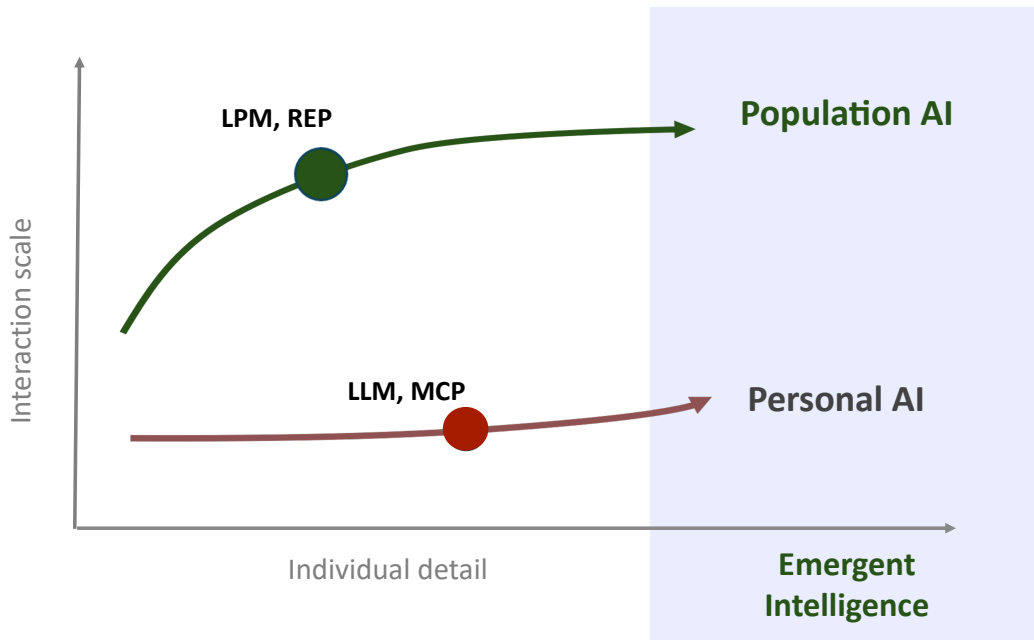


Figure 1.1: Personal AI focuses on improving individual agent capability. Population AI models how interactions between agents produce collective outcomes that no individual-level model can capture.

The dominant paradigm in contemporary AI focuses on improving individual agents. Large language models increase the ability of an agent to reason, plan, and act in complex environments. This enables agents to solve more sophisticated tasks and adapt to novel situations.

Improving individual capability does not guarantee coordinated behavior across a population. Fishbanks illustrates this directly. Participants understood the system dynamics and the long-term consequences of their actions. The outcome arose from the inability to coordinate decisions across agents over time. A more capable individual participant would not have changed the result. The bottleneck was the interaction structure.

The same gap appears across domains. A pandemic is shaped by patterns of interaction across populations: contact networks, behavioral responses, and collective timing of interventions. A labor market transition unfolds through dependencies between skills, firms, and regions. These systems require models that represent how agents interact and how those interactions propagate across the population.

This thesis introduces **Population Intelligence**: a paradigm focused on modeling the interaction structures through which individual behavior becomes collective outcome. Where Personal Intelligence builds more capable individuals, Population Intelligence models the

interactions through which individuals collectively produce outcomes no individual could produce alone.

Population Intelligence applies at any scale. Even small populations exhibit collective behavior when interactions are modeled explicitly. A population of six fishing companies, five hundred patients in a clinical trial, or a thousand workers in a regional labor market can reveal collective phenomena that no individual-level model ever could. The key requirement is to represent agents as interdependent rather than isolated, and to model the interactions through which their decisions jointly produce outcomes.

This thesis develops **Large Population Models** (LPMs): a framework that represents populations as collections of interacting agents, learns from heterogeneous real-world data, and scales from hundreds to millions of agents on commodity hardware. The open-source implementation is **AgentTorch** [7], deployed in biosecurity monitoring, labor market analysis, and policy evaluation across multiple countries.

Agents influence one another through signals that propagate across the population when any agent’s behavior changes. We call this a ripple. Chapter 3 formalizes this intuition as the *sensitivity* primitive: the signal that connects local behavior to population-level outcomes and enables learning from data. It is the unifying thread across all the algorithms developed in this thesis.

1.4 Publications Associated with this Thesis

The work presented in this thesis is based on the following publications and manuscripts:

Methodological Contributions

- **Ayush Chopra**, Shashank Kumar, Nurullah Giray-Kuru, Ramesh Raskar, and Arnau Quera-Bofarull. *"On the Limits of Agency in Agent-Based Models"*. Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), **2025**.
- **Ayush Chopra**, Alexander Rodríguez, Jayakumar Subramanian, Arnau Quera-Bofarull, Balaji Krishnamurthy, B. Aditya Prakash, and Ramesh Raskar. *"flame: A Framework for Learning in Agent-Based Models"*. Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), **2024**.
- **Ayush Chopra**, Arnau Quera-Bofarull, Nurullah Giray-Kuru, Michael Wooldridge, and Ramesh Raskar. *"Private Agent-Based Modeling"*. Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), **2024**.
- **Ayush Chopra**, Alexander Rodríguez, Jayakumar Subramanian, Arnau Quera-Bofarull, Balaji Krishnamurthy, B. Aditya Prakash, and Ramesh Raskar. *"Differentiable Agent-Based Epidemiology"*. Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), **2023**. **Best Paper Award, ICML 2022 Workshop on AI for Agent-Based Models**.

- Arnau Quera-Bofarull, **Ayush Chopra**, Joseph Aylett-Bullock, Carolina Lazaro, Ani Calienscu, Ramesh Raskar, and Michael Wooldridge. *"One-shot Sensitivity Analysis of Agent-Based Models via Automatic differentiation"*. Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), **2023**.
- **Ayush Chopra**, Aman Sharma, Feroz Ahmad, Luca Muscariello, Vijoy Pandey, and Ramesh Raskar. *"Ripple Effect Protocol: Coordinating Agent Populations"*. Under review.

Applications

- Abhijin Adiga^{*}, **Ayush Chopra**^{*}, Mandy L. Wilson^{*}, S. S. Ravi, Dawen Xie, Samarth Swarup, Bryan Lewis, Andrew Warren, John Barnes, Ramesh Raskar, and Madhav V. Marathe. *"A High-Resolution, US-scale Digital Simulation of Interacting Livestock, Wild Birds, and Human Ecosystems with Applications to Multi-host Epidemic Spread"*. Proceedings of the National Academy of Sciences (**PNAS**), **2026**.
- Mark P. Khurana^{*}, Joseph L.-H. Tsui^{*}, Bernardo Gutierrez^{*}, **Ayush Chopra**, and others. *"Global Approaches to Infectious Disease Surveillance and Modelling"*. **Nature Medicine**, **2026**.
- **Ayush Chopra**, Santanu Bhattacharya, DeAndrea Salvador, Ayan Paul, Teddy Wright, Aditi Garg, Feroz Ahmad, Alice C. Schwarze, Ramesh Raskar, and Prasanna Balaprakash. *"The Iceberg Index: Measuring Skills-centered Exposure in the AI Economy"*. Under review.

^{*}Equal contribution.

Chapter 2

Background: Modeling Agent Populations

To understand collective phenomena such as pandemics, market cascades, and labor disruptions, we must model the populations that produce them. This requires answering two questions: how do we represent agents and their interactions, and how do we learn from real-world data to make those representations accurate? This chapter traces how researchers have approached both questions and identifies the gaps that remain.

The central tension organizing this history is captured in Figure 2.1. Modeling a population of agents involves two dimensions: *scale* (how many agents are represented) and *detail* (how rich and adaptive each agent’s behavior is). Every approach in this field occupies a position in this space, and scale and detail have historically been in tension: achieving one has almost always required sacrificing the other. This thesis works toward the top-right corner of this space, where high agent count and rich behavior coexist. We call this region Population AI.

2.1 Mean-Field Systems

The earliest formal approaches to modeling large populations did not represent individuals at all. They described populations through aggregate distributions and differential equations governing how those distributions evolve. Compartmental models in epidemiology (SIR, SEIR, and their variants) are the canonical example: entire populations are collapsed into a handful of compartments (Susceptible, Infected, Recovered), and the dynamics are governed by equations describing flows between them [8]. The individual disappears entirely; only the aggregate distribution remains.

Mean-field games (MFGs), introduced formally by Lasry and Lions [9] and independently by Huang, Malhamé and Caines [10], extended this tradition by allowing agents to make strategic decisions while interacting through a shared mean-field: the aggregate distribution of the population’s state. Each agent solves an individual optimization problem, but the coupling to others occurs only through this aggregate quantity, not through direct pairwise interactions. This makes the problem tractable: instead of tracking N^2 pairwise relationships, the model tracks a single evolving distribution. MFGs have been applied to epidemiological policy [11,12], financial markets, crowd dynamics, and opinion formation.

Agent-based Models: Scale vs Detail Challenge

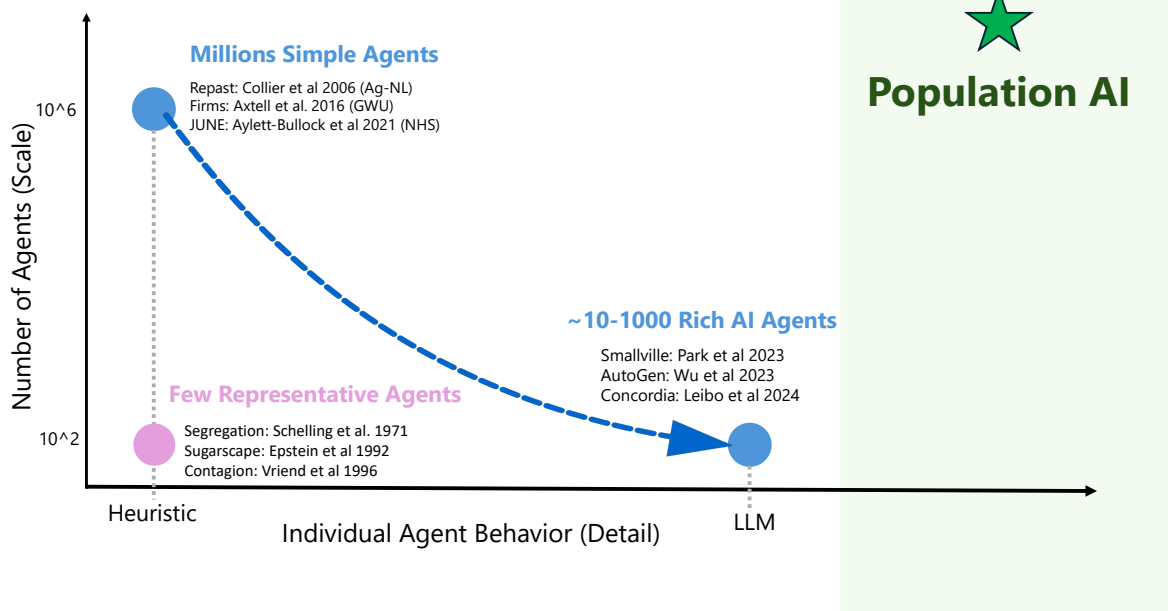


Figure 2.1: The scale-detail frontier in agent-based modeling. Mean-field systems collapse agent heterogeneity to gain tractability. Large-scale heuristic agents achieve population scale at the cost of behavioral realism. Rich data-driven agents achieve behavioral expressiveness at the cost of scale. This thesis aims to occupy the top-right: populations of agents with realistic, learned behavior at true societal scale.

The tractability of mean-field systems comes at a fundamental cost. They assume agents are exchangeable and interact only through aggregate quantities. This eliminates the features that make collective phenomena interesting: individual heterogeneity, local network structure, and the feedback loops between specific agent behaviors and population-level outcomes. A mean-field epidemic model cannot represent the difference between a superspreader event at a crowded venue and diffuse community transmission, because it has no individuals, no venues, and no contact network. It has only rates. Parameters in these models are typically fit by hand from aggregate observations, with no principled mechanism to learn from heterogeneous data streams or update as new observations arrive.

2.2 Large-Scale Heuristic Agents

Agent-based models (ABMs) emerged as a response to this limitation. Rather than collapsing populations into distributions, ABMs represent each individual explicitly as an autonomous agent with its own state, behavioral rules, and local interactions. The intellectual roots of this approach trace to Thomas Schelling’s segregation model [13], one of the earliest demonstrations that individually tolerant agents, each following a simple local preference rule, reliably produce globally segregated outcomes. Schelling originally used coins and graph paper rather than computers, but the core insight was already present: collective

outcomes are not reducible to individual intentions. John Conway’s Game of Life [14] made the same point in a different register: four simple rules applied to cells on a grid produce patterns of extraordinary complexity, none of which were programmed in. Epstein and Axtell’s Sugarscape [15] extended this tradition to social phenomena including migration, trade, disease spread, and cultural formation, demonstrating that simple agent rules could generate phenomena resembling those observed in real human societies.

The ABM tradition has also made significant contributions in economics and finance. Axtell demonstrated that 120 million agents interacting through simple firm formation rules self-organize into a firm size distribution matching US Census data [16]. Farmer and Axtell survey a broad range of economic phenomena, from housing markets to financial crises, where agent-based approaches have produced insights that equation-based models cannot [17]. These models share a common structure: heterogeneous agents interacting through local rules, producing emergent aggregate dynamics.

Over subsequent decades, operational ABMs were built at increasingly large scales, particularly in epidemiology and public health. JUNE [18] constructed individual-level models of England’s population for biosecurity response. EpiHiper [19] modeled state-level populations in the United States for pandemic preparedness. Covasim [20] provided a flexible agent-based model of disease transmission deployed across multiple countries. Repast [21] provided general-purpose infrastructure for large-scale agent-based modeling, deployed across national laboratories and government agencies. These systems demonstrated that ABMs could operate at population scale, preserving individual heterogeneity and local network structure that mean-field models discard.

The limitation of large-scale heuristic ABMs spans three dimensions. First, agent behavior is simple by design: contact rates, compliance probabilities, and mobility patterns are hand-specified by domain experts and fixed throughout execution. These rules do not learn or adapt, and even with simple rules, scaling to millions of agents historically required large supercomputing infrastructure, limiting practical deployment. Second, learning algorithms for ABMs have relied on building offline surrogate models trained on thousands of model runs [22,23], or on REINFORCE-based policy optimization, both of which are extremely sample-inefficient and computationally prohibitive at population scale [24,25]. Simulation-based inference methods [26] have extended this further using neural density estimators, but the fundamental problem remains: learning requires treating the model as a black box and running it many thousands of times before any gradient information is available. Sensitivity analysis, counterfactual evaluation, and parameter fitting are all slow and imprecise under these approaches. Third, ABMs face a persistent gap between model and reality: they are built from aggregate, anonymized, or synthetic data rather than from real-time individual-level data, because centralizing sensitive data about health, location, and behavior creates privacy risks that prevent adoption [27].

2.3 Rich Data-Driven Agents

A third tradition has pursued the opposite trade: sacrificing scale to achieve more realistic, adaptive, and data-driven agent behavior.

Reinforcement learning offered the first principled approach to learned agent behavior in

multi-agent systems. Rather than specifying behavioral rules by hand, RL agents learn policies by interacting with an environment and optimizing for reward. Multi-agent reinforcement learning (MARL) extended this to populations of interacting agents, with applications to game playing, resource allocation, and policy design [25,28]. The AI Economist [24] used two-level RL to model agents and a social planner simultaneously, learning tax policies that balance productivity and equality. These approaches demonstrated that learned agent behavior could produce emergent economic dynamics more realistic than hand-specified rules. MARL faces fundamental scaling challenges: the joint state-action space grows exponentially with the number of agents, making training intractable beyond hundreds of agents in most settings. Recent work has pushed MARL toward more general settings, including n-player general-sum games with heterogeneous players and complex visuomotor skills [29], but scaling to realistic human population sizes remains out of reach. Beyond the computational challenge, MARL requires specifying reward functions that capture what agents are optimizing, a requirement that is easy to satisfy in synthetic environments but deeply difficult in real human populations, where motivations are heterogeneous, partially observed, and context-dependent.

Large language models (LLMs) opened a different path to behavioral realism. Rather than learning behavior through trial and error, LLM-based agents reason from context, drawing on the vast behavioral knowledge encoded in pre-trained models. This approach relies on behavioral knowledge encoded during pre-training rather than on fitting agent behavior to observed population data. Smallville [30] demonstrated that 25 LLM-powered agents, given memories, goals, and the ability to interact, produced emergent social behaviors resembling those observed in human communities: information spread, relationship formation, and coordinated activity. Concordia [31] generalized this framework to diverse physical, social, and digital environments. These systems demonstrated behavioral richness and contextual adaptability that heuristic ABMs cannot match.

These systems operate at radically limited scale. Simulating 25 to 1,000 LLM-powered agents requires thousands to millions of model queries per time step, which is computationally and economically infeasible at population scale. More fundamentally, these systems cannot predict real-world population dynamics: agent behavior is generated from pre-trained priors rather than from observed population data, and there is no mechanism to update that behavior as real-world conditions change. Without grounding in observed outcomes, there is no principled way to assess whether their emergent dynamics match what actually happens in human populations. Together, these limitations mean that rich data-driven agents demonstrate behavioral plausibility but cannot serve as predictive models of collective phenomena.

2.4 Research Challenges

The three traditions surveyed above have each made real progress. Mean-field systems provide tractable models of aggregate dynamics. Large-scale heuristic ABMs demonstrate that populations of millions of agents can be executed efficiently. Rich data-driven agents show that realistic, adaptive behavior is achievable. Yet each tradition has reached a frontier it cannot cross alone. Three problems remain open.

Challenge 1: Scale and behavioral realism together. Real populations interact through multiple coupled channels simultaneously. A person spreads disease through physical contact, responds to economic incentives, and adapts behavior based on social influence, all at once. These channels are coupled: what happens in one affects the others. No existing approach models this at realistic population scale with agents that learn and adapt. Mean-field systems lose the individual entirely. Large-scale heuristic ABMs use fixed rules that do not adapt. Rich data-driven agents capture behavioral realism but only for small populations of hundreds of agents. The field has produced approaches that occupy different corners of the scale-detail space but none that occupy the top-right.

	Mean-Field		Heuristic ABM		Rich Agents	LPM
Agent representation	Aggregate	dis-tribution	Fixed	behav-ioral rules	Learned policy	State and sensi-tivity
Interaction dynamics	Single	aggre-gate	Single	channel, discrete-time	Single channel, learned	Multiplexed, heterogeneous
Population scale	Unlimited		Millions, simple		Hundreds, rich	Millions, rich
Learning from data	None		Surrogate, sample-inefficient		Reinforcement learning or pre-trained priors	Direct from population observations
Deployment	Centralized, of-fine		Centralized, of-fine		Centralized, of-fine	Vectorized, distributed, networked
Privacy	None		None		None	Secure coordi-nation

Table 2.1: Comparison of existing approaches to modeling populations of agents. Mean-field systems, heuristic ABMs, and rich data-driven agents each address different parts of the problem but none addresses all three open problems simultaneously. Large Population Models are introduced in Chapter 3 as a framework that addresses all three.

These three problems define the research agenda of this thesis. Each is addressed by a different component of the framework introduced in Chapter 3.

Challenge 2: Learning from real population data. A population model is only useful if its agents behave like real people. This requires fitting agent behavior and interaction dynamics to observed data. Existing methods are either too slow (surrogate models require hundreds of thousands of runs before any learning begins), too small (reinforcement learning approaches do not scale beyond hundreds of agents), or not grounded in observed outcomes at all (language model agents rely on pre-trained priors rather than real population data).

There is no efficient, principled way to learn agent behavior and interaction dynamics directly from real population observations at scale.

Challenge 3: Coordinating across distributed and private populations. Existing approaches treat population models as centralized offline tools: all agents run on shared infrastructure, all data is accessible to a central modeler, and learning happens after data is collected. Real populations do not work this way. Agents are distributed across institutions and edge devices. Data about individual health, location, and behavior is sensitive and cannot be centralized. Coordination must happen in real time, across organizational boundaries, without any agent or institution revealing private state to others. No existing approach supports populations of agents that coordinate, learn from each other, and act collectively under these constraints.

Table [2.1](#) summarizes how existing approaches address each dimension and where the gaps lie.

Chapter 3

Large Population Models

The preceding chapter identified three gaps that no existing approach fully addresses: modeling populations of agents with realistic behavior, learning from real-world data, and deploying across privacy and infrastructure boundaries. This chapter introduces the formal framework that addresses all three. We call this framework **Large Population Models** (LPMs). The name is intentionally general: an LPM can represent a population of six fishing companies, five hundred patients in a clinical trial, or eight million residents of a city. What defines an LPM is not the number of agents but the commitment to modeling agents as autonomous entities with local states, behavioral rules, and structured interactions, and to learn how agents behave and how their environment responds from real-world population data.

We use *population* to refer to the formal object the framework operates over: a set of N agents indexed by $a \in \mathcal{A}$, with their states, decisions, and interactions. We use *society* to refer to the real-world system the population represents: a structure of relationships and institutions. LPMs are defined over populations; the societies they model emerge from how those populations interact.

3.1 Formal Definition

Consider a population of N agents, indexed by $a \in \mathcal{A}$, evolving over discrete timesteps $t = 0, 1, \dots, T$. Each agent maintains a **state** $s_t^a \in \mathcal{S}$ that captures everything the agent knows about itself at time t : its current condition, its history of past interactions, and any experience accumulated over time. In a biosecurity setting, s_t^a might encode an agent’s health status, employment condition, and memory of recent exposure events. In a supply chain setting, s_t^a might encode inventory levels, recent order history, and observed demand signals from past rounds.

Agents exist within a shared environment governed by interaction parameters $\Phi = \{\phi_j\}_{j \in J}$, where each ϕ_j parameterizes a distinct channel of interaction $j \in J$. These parameters are shared across the population. They belong to the interaction structure, not to any individual agent. In a biosecurity setting, ϕ_{disease} might govern transmission and recovery dynamics, while ϕ_{mobility} governs how agents move and contact one another.

Each agent also tracks a **sensitivity** σ_t^a that captures how its current state responds to changes in the shared interaction parameters:

$$\sigma_t^a = \frac{\partial s_t^a}{\partial \Phi} \quad (3.1)$$

The full agent state in an LPM is therefore the pair $\tilde{s}_t^a = (s_t^a, \sigma_t^a)$. Sensitivity is the connective tissue that links local agent behavior to population-level outcomes: it tells the agent how its trajectory would shift under variations in the conditions it shares with others, and it is the signal through which populations can be calibrated to real-world observations.

3.1.1 The Cognition Module

Each agent applies a **cognition module** $\mathcal{C}_{\varphi_C}(s_t^a)$ that maps its current state to a decision:

$$d_t^a = \mathcal{C}_{\varphi_C}(s_t^a) \quad (3.2)$$

The cognition module is local: it depends only on the agent’s own state s_t^a . It is parameterized by φ_C , which may encode a fixed mechanistic rule, a learned neural policy, or a frozen language model conditioned by a structured prompt. The parameters φ_C are agent-local: they govern how an agent reasons from its own state, distinct from the shared interaction parameters Φ that govern how agents influence one another.

Interaction effects from neighbors influence the agent’s decisions indirectly. They update s_t^a through state evolution, and s_t^a carries the memory of past coordination signals forward in time. An agent in Fishbanks does not respond instantaneously to a neighbor’s signal. It accumulates experience across rounds in s_t^a and reasons from that accumulated state.

3.1.2 The Coordination Module

Each agent receives signals from its neighbors $\mathcal{N}(a) \subseteq \mathcal{A}$ through a **coordination module** $\mathcal{K}_{\varphi_K}(s_t^a, \sigma_t^a, \mathcal{N}(a))$. Agents may interact through multiple channels simultaneously (disease contact, economic exchange, social influence) and each channel $j \in J$ produces a message from neighbor n to agent a :

$$M_{an}^j(s_t^n, \sigma_t^n, d_t^n, \phi_j) \quad (3.3)$$

The message depends on the neighbor’s state s_t^n , the neighbor’s sensitivity σ_t^n (how the neighbor’s decision would shift under variations in shared conditions), the neighbor’s decision d_t^n , and the interaction parameters ϕ_j of channel j . Sharing σ_t^n alongside d_t^n is essential: agents can coordinate effectively only when they know not just what their neighbors decided but how those decisions would change under variation. This is the unifying coordination primitive across all instantiations developed in Part II.

The full coordination signal aggregates across neighbors and composes across channels:

$$\mathcal{K}_{\varphi_K}(s_t^a, \sigma_t^a, \mathcal{N}(a)) = \prod_{j \in J} \bigoplus_{n \in \mathcal{N}(a)} M_{an}^j(s_t^n, \sigma_t^n, d_t^n, \phi_j) \quad (3.4)$$

where $\bigoplus$ denotes aggregation over neighbors within each channel and $\prod$ denotes composition across channels. The coordination module is itself parameterized by φ_K , which may be the identity (passing aggregated messages through unchanged), a fixed cryptographic

protocol, or a trainable neural network operating over numerical tensors or natural-language tokens. The form of M_{an}^j , $\oplus$, and the parameterization φ_K defines the interaction mechanism: what agents share, how those signals are combined, and whether the coordination logic itself is learned. This is the component that varies across the algorithms developed in Part II.

Joint Evolution of State and Sensitivity

The agent’s state and sensitivity evolve jointly under a single update rule that integrates cognition and coordination:

$$(s_{t+1}^a, \sigma_{t+1}^a) = f(s_t^a, \mathcal{C}_{\varphi_C}(s_t^a), \mathcal{K}_{\varphi_K}(s_t^a, \sigma_t^a, \mathcal{N}(a))) \quad (3.5)$$

Equation 3.5 is the formal definition of a LPM agent. State and sensitivity update together: changes in Φ influence each agent through its own reasoning ($\mathcal{C}_{\varphi_C}$) and through the coordination signal it receives ($\mathcal{K}_{\varphi_K}$), and these influences accumulate into σ_{t+1}^a via the chain rule.

A Large Population Model takes a population of such agents $\{\tilde{s}_t^a\}_{a \in \mathcal{A}}$, applies cognition and coordination, and returns an updated population $\{\tilde{s}_{t+1}^a\}_{a \in \mathcal{A}}$ of the same form. Three classes of parameters can be learned from real-world observations: the shared interaction parameters Φ governing how agents influence one another, the cognition parameters φ_C governing how each agent reasons from its state, and the coordination parameters φ_K governing how neighbor signals are combined into a coordination signal. Different chapters in Part II focus on learning different subsets of these.

Example. Consider a biosecurity agent population. Each agent a maintains a state s_t^a encoding health status, employment condition, and mobility history. The coordination module has two channels: a disease transmission channel with parameters $\phi_{\text{disease}} = \{\beta, \gamma\}$ governing infection and recovery rates, and a mobility channel with parameters ϕ_{mobility} learned from observed movement data. The message $M_{an}^{\text{disease}}(s_t^n, \sigma_t^n, d_t^n, \phi_{\text{disease}})$ encodes neighbor n ’s infection status, compliance decision, and how that decision would shift under changes in transmission parameters. The agent tracks sensitivity $\sigma_t^a = \frac{\partial s_t^a}{\partial \Phi}$ across both channels. Learning how these interaction parameters shape population dynamics from observed data is the problem that Chapter 4 addresses.

3.2 Simulation and Design with LPMs

The LPM formalism supports two epistemically distinct modes of use, corresponding to two long-standing traditions in the study of multi-agent populations.

In a *simulation* mode, agents represent real-world entities and the framework’s objective is to learn dynamics that reproduce observed population behavior. Coordination structure is treated as a structural assumption derived from domain knowledge or external taxonomies (network topologies, message-passing rules, taxonomies of skills or species). What is calibrated to data are the shared interaction parameters Φ . The success criterion is fit to observed population statistics. This is the agent-based modeling tradition, where heterogeneous agents are simulated to study how individual behavior produces collective phenomena [15,32,33].

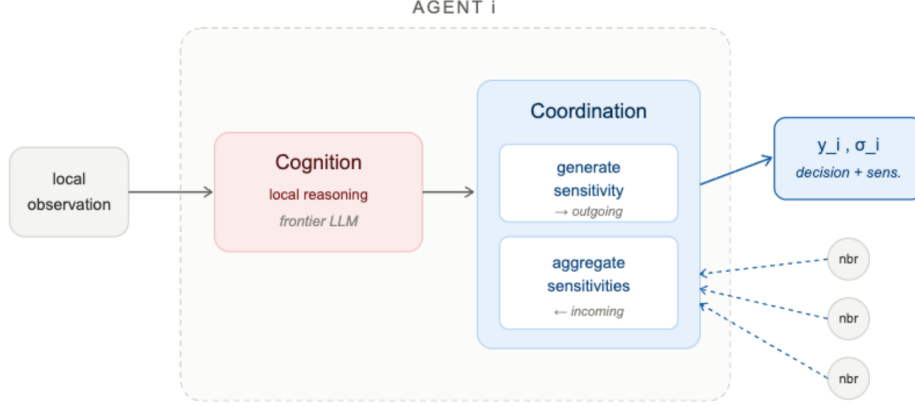


Figure 3.1: A single agent in a Large Population Model. The cognition module $\mathcal{C}_{\varphi_C}$ maps the agent’s state s_t^a to a decision y_i . The coordination module $\mathcal{K}_{\varphi_K}$ generates an outgoing sensitivity message and aggregates incoming messages from neighbors $\mathcal{N}(i)$. The agent emits (y_i, σ_i) at each timestep; the next state $(s_{t+1}^a, \sigma_{t+1}^a)$ is computed via the joint update in Equation 3.5.

In a *design* mode, agents are AI participants whose interactions are the target of optimization. The framework’s objective is to develop coordination protocols that produce desired collective outcomes. Here the coordination parameters φ_K are learned, and the calibration target is collective performance on a task rather than fit to observed data. This is the multi-agent systems and multi-agent learning tradition, where coordination protocols are designed and learned to enable populations of agents to act effectively together [30,34,35].

Both modes share the same formalism (Equation 3.5) and the same core primitive (sensitivity sharing). They differ in what is calibrated to data and what is the target of optimization:

- Simulation mode: Φ learned from observed data; φ_K structural; success measured against population statistics.
- Design mode: φ_K learned from collective performance; Φ may be specified by the modeler; success measured against task outcomes.

Given population-level observations y and population-level statistics $x(\cdot)$ produced by the LPM, the learning problem in either mode is to determine some combination of $\{\Phi, \varphi_C, \varphi_K\}$ that minimizes a loss $\mathcal{L}(x, y)$. In simulation mode, y are observed population statistics and $\mathcal{L}$ measures fit. In design mode, $\mathcal{L}$ is replaced by a reward $\mathcal{R}(x(\cdot))$ on collective outcomes. Different chapters of Part II hold different subsets of $\{\Phi, \varphi_C, \varphi_K\}$ fixed and learn the rest, depending on what is observable in each setting.

A natural question from machine learning practice is whether calibrated parameters generalize beyond the data on which they were fit. The chapters in this thesis treat this differently depending on what each domain’s data permits. Chapters 4 and 5 report held-out validation: parameters are fit on a calibration window and predictive performance is evaluated against subsequent observed dynamics. Chapter 8 validates against reported H5N1

incidence using recall metrics. Chapter 9 validates correlationally against career transitions and AI usage patterns rather than against displacement outcomes, which are not yet observed at population scale. Cross-validation protocols and shared benchmarks for learned agent populations remain an open problem the framework opens for future work, discussed further in Section 10.2.

In all settings, learning proceeds via variants of gradient-based optimization. The gradient flows from the objective back through the coordination module $\mathcal{K}_{\varphi_K}$, through the message functions M_{an}^j , the aggregation operator $\oplus$, and the sensitivity σ_t^a , to the parameters being learned. Each agent tracks σ_t^a as a local signal of how its state responds to the shared interaction parameters Φ . Throughout Part II, we explore different ways of learning from these sensitivity signals across varying population sizes, objective functions, and reasoning modalities.

Most chapters of this thesis operate in simulation mode: Chapter 4 (multiplexed dynamics), Chapter 5 (LLM cognition at scale), Chapter 7 (private coordination), Chapter 8 (biological disruption), and Chapter 9 (labor market disruption). Chapter 6 operates in design mode and lays foundations for the populations of AI agents that Chapter 10 considers.

Setting	How M_{an}^j is computed	How σ_t^a is shared	Learning method
Centralized shared dynamics	Numerical: discrete-time, continuous-time, neural, or discrete-event	Gradients flow through joint compute graph	Gradient-based optimization via backprop through $\mathcal{K}_{\varphi_K}$
Decentralized, private states	Numerical: same as centralized, messages not revealed in plaintext	Each agent computes σ_t^a locally, shared without revealing individual values	Gradient-based optimization via secure aggregation of σ_t^a
Decentralized, natural language	Structured natural language: decision and sensitivity description	Sensitivity expressed as natural language, aggregated by trainable $\mathcal{K}_{\varphi_K}$ (e.g., a small language model)	Gradient-based optimization via TextGrad over φ_K OR supervised fine-tuning on simulated data

Table 3.1: Coordination substrates across LPM instantiations. Each agent tracks σ_t^a as a local sensitivity signal. What varies across settings is how the message M_{an}^j is computed, how σ_t^a is shared, and which parameter class is learned. Each row corresponds to a chapter in Part II.

3.3 AgentTorch: Open-Source Framework for Building LPMs

AgentTorch is an open-source framework for building and deploying Large Population Models. Think PyTorch but for LPMs. Developed as part of this research, it has since been adopted by governments, national laboratories, and Fortune 500 companies globally. AgentTorch provides the infrastructure and algorithms that make LPMs practical at the scale of millions of autonomous agents, from commodity laptops to supercomputers [7]. We highlight three capabilities.

Table 3.2 compares AgentTorch against existing frameworks across the capabilities that matter for building LPMs.

Feature	AgentTorch	Concordia	FLGPU	Mesa
GPU Execution	✓	✓	✓	×
Million-agent Populations	✓	×	✓	~
Differentiable Interactions	✓	×	×	×
Mechanistic Environments	✓	×	✓	✓
LLM Integration	✓	✓	×	×
Neural Composition	✓	~	×	×

Table 3.2: AgentTorch compared to existing frameworks for building agent populations. AgentTorch is the only framework that simultaneously supports differentiable environments, million-agent populations, LLM integration, and neural composition, the combination required to build LPMs at scale.

Capability 1: Efficient Runtime for Millions of Agents. Scaling LPMs to millions of agents requires executing agent state updates (each involving stochastic interactions across sparse network topologies) efficiently and in parallel. AgentTorch provides a runtime that executes all agent updates simultaneously via vectorization and just-in-time compilation on commodity GPU hardware, the default mode for populations up to several million agents, used in Chapters 4 and 5. Because LPMs are differentiable, gradient-based optimization runs directly through the LPM without building a surrogate model as traditional ABMs require, and without the high variance of REINFORCE-based policy optimization. The same defined LPM can then be automatically distributed across multiple GPU nodes via data and model parallelism, or deployed over a decentralized mesh where agents communicate directly over a network. Chapter 9 demonstrates distributed execution at national scale, running 151 million agents representing the US workforce on the Frontier supercomputer at Oak Ridge National

Laboratory. Chapter 6 demonstrates networked execution, deploying agents as independent processes communicating over gRPC.

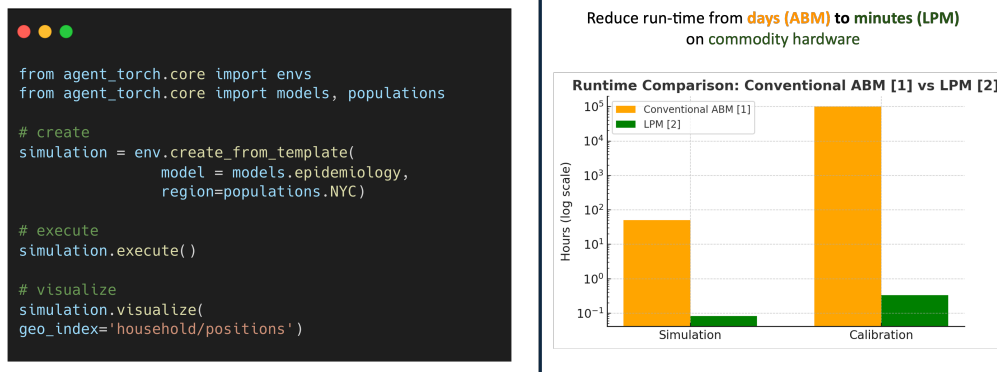


Figure 3.2: Large Population Models achieve dramatic performance improvements over conventional agent-based models. (Left) Simple AgentTorch API enables million-scale simulations with just a few lines of Python code, demonstrated here with an 8.4 million agent NYC epidemiology model. (Right) Runtime comparison shows LPMs reduce simulation time from days to minutes and calibration time from weeks to hours compared to conventional ABMs [36], enabling real-time population modeling on commodity hardware. The differentiable architecture and tensorized execution provide orders-of-magnitude speedups while preserving behavioral richness.

Capability 2: Differentiable Operators for Discrete Agent Interactions. Agent behavior and interactions in LPMs are stochastic and conditional. Agents make discrete choices, contact events are probabilistic, and state transitions depend on neighborhood signals aggregated across multiple interaction channels. These operations break standard automatic differentiation: gradients do not propagate through discrete sampling, and score-function estimators such as REINFORCE introduce prohibitively high variance. Continuous relaxations such as Gumbel-Softmax degrade under the composition of many discrete operations across agents and timesteps.

AgentTorch provides a library of differentiable operators tailored for these settings. These operators preserve discrete stochastic semantics in the forward pass while defining structured backward rules that enable gradient propagation through sampled events (Figure 3.3). For example, `agent_torch.distributions.Categorical` builds on recent work on differentiable programs with discrete randomness for Julia [37], adapting these ideas to support large-scale, vectorized execution across millions of agents. It performs exact sampling during execution and defines a custom backward pass that propagates gradients to the underlying distribution parameters. This enables gradient-based learning through discrete agent interactions without relaxing the dynamics or relying on high-variance score-function estimators.

Empirically, this leads to stable optimization under increasing discrete complexity. Figure 3.4 shows results on two representative tasks. On a Random Walk task, the estimator achieves substantially lower test loss (403× in this setting) compared to Gumbel-Softmax. On a Mixed Pendulum control task, performance is evaluated as the number of discrete choices

increases. Continuous relaxations degrade as the action space grows, while the AgentTorch estimator remains stable, enabling consistent optimization across composed discrete decisions.

```

# Standard PyTorch - no gradient through sampling
from torch.distributions import Categorical

torch_dist = Categorical(logits=agent_logits)
action = torch_dist.sample() # gradient STOP here !!
action.backward() # [None,]

# AgentTorch
from agent_torch.distributions import Categorical

at_dist = Categorical(logits=agent_logits)
action = at_dist.sample() # gradient FLOWS
action.backward() # [0.3, 0.6, 0.8, 0.2]

```

Figure 3.3: `agent_torch.distributions.Categorical` provides gradient flow through discrete randomness, in contrast to Pytorch Categorical distribution. AgentTorch provides a library of distribution helpers that provide low-variance backward estimates while providing hard rules in forward pass.

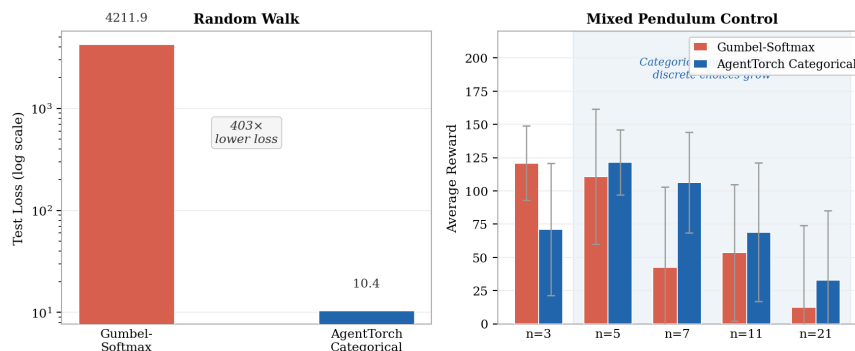


Figure 3.4: `agent_torch.distributions.Categorical` enables stable gradient-based optimization through discrete stochastic operations. Left: on a Random Walk task, the estimator achieves substantially lower test loss (403× in this setting) compared to Gumbel-Softmax. Right: as the number of discrete choices increases in a Mixed Pendulum control task, Gumbel-Softmax degrades, while the AgentTorch estimator remains stable.

Capability 3: Scalable Optimization across Multiplexed Agent Dynamics. LPMs evolve through multiplexed interaction channels at every timestep, where neural modules, mechanistic dynamics, and stochastic operators are composed to update agent states. When unrolled over time, this produces a computation graph that is both wide (across agents and interaction channels) and deep (across timesteps). For a population of N agents interacting over T timesteps through $|J|$ channels, the resulting computation graph contains $O(N \cdot T \cdot |J|)$ operations. Standard reverse-mode automatic differentiation requires storing intermediate activations across this graph, leading to memory that grows linearly with T . As shown in

Figure 3.5, this quickly becomes prohibitive at population scale, limiting optimization over the long time horizons required to capture collective dynamics.

AgentTorch addresses this using a hybrid differentiation strategy aligned with the structure of LPMs. Forward-mode differentiation is applied through the temporally unrolled simulation, propagating sensitivities alongside agent states without storing the full computation graph. Let s_t denote the agent state at timestep t , and define the sensitivity $\sigma_t = \frac{\partial s_t}{\partial \Phi}$. Given the transition $s_{t+1} = f_t(s_t, \Phi)$, sensitivities evolve as:

$$\sigma_{t+1} = \frac{\partial f_t}{\partial s_t} \sigma_t + \frac{\partial f_t}{\partial \Phi} \quad (3.6)$$

This recursion computes Jacobian-vector products at each timestep, allowing sensitivities to be propagated through time without materializing the full computation graph.

Within each timestep, parameterized components such as neural modules are differentiated using reverse-mode automatic differentiation, where backpropagation is efficient. Together, this yields end-to-end gradients with memory that is constant in the number of timesteps.

This strategy achieves memory usage that is constant with respect to the number of timesteps, making gradient-based optimization tractable for LPMs with millions of agents interacting across long horizons. It is implemented in AgentTorch as HybridAD.

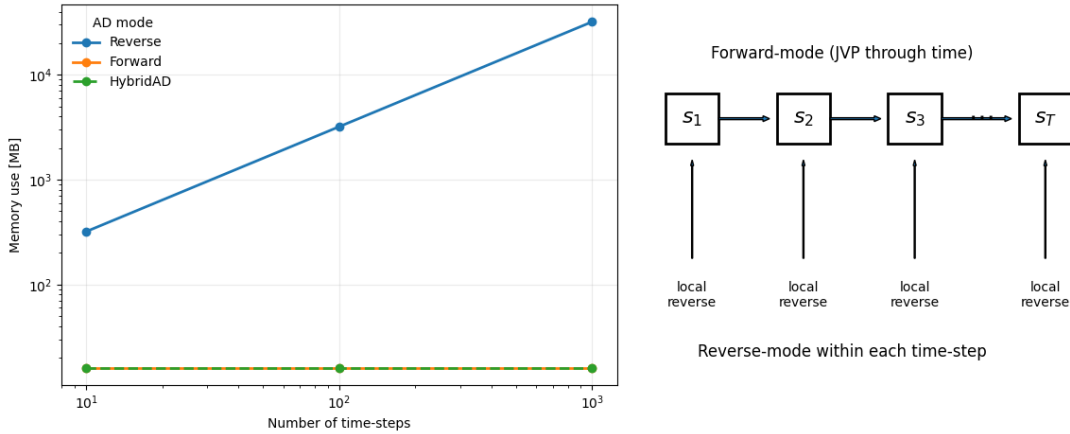


Figure 3.5: Hybrid differentiation in AgentTorch. Left: memory usage as a function of the number of timesteps. Reverse-mode automatic differentiation grows linearly with the time horizon, while forward-mode remains constant. HybridAD follows forward-mode scaling by avoiding storage of the full temporal computation graph. Right: HybridAD aligns differentiation with the structure of LPMs. Forward-mode propagates sensitivities (Jacobian-vector products) through the temporally unrolled simulation, while reverse-mode is applied to parameterized components within each timestep.

3.4 Motivating Case Studies

The algorithmic contributions of this thesis are developed and validated in two canonical domains: biological disruption and labor market disruption. Both share the structure

identified in Chapter 1: they emerge from populations of interacting agents, they resist analysis by individual-level models, and they require both scale and behavioral realism to understand. Part II uses these domains to validate each algorithmic contribution. Part III deploys the full framework at national scale in each domain.

3.4.1 Biological Disruption

Biological disruptions (zoonotic spillovers, outbreak cascades, pandemic waves) are canonical population-level phenomena. They emerge not from any single pathogen or individual, but from the interaction networks of species, populations, and behaviors through which pathogens propagate. Understanding them requires modeling heterogeneous agents interacting across overlapping contact networks, with behavioral responses that feed back into transmission dynamics.

Three features make biological disruption a demanding testbed for LPMs. First, the agent population is large and heterogeneous: realistic models require millions of agents with different demographic, behavioral, and biological attributes interacting across household, workplace, and community networks simultaneously. Second, the interaction structure is multiplexed: agents interact through disease contact, mobility, and economic networks that are coupled to one another, and disruption in one propagates through the others. Third, the data is heterogeneous and siloed: surveillance, mobility, and behavioral data arrive from different sources at different granularities, often under privacy constraints that prevent centralization.

The algorithmic chapters of Part II validate each contribution against real biosecurity data. Chapter 8 then deploys the full framework at national scale, modeling the interaction networks of livestock, wild birds, and human populations across the contiguous United States for zoonotic spillover risk assessment [38].

3.4.2 Labor Market Disruption

Labor market disruptions driven by artificial intelligence share the same analytical structure at a different scale and through different interaction channels. When AI capabilities change the demand for particular skills, the consequences propagate through skill interaction networks, occupational dependencies, and geographic labor markets in cascades that no individual-level analysis can capture. Workers, firms, and institutions are the agents; skills, occupations, and geographic proximity define the interaction topology; and the collective outcome (which regions and occupations are exposed, which are insulated, how disruption ripples through the full economy) is an emergent property of the interaction structure.

Three features make labor market disruption a demanding testbed for LPMs. First, the agent population is large: the US labor market comprises 151 million workers across thousands of occupations and counties, requiring national-scale execution on distributed infrastructure. Second, the interaction structure is complex: skills interact through occupational task structures, occupations interact through industry supply chains, and geographic labor markets interact through mobility and trade. Third, the disruption is largely structural: the exposure of workers to technology displacement is not directly observable in standard economic data, precisely because it operates through interaction networks that aggregate statistics cannot capture.

Chapter 9 deploys LPMs at national scale in this domain, representing the full US workforce as a population of interacting agents and estimating AI exposure across the full interaction network under stated structural assumptions [1].

3.5 Contributions and Roadmap

The formal framework and case studies above motivate four algorithmic contributions and two application contributions. Part II develops four algorithmic contributions, each an instantiation of a specific component of the LPM framework. Part III deploys the full framework at national scale, combining these contributions to address two classes of systemic risk. Each algorithmic contribution instantiates a different component of Equation 3.5 under a different setting, and each is validated against the motivating domains above.

Learning multiplexed agent interactions. We introduce a framework for specifying agent interactions as compositions of discrete-time, neural, and discrete-event dynamics on a centralized vectorized runtime. This instantiates the coordination module $\mathcal{K}_{\varphi_K}$ with $\varphi_K = \emptyset$ (no learnable coordination parameters); learning concentrates on the shared interaction parameters Φ from population data. Developed in Chapter 4 [39,40].

Scaling agent cognition to population scale. We study how the design of the cognition module $\mathcal{C}_{\varphi_C}$ interacts with population scale to determine collective outcomes. We introduce LLM archetypes, an agent architecture that enables rich cognition at population scale through representative agent groupings, and show that neither scale alone nor cognitive richness alone is sufficient: both together produce the best collective outcomes. Developed in Chapter 5 [41].

Learning to coordinate in natural language. We introduce an algorithm for populations of language model agents to coordinate via structured sensitivity messages. This instantiates $\mathcal{K}_{\varphi_K}$ as a trainable small language model that ingests structured natural-language sensitivity messages and produces aggregated coordination signals; φ_K is learned via supervised fine-tuning, enabling coordination across decentralized networked agents. Developed in Chapter 6 [42].

Learning without observing each other. We introduce an algorithm for agents to coordinate with neighbors without directly exchanging individual messages or neighborhood identities in plaintext. This instantiates $\oplus$ in $\mathcal{K}_{\varphi_K}$ as a secure aggregation operator, preserving the learning objective while hiding individual contributions. Developed in Chapter 7 [43].

Biological disruption at national scale. We deploy LPMs for biosecurity risk assessment, modeling interacting populations of livestock, wild birds, and humans across the contiguous United States. This is the first national-scale digital twin for zoonotic spillover risk assessment, enabling commodity-specific spatiotemporal analysis for biosecurity preparedness. Developed in Chapter 8 [38].

Labor market disruption at national scale. We deploy LPMs at national scale to simulate AI adoption cascades across a skill-skill hypergraph spanning 151 million US workers, 923 occupations, and 3,000 counties. The Iceberg Index is computed from the simulation as a forward-looking measure of workforce exposure, projecting risk patterns fivefold larger than what aggregate statistics detect, distributed across geographies and occupations invisible to traditional measurement. The Index has been formally adopted by multiple US states, monitoring over \$1T of labor exposure as of thesis writing. Developed in Chapter 9 [1].

These six contributions distribute across the two modes introduced in Section 3.2. Five operate in simulation mode (Chapters 4, 5, 7, 8, and 9), where agents represent real-world entities and the learning target is fit to observed population data. One operates in design mode (Chapter 6), where agents are AI participants and the learning target is collective performance on a task. Together, the contributions develop the LPM framework along both modes: the simulation-mode chapters establish the framework as a tool for measuring and modeling existing populations, while the design-mode chapter lays foundations for the populations of AI agents that Chapter 10 argues are now beginning to act in the world.

The contributions also vary in the substrate over which agent coordination operates and in the neural module trained on real-world data through the sensitivity primitive σ_t^a . Table 3.3 summarizes this structure as a roadmap for Parts II and III.

The chapters that follow study different instantiations of $\mathcal{C}_{\varphi_C}$ and $\mathcal{K}_{\varphi_K}$ under varying constraints, including scalability, privacy, and decentralized communication. Together, they show how structured interactions and adaptive agent behavior can be composed to model and learn population-level dynamics across both human and AI agent populations.

Chapter	Coordination strate	sub-	Neural module trained on data
Ch. 4: Multiplexed interactions	Numerical graph over channels	compute multiplexed	CalibNN predicts Φ from heterogeneous context (CDC, mobility, surveys); trained end-to-end via MSE on case counts
Ch. 5: Scaling cognition	Numerical graph with LLM cognition	compute prompted	CalibNN over $(\beta, \gamma_0, \gamma_1)$; LLM cognition prompted from priors, not trained
Ch. 6: Natural-language coordination	Natural-language sensitivity messages	sensi-	Qwen-1.5B post-trained via SFT for sensitivity generation and aggregation
Ch. 7: Private coordination	Encrypted messages via SecureSum	numerical	β recovered via secure gradient flow on Oxford contact graph
Ch. 8: Biological disruption	Mechanistic and HMM over month series	spillover county-	$\alpha(s)$ via Bayesian optimization; HMM parameters from H5N1 incidence
Ch. 9: Labor market disruption	Skill-skill over occupations and regions	hypergraph and	LPM simulates adoption cascades across 151M workers; Iceberg Index is computed from dynamics, validated against career transitions and AEI adoption

Table 3.3: Each chapter instantiates the LPM coordination module $\mathcal{K}_{\varphi_K}$ over a different substrate, paired with a neural module trained on real-world data via the sensitivity primitive σ_t^a . Chapters 4–6 share a common pattern: a neural network is trained end-to-end through the differentiable LPM. What varies is the substrate over which coordination operates (compute graph, language tokens, encrypted aggregation) and what the neural module learns. Chapters 8 and 9 deploy the framework at national scale with domain-specific calibration and validation strategies.

Part II

Algorithms

Chapter 4

Learning Multiplexed Agent Interactions

4.1 Motivation and LPM Instantiation

Recall the Large Population Model formulation from Chapter 3. Each agent a maintains a state s_t^a and a sensitivity $\sigma_t^a = \partial s_t^a / \partial \Phi$. The state and sensitivity evolve jointly under

$$(s_{t+1}^a, \sigma_{t+1}^a) = f(s_t^a, \mathcal{C}_{\varphi_C}(s_t^a), \mathcal{K}_{\varphi_K}(s_t^a, \sigma_t^a, \mathcal{N}(a))) \quad (4.1)$$

where $\Phi = \{\phi_j\}_{j \in J}$ are the shared interaction parameters, $\mathcal{C}_{\varphi_C}$ is the cognition module parameterized by φ_C , and $\mathcal{K}_{\varphi_K}$ is the coordination module parameterized by φ_K . In this chapter, $\mathcal{K}_{\varphi_K}$ takes the multiplexed form

$$\mathcal{K}_{\varphi_K}(s_t^a, \sigma_t^a, \mathcal{N}(a)) = \prod_{j \in J} \bigoplus_{n \in \mathcal{N}(a)} M_{an}^j(s_t^n, d_t^n, \phi_j) \quad (4.2)$$

where j indexes interaction channels and $\bigoplus$ aggregates messages across neighbors within a channel.

In real-world populations, agents participate in multiple interaction channels simultaneously, each following fundamentally different dynamics. Disease transmission events propagate across contact networks through discrete-time stochastic rules. Individual viral evolution unfolds through continuous-time differential equations. Mobility patterns are learned via neural networks fitted to observed movement data. Stimulus interventions arrive as discrete-event logic that conditions agent behavior at specific points in time. These diverse dynamics must be coordinated across individual, household, neighborhood, and city-level scales at each timestep, and they are coupled: what happens in one channel reshapes the conditions under which agents act in every other.

This chapter studies how to represent and learn multiplexed interactions within the LPM framework. The learning objective is to recover the shared interaction parameters Φ from population data, where interaction channels $j \in J$ are specified through their corresponding message functions M_{an}^j , allowing heterogeneous interaction dynamics (discrete-time stochastic, continuous-time differential, event-driven, neural-parameterized) to be composed within the product over channels.

Learning across this heterogeneity requires a composing mechanism that is agnostic to the internal form of each channel's dynamics. Differentiability provides exactly this: if the

full product $\prod_{j \in J}$ is differentiable, then the sensitivity σ_t^a propagates through every channel jointly, whether that channel’s dynamics are discrete-time, continuous-time, event-driven, or neural-parameterized. Gradients are the composing mechanism because they are agnostic to the internal form of each M_{an}^j as long as the composition is differentiable end-to-end.

In LPM terms, this instantiation specifies how the coordination module $\mathcal{K}_{\varphi_K}$ is realized as a multiplexed product over interaction channels. Each channel contributes through its message function $M_{an}^j(s_t^n, d_t^n, \phi_j)$, and the full coordination signal aggregates these contributions across neighbors and channels simultaneously. Critically, $\varphi_K = \emptyset$ in this chapter: the coordination rule itself is structural rather than learned. What is learned is Φ , recovered from population data via gradient flow through the differentiable product. The cognition module $\mathcal{C}_{\varphi_C}$ in this chapter is either hand-specified mechanistic rules or a neural policy with parameters absorbed into Φ , and the aggregation operator $\oplus$ and the interaction structure $\mathcal{N}(a)$ remain as defined in Chapter 3.

The content of this chapter draws from the following publications:

- Chopra et al. *flame: a framework for learning in agent-based models*. AAMAS 2024 [39]
- Chopra et al. *Differentiable Agent-based Epidemiology*. AAMAS 2023 and ICML-W 2022 (Best Paper Award) [40]
- Quera-Bofarull, Chopra et al. *Don’t Simulate Twice: One-Shot Sensitivity Analyses via Automatic Differentiation*. AAMAS 2023 [44]
- Khurana, M.*, Tsui, J.*, Gutierrez, B.*, Chopra, A., (40 others) et al. Kraemer, M. *Global Approaches to Infectious Disease Surveillance and Modelling*. Nature Medicine 2026 [45]

4.2 Background

Agent-based models represent populations as collections of autonomous agents, each maintaining local state and interacting with neighbors through structured rules. This granularity has made them productive tools across biological, physical, digital, and financial domains: from modeling cells in tumor micro-environments [46] and disease spread through contact networks [20] to opinion dynamics on social platforms [47] and firm behavior in financial markets [48]. The expressiveness that makes agent populations useful also makes them computationally demanding. Executing millions of agent state updates across coupled interaction channels, and learning the parameters that govern those interactions from real population data, requires two capabilities that most ABM frameworks were not designed to provide: GPU vectorization for parallel execution across the full agent population, and differentiable composition so that gradient signals can propagate through heterogeneous interaction dynamics jointly. Conventional frameworks such as Mesa and NetLogo [49,50] are neither vectorized for GPU execution nor differentiable for gradient-based optimization, making them incompatible with learning multiplexed agent interactions at scale.

The intellectual path toward differentiable agent population models ran first through scientific computing. Frameworks such as JAX.MD [51] and TorchMD [52] for molecular dynamics, TorchDyn [53] for neural differential equations, and JAXFluids [54] for fluid dynamics

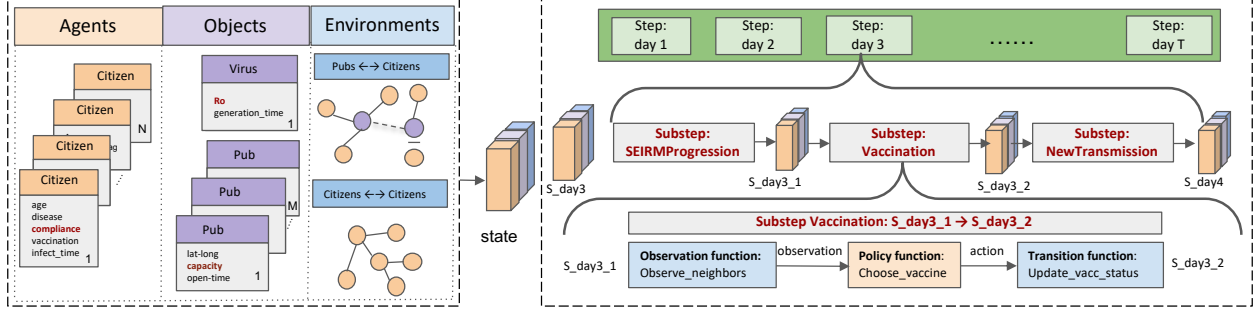


Figure 4.1: **Defining a flame model for spatial epidemiology.** The simulation has N citizens (Agents) that interact through direct mobility and co-locate across pubs (Object) to spread the virus (Object). The simulator state is a collection of properties that describe each of these entities, is initialized once, and transformed during T simulation steps. Each step models the disease progression of infected agents (**SEIRMPgression**), vaccination of susceptible agents (**Vaccination**), and transmission of new infections (**NewTransmission**) to recursively transform the simulation state over these substeps. **flame** is designed to ensure gradient flow through all simulation steps and enables automatic differentiation of any state property or substep function.

demonstrated that equipping domain-specific simulators with automatic differentiation and GPU execution could transform entire fields, enabling gradient-based calibration, neural composition, and sensitivity analysis at scales previously out of reach [55–58]. These results motivated a parallel effort in agent-based modeling.

Several works demonstrated the utility of automatic differentiation for agent populations: accelerating execution on GPUs [59], improving calibration by composing agent populations with deep neural networks [40,60], conducting one-shot sensitivity analyses with gradients [44], and replacing hand-specified rules with learned neural transition functions [61,62]. Each of these was a focused proof of concept in a single domain with a single interaction channel. No general framework existed that could express the full range of agent population dynamics, including discrete-time rules, continuous-time differential equations, neural substeps, and stochastic interventions, in a unified differentiable form that scales to millions of agents. **flame** addresses this gap.

4.3 Differentiable Agent-based Population Models

flame represents agent populations and their interactions over large-scale populations and enables learning using automatic differentiation. First, we define a differentiable agent population model. Second, we introduce the design specification to build an agent population using **flame**. Third, we provide a proof sketch and implementation primitives to validate the differentiability of each **flame** model. Finally, we describe learning with **flame**, which enables building hybrid DNN-agent population pipelines.

4.3.1 Agent Population as a Differentiable Program

Consider an agent population of N agents evolving over T timesteps under shared interaction parameters Φ . The population is differentiable if, given a smooth objective $Y = c(\{s_T^a\}_{a \in A})$, the gradients $\frac{\partial Y}{\partial \Phi}$ and $\frac{\partial Y}{\partial \{s_0^a\}}$ can be computed using automatic differentiation. These gradients are useful for calibrating Φ to population data [40], conducting one-shot sensitivity analyses [44], and composing agent populations with deep neural networks [61]. Achieving this in practice is constrained by two factors: slow sequential execution across large agent populations, and non-differentiable operators in the mechanistic dynamics and interventions that govern agent behavior. **flame** resolves both by providing a composable agent population definition with utilities to implement it via differentiable operators and execute it via fast tensorization on GPU hardware.

4.3.2 Definition of a flame model

Definition 4.1 (flame model). A flame model is defined by the following tuple: $\langle \mathcal{S}, \mathcal{G} \rangle$. These are defined as given below. When reading the terminology below, consider the example in Figure 4.1, which simulates the spread of an infectious disease (like COVID-19).

1. $\mathcal{S} = \langle \mathcal{S}_{Ag}, \mathcal{S}_{Ob}, \mathcal{S}_{Env} \rangle$, represents the set of states of the three kinds of entities in **flame**, which are:
 - (a) *Agents* which observe, act, and interact within a computational world. For instance, these can be infected citizens that spread diseases (with properties like $\{\text{age, disease_stage}\}$).
 - (b) *Objects* which interface between agents but don't have the agency to act. For instance, these can be a virus that carries infection (with properties like $\{\text{RO}\}$) or a pub where citizens co-locate (with properties like $\{\text{lat-long, capacity}\}$).
 - (c) *Environments* which facilitate the interactions of agents with other agents or objects. The interaction graphs are of two types: *agent-agent* and *agent-object*. For instance, a *citizen-pub* (agent-object) graph can represent the interaction of citizens across different pubs in a geo-locality.

Each state property is initialized once to define the initial state and may be transformed during the simulation. This transformation is managed by the substeps defined below.

2. Each episode or rollout of the simulation is assumed to run for T steps, where each step $t \in \{1, \dots, T\}$ comprises multiple substeps. Thus, in **flame**, a substep is the main repeating operational unit, and $\mathcal{G}$ represents the set of substeps within each step. Each substep is composed of the three functions:
 - (a) *Substep Observation* which uses the current state and returns an observation for the agents: $o : \mathcal{S} \rightarrow \mathcal{O}$, where $\mathcal{O}$ is the space of all observations. For instance, an agent can receive an observation regarding the state of infection and vaccination status of its immediate neighbors (`observation = observe_neighbors(state)`).

- (b) *Substep Policy* which uses this generated observation, along with the entire history of earlier observations (compiled as `observation_history`) to yield the agents' actions: $\pi : \mathcal{H} \rightarrow \mathcal{A}$, where $\mathcal{H}$ is the set of trajectories of all historical observations for the agent and $\mathcal{A}$ is the set of all actions (over all agents). For instance, an agent uses its current observations of vaccination by neighbors and historical deaths to decide whether to vaccinate itself
(`action = choose_vaccine(observation_history)`).
- (c) *Substep Transition* which uses the current state and agent actions to generate the next state: $t : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$. For instance, a non-vaccinated agent may choose to accept the dose (`next_state = update_vacc_status(state, action)`).

Note that in each substep, one or more classes of agents may interact with each other and the environment.

4.3.3 Differentiability of a flame model

Proof Sketch Each flame ABM is a finitely iterated, nested composition of differentiable functions where differentiability follows from the chain rule. From the definition above, ABM is composed of repeated simulation steps of function f , such that $X_{t+1} = f(X_t; \theta)$. f transforms X_t by composing k structured substeps invoked in sequence. Each substep i ($=1$ to k) takes the state $(X_{t,i})$ to produce a new simulation state $(X_{t,i+1})$ by composing observation (o_i), policy (p_i) and transition (g_i) functions, all substep functions (o_i, p_i, g_i) are constructed via differentiable (mechanistic and neural) torch and flame operators (section 4.3.6) and, hence are differentiable w.r.t their parameters and inputs. The differentiability of substep functions, via the chain rule, invokes differentiability of each substep and which implies differentiability of step function f . Since f is differentiable, an objective $Y = c(X_N, \theta)$ is differentiable with respect to the parameters θ , state X_N and by chain rule with respect to initial state X_0 .

Implementation flame is built using modern autograd libraries that are compatible with hardware accelerators (GPU/TPU) to ensure differentiability and performance efficiency. In this paper, we use abstractions underlying PyTorch [63], which is a popular autograd framework. Specifically, the simulation state is described using `nn.ParameterDict` and the functions in each substep are defined using the component `nn.ModuleDict` from PyTorch. This design choice allow us to leverage in-built support for tensorized GPU execution (via BLAS/CUDA) and autograd when simulating ABMs. Furthermore, it also streamlines integration of mechanistic ABMs with DNNs to build hybrid DNN-ABM pipelines. Specifically, any substep function, independently of others, can be specified with mechanistic rules or with a deep neural network (DNN). The learning-first design of flame enables calibrating arguments of any rule-based mechanism using autograd and learning parameters of neural specification without affecting the rest. Each substep extends `nn.ModuleDict` class. This design helps flame simulations ensure gradient flow (and parameter tracking) through each substep and across all the steps of a simulation.

4.3.4 High-level modules and API

`flame` has multiple high-level modules to define models, execute simulations and track variables. `Config` and `Runner` are exposed to the user for defining the model and executing simulation episodes. Internally, these interface with `Controller` to initialize the simulator state, register control flow of substeps and track variables.

The model is defined by creating a `config = Config()` object. This `config` enables adding agents and objects, inserting interaction environments, defining metadata and creating simulation substeps. The code listing below uses the `config` to create infectious citizens agents (line 3) and infecting virus object (line 4), define citizen mobility networks (line 6), describe a infection transmission substep (line 7) and execute the simulation for 10 episodes (line 5).

```
1 from flame import Config
2 config = Config()
3 config.add_agents(name="citizens", num_citizens, prop_list)
4 config.add_objects(name="virus", num_strains, prop_list)
5 config.add_metadata("num_episodes", 10)
6 config.add_environment(type="agent-agent", src="from_file", path="
    citizen_citizen.networkx")
7 config.add_substep(name="NewTransmission", active_agents="citizens")
```

Listing 4.1: Using the `flame` `Config` API to define a model

The simulation is executed by creating `runner = Runner(config)` which links with the `Controller`. The user can invoke various `runner` functions which are described below:

- First, `runner.init()` is used to initialize all state properties and create a tracking registry of substep functions (`registry_dict`) via `controller.initialize()`
- Second, `runner.step()` is used to run all steps in a simulation episode. Each episode step invokes all substeps in sequence via `controller.execute_substep(state, registry_dict)`.
- Third, `runner.reset()` is used to reinitialize the state of the simulator before the start of subsequent episodes. While the default is to just use `runner.init()`, this function is often overloaded to specify custom reset functions (as in Case Study 1 using state from prior episodes).
- Fourth, `runner.parameters()` tracks and return all learnable parameters in the simulation episode. These parameters can be properties of the simulator state or arguments of substep functions; and are used when defining optimizers.
- Finally, `runner.trajectory` tracks the simulation state across multiple steps and episodes and, is used to define loss functions and plot outputs.

The code listing below demonstrates use of `flame` to define, simulate and optimize an LPM. Further, (lines 19 and 34-37) shows how `flame` can interface with `torch` to define custom optimizers for LPM parameters and minimize user-specified objective functions.

```
1 from flame import Config, Runner
2 from torch import optim
```

```

3
4 # Step 1: define entities, metadata and substeps
5 config = Config()
6 config.add_agents(...)
7 config.add_objects(...)
8 config.add_environments(...)
9 config.add_metadata(...)
10 config.add_substeps(...)
11
12 # Step 2: create simulation instance
13 runner = Runner(config)
14
15 # Step 3: initialize simulation state and create registry
16 runner.init()
17
18 # Step 4: create optimizer using learnable simulation parameters
19 opt = optim.SGD(list(runner.parameters()), lr=config_lr)
20
21 for episode in range(num_episodes):
22     opt.zero_grad()
23
24     # Step 5: reset state before each episode
25     runner.reset()
26
27     # Step 6: execute all substeps in sequence
28     runner.step(num_steps)
29
30     # Step 7: read the trajectory to extract output
31     trajectory = runner.trajectory
32     output = generate_output(trajectory)
33
34     # Step 8: compute loss and optimize parameters
35     loss = loss_fn(output, ground_truth)
36     loss.backward()
37     opt.step()

```

Listing 4.2: Using `flame` to define simulate and optimize LPMs

4.3.5 Differentiating with `flame`

As stated earlier, using autograd primitives to design `flame` allows to represent an ABM with substeps of dynamics and interventions on a compute graph, which streamlines back-propagation. However, unlike DNNs on computation graphs, defining ABM substeps requires mechanistic operators (such as `torch.max`, `torch.compare`) which are conventionally non-differentiable and can cause incompatibility with autograd. Recent differentiable ABM have used straight-through estimators for discrete distributions [64] to represent stochastic dynamics [40]. However, this is insufficient to differentiate through interventions (eg: offer a vaccine if age < 60; purchase product with min price etc.). For `flame`, we generalize the straight-through-trick [65] to build a library of foundational operator such as `flame.compare`, `flame.max`, `flame.logical_and` etc. For each operator, we define a smooth approximation to obtain the gradient while using the exact function for computation. One such

implementation of `flame.compare`, is given below.

```
1 @flame.helpers
2 def compare(a, b):
3     '''return 1 if a>b; 0 otherwise'''
4     def compare_soft(a, b, hardness=0.8):
5         # approximate gradient
6         return torch.sigmoid(hardness * (a - b))
7
8     def compare_hard(a, b):
9         # exact computation
10        return (a > b).float()
11
12    soft = compare_soft(a, b)
13    return compare_hard(a, b) + soft - soft.detach()
```

Listing 4.3: Differentiable operators like `flame.compare` help define autograd-compatible dynamics and interventions.

4.3.6 Gradient-based Optimization

As explained in the previous section, by only using differentiable operators, `flame` ensures gradient-flow through all substeps of the simulation and thus, is compatible with automatic differentiation. This allows using gradient-based learning to update properties of the state or arguments of any substep function. All learnable parameters across the simulator can be accessed via `runner.parameters()` and used in `torch.optim` to define custom optimizers, which can then be used to optimize any user-specified objective function. `flame` supports both supervised learning (SL) and reinforcement learning (RL) using first-order gradient estimates [66] and leverages the PyTorch API for optimization. There are three modes:

- **Mode 1:** *Optimize scalar/tensor parameters.* For instance, this may involve calibrating the R0 parameter of a virus to death statistics using SL (sec 4.4.2) or learning a purchase policy to maximize expected utility using RL (sec 4.4.3). The optimizer is defined as `torch.optim.SGD(runner.parameters())`.
- **Mode 2:** *Embed DNN inside ABM to learn substep functions.* Often the substep functions in any conventional ABM are defined as simple mechanistic functions such as cellular automata rules. The purpose of the ABM is then to simulate the effect of these simple functions when applied repeatedly. However, this approach cannot facilitate learning structurally unknown substep functions, which can, in-principle be modeled using any universal function approximator. For instance, an unknown function in a substep can be parameterized with a neural network without affecting any other components of the simulator; and then the parameters of this neural network can be learned to reproduce observed simulation output using SL (sec 4.4.1). This neural substep function is defined in the `config` and its parameters tracked in `runner.parameters()`. The optimizer is defined as `torch.optim.SGD(runner.parameters())`.
- **Mode 3:** *Integrate ABM with DNN pipelines.* Instead of optimizing components of the simulation, a `flame` model can become an objective function and provide gradients to learn an

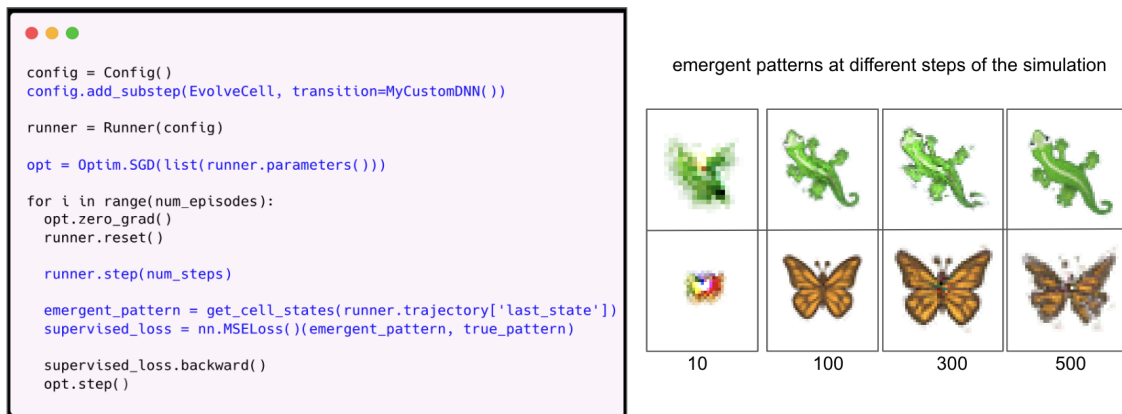


Figure 4.2: **C1: Embed DNN inside ABM with flame.** [61] uses cellular automata to simulate morphogenesis and parameterizes the update rules with a CNN. `flame` is used to learn these rules by representing the substep transition function with a DNN (`MyCustomDNN()`) and optimizing with an MSE loss (`supervised_loss`). Enabling this requires only a few additional lines, as shown in the pseudocode on the left, and is an instance of **Mode 2** in sec 4.3.6. Result, on the right, shows the emergent pattern for two shapes (lizard and butterfly) at different steps along the simulation.

external black-box models (`external_nn`). For instance, they may be used to jointly forecast infections across multiple counties with distinct simulators via SL (sec 4.4.2). The hybrid optimizers can be defined with `torch.optim.SGD(list(external_nn.parameters() + runner.parameters()))` or by using multiple optimizers for the various groups of learnable parameters.

4.4 Experiments and Analysis

Here, we present diverse case studies to show the flexibility of `flame` in definition, simulation, and optimization. These case studies span digital, physical, and biological realms; scale from a few hundred agents in synthetic grid spaces to millions of agents over city-scale contact networks; and involve learning simulation parameters, agent policies, and transition rules. Specifically, these include cells in a bio-electric micro-environment assembling organs, human citizens in a physical environment spreading infections, and avatars in a digital environment sharing opinions. The flexibility in design is coupled with computational benefits realized by tensorization, GPU execution, and support for automatic differentiation which unlocks new capabilities via seamless (end-to-end) integration with deep neural networks. For this analysis, we implement previously introduced simulators using `flame`. The key objective is to demonstrate the capabilities of the design of `flame`, which allows the specification of diverse multi-agent scenarios, executes million-scale simulations, conducts gradient-based optimization, and evaluates interventions in agent-based models.

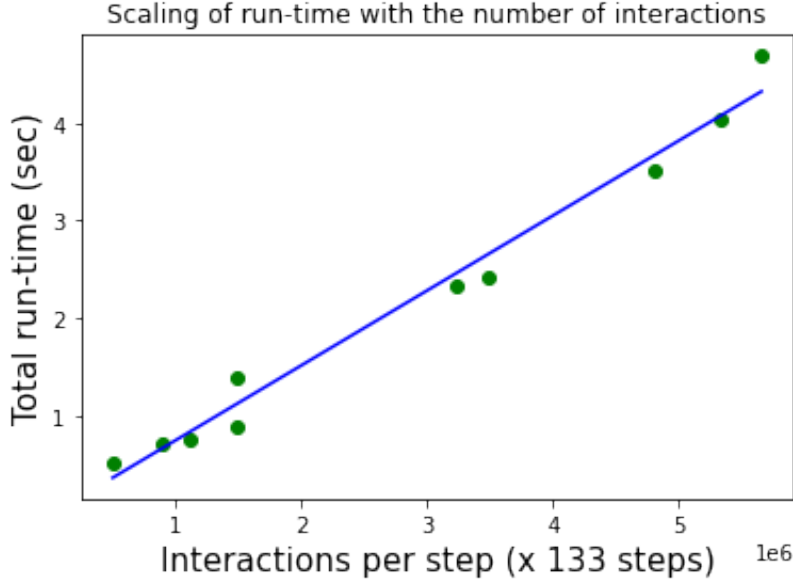


Figure 4.3: Benchmarking run-time performance for `flame` simulations. `flame` can simulate millions of interactions in a few seconds on commodity GPUs which demonstrates real world utility. This plot follows from section 4.4.2

4.4.1 Morphogenesis via Neural Cellular Automata

Morphogenesis is the process of an organism’s shape development where cells interact over bio-electric networks to self-assemble into tissues and organs. The process is extremely robust to perturbations where several species have the ability to regenerate entire organs by repairing damage (to intermediate states) or produce viable organs even from atypical initial states. Understanding the mechanism behind morphogenesis is an active area of research and key to progress in regenerative medicine. We follow from [61], which extends cellular automata (CAs) to identify cell-level rules that result in adaptive and robust morphogenesis. CAs consist of a grid of cells that are iteratively updated with the same set of rules applied to each cell at every step. The new state of a cell depends only upon the state of a few cells in its immediate neighborhood. Conventionally, the rules in CAs are fairly simple deterministic rules. The complexity in the system modeled using CAs is an emergent property of these simple rules. However, in several real-world use cases, we may not know these rules, or these rules may themselves be fairly complex, requiring function approximation with a neural network to track state evolution using these rules. `flame` allows us to define rules of arbitrary complexity using our modular substep architecture, where the function within a rule can be a deep neural network (DNN) as well. In this case study, the goal of the simulation is twofold: a) learn the cell-level mechanism by representing it with a DNN, and b) validate the robustness of the learned mechanism to perturbations in the initial state.

Following [61], in the simulation, agents are cells with a 16-dim property state. Agent-agent interactions are described over a 2D grid space environment. The state is initialized with a single active agent, producing a multi-cellular pattern through simulation. The simulation has a single substep (`EvolveCell`) with a transition function that is parameterized with

	ABM Calibration algorithm	COVID-19			Influenza		
		ND	RMSE	MAE	ND	RMSE	MAE
a	Direct autograd on parameters.(Mode 1)	2.39 $\pm$ 0.39	205.14 $\pm$ 42.56	73.6 $\pm$ 10.8	0.88 $\pm$ 0.14	2.97 $\pm$ 0.44	2.64 $\pm$ 0.43
b	Compose DNN with ABM (Mode 3)	1.15 $\pm$ 0.24	67.09 $\pm$ 23.89	35.5 $\pm$ 7.36	0.50 $\pm$ 0.19	1.78 $\pm$ 0.62	1.50 $\pm$ 0.57
c	Compose DNN with multiple ABM (Mode 3)	0.97 $\pm$ 0.18	50.99 $\pm$ 12.12	30.1 $\pm$ 5.60	0.41 $\pm$ 0.02	1.47 $\pm$ 0.06	1.22 $\pm$ 0.06

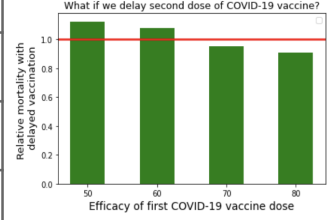


Figure 4.4: **C3-C5: flame** simulate ABMs with millions of agents, build hybrid DNN-ABM pipelines, and jointly optimize multiple ABMs by changing only a few lines of code. **flame** is used to calibrate ABM on 6.5 million agents and forecast the spread of two diseases - COVID-19 and Influenza over different learning situations denoted by (a), (b), and (c). This calibrated model answers policy questions, as shown on the right. The setup follows from [40].

a convolutional neural network (CNN) and describes how cells interact with neighbors to update their state. The simulation output is a 2D grid pattern of all cell states (denoting organism shape), and the learning objective is supervised mean-squared error loss with respect to a pre-specified shape (or grid pattern). The goal is to learn a transition function robust to perturbations in the initial state and involves jointly optimizing over multiple simulations with varying initial states.

flame demonstrates two key capabilities in this case study:

- **Capability 1:** **flame** allows to embed DNN inside an ABM. Here, the transition function of `EvolveCell` substep is parameterized with a CNN. This is captured by `runner.parameters()` and can be used with an optimizer as shown by pseudocode in Figure 4.2.
- **Capability 2:** **flame** enables joint optimization and parameter sharing between multiple ABMs. Here, multiple **runner** objects created for different initial states utilize a shared optimizer.

4.4.2 Spatial Infectious Disease Epidemiology

Infectious diseases spread through contact with infected agents and have two phases: transmission to new agents and disease progression in infected agents. Modeling both phases is crucial for designing effective interventions. For COVID-19, this involved deciding lockdowns, vaccination schedules, and testing strategies. These decisions are complex, requiring consideration of population scale, individual behavior, and intervention properties. Decision-making is further complicated by delayed feedback from interventions and their non-linear interactions. Examples of such interventions include formulating and implementing effective public health policies during COVID-19, including decisions such as delaying the administration of the second vaccine dose [67], prioritizing test speed over specificity [68] etc. Evaluating these decisions in-silico requires granular and data-driven simulations, fast calibration, and sensitivity analysis. For this case study, the goals of this simulation are: a) recreate infection dynamics

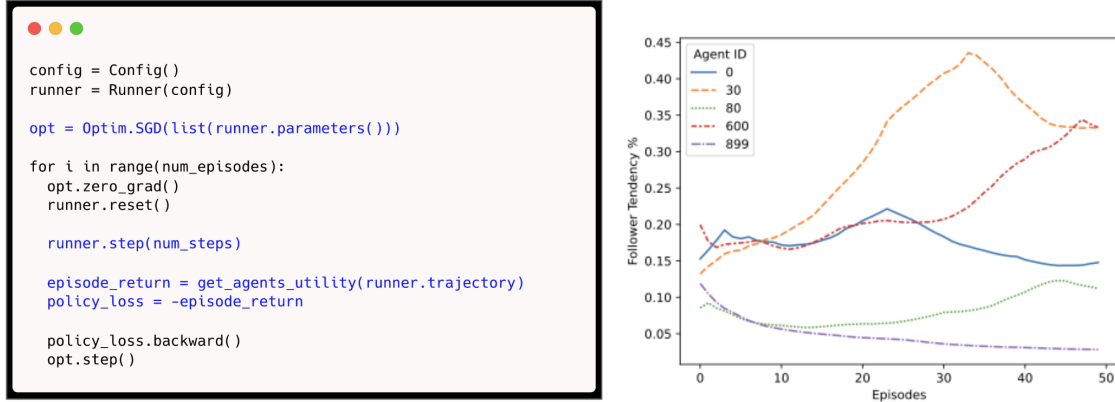


Figure 4.5: **C6 - Policy optimization with flame**. In social opinion dynamics [47], purchase behavior is parameterized with a follower tendency, and agents learn personalized optimal follower tendencies to maximize utilities over timesteps (`episode_return`) by minimizing `policy_loss`. This policy learning is enabled easily in `flame` with parameters in `runner` as shown in the pseudocode on the left. The learning curve with follower tendencies versus episodes for a few sample agents is shown on the right. The experimental setup primarily follows from [47], except for the learning dynamics. More details are given in the appendix.

in real-world million-scale populations, b) improve the calibration of simulation parameters using DNNs, c) analyze the sensitivity of diverse interventions. Specifically, we model spread of COVID-19 and Influenza over 6.5 million people across 12 counties in Massachusetts, calibrate using real CDC data and validate performance by forecasting county and state-level infections consistent with CDC guidelines.

Following from [40,59,67,69], in the simulation, agents are citizens with 5-dim property state (age, occupation, disease-stage, infected-time, vaccine-status) that spread Covid-19 infection. Objects include both the infecting virus and co-location centers like pubs, schools, and care homes. Environments are obtained using real-world contact graphs and describe interactions in citizen-citizen mobility networks and citizen-pub co-location networks. The simulator state is initialized with a few infected agents. Each simulation step has two substeps `InfectionTransmission` and `SEIRMPgression`, which describe the transmission of new infections and a SEIRM progression of previously infected agents, respectively. Discrete stochasticity in the simulation is handled by reparameterizing with Gumbel-softmax gradient estimator to ensure differentiability. The simulation output is the histogram of citizen disease stages, and the learning objective for calibration is a supervised loss against ground truth case statistics (from CDC).

`flame` demonstrates the following capabilities in this case study:

- **Capability 3:** `flame` enables realistic simulations with million-scale populations and real-world contact networks, all while abstracting away the engineering complexity from the modeler. The same API scales to millions of agents and can support real-world contact graphs. The run-time performance is benchmarked in Figure 4.3.
- **Capability 4:** `flame` enables composing ABMs and DNNs end-to-end. Here, this pipeline calibrates simulation parameters using gradient-based learning by designing hybrid opti-

mizers. This is evaluated using forecasting results in figure 4.4(left).

- **Capability 5:** `flame` allows flexible experimentation through its modular design. First, `flame` can evaluate policy interventions through white-box scenario analysis. Figure 4.4 (right) shows results that evaluate the efficacy of a delayed vaccination schedule. Second, `flame` can generalize across simulation assumptions by changing a few lines of code. Figure 4.4 (left) shows a model built for COVID-19 can be adapted to Influenza by just replacing a single substep (`SEIRMPrediction` with `SIRSPrediction`). More details about these experiments are in the appendix.

4.4.3 Social Opinion Dynamics

Digital interactions are already ubiquitous and have become increasingly relevant with the advent of autonomous agents. Such agents, trained to act strategically, will become integral to society and business as they redefine interfaces with humans to mitigate vaccine hesitancy against diseases, advertise new products in competitive markets etc. Typically, agents in these systems interact in two ways – directly via communicating with each other (influenced by their individual follower tendencies) and indirectly via affecting the environment or objects. We implement a standard opinion dynamics model, focusing on direct interactions, used in literature to simulate effects such as lock-in of consumer behavior to a particular product or service. In this case study, the goal of this simulation is: a) learn agent policies that maximize utility over time horizon.

Following from [47], in the simulation, consumers and marketers are the two types of agents. Objects include products that the marketers advertise to consumers. We specifically consider a duopoly with two products. Environment describes interactions through consumer-consumer networks defined using the simulation’s grid graphs. The simulation has a single substep `PurchaseProduct` where agents observe the purchase behavior of neighbors to make a discrete purchase decision. The simulation output is the observed utility for all agents, and the learning objective is to maximize each agent’s expected utility over the finite time horizon. Specifically, the agents wish to optimize their cumulative experienced product quality. To achieve this, they need to balance exploitation and exploration in their product purchase choices. Here, exploitation corresponds to the agents sticking to their own opinions based on their historical product usage experience, and exploration corresponds to modifying their opinions about the products using the opinions of their neighboring agents. In terms of learning, in this simulation, we make the follower tendency, i.e., the degree of exploration of the agents, learnable parameters for each agent in the system, which was not explored in the reference paper [47]. The discrete stochasticity in the simulation, arising from agent purchase behavior, is reparameterized with the Gumbel softmax gradient estimator and also other straight-through differentiable approximators of `max`, `min` operators in `flame`. This allows automatic differentiation, with first-order gradient estimates, through time. In principle, a score-function gradient estimate (variant of REINFORCE) can also be used, but are less performant in practice [70,71]. The goal here to demonstrate the utility of `flame` for sequential decision-making, independent of the specific algorithm.

- **Capability 6:** `flame` allows learning agent policies by specifying custom reward functions. End-to-end differentiability of `flame` enables reinforcement learning with first-order policy

gradients [71]. Pseudocode and specific example in the context of opinion dynamics are shown in Figure 4.5.

4.5 Application: Federated Surveillance for Global Outbreak Response

The contributions in this section are based on the following publication:

Khurana, M.P., Tsui, J.L.-H., Gutierrez, B., Chopra, A., et al. “Global Approaches to Infectious Disease Surveillance and Modelling.” *Nature Medicine*, 2026. [45]

The author contributed the differentiable agent-based modeling components to the Nature Medicine work.

4.5.1 The Global Surveillance Problem

Infectious disease outbreaks do not respect institutional or national boundaries, but the data required to model them does. The data generated during an outbreak spans multiple institutions and modalities: patient-level clinical records reside in hospitals, genomic sequences are held in laboratories, case surveillance data is managed by health ministries, and mobility and contact patterns are owned by private companies. Each of these datasets is subject to distinct regulatory frameworks — GDPR, HIPAA, national biosecurity laws — and releasing them, even for scientific purposes, carries real risks of privacy breach and re-identification [72].

The consequence is that most epidemiological models are calibrated on sparse summary statistics rather than the granular individual-level data that would most improve their accuracy. During the COVID-19 pandemic, the mpox outbreak, Ebola, Marburg, and the ongoing H5N1 avian influenza crisis, this pattern has repeated: difficulties in timely data access and interoperability have consistently impeded the timeliness and effectiveness of public health responses [45]. As Khurana et al. describe, despite ongoing efforts to adopt global health data infrastructures, “there remain regulatory, logistical, human resources and computational barriers to data sharing” that no existing epidemiological modeling framework has resolved [45].

The infectious disease modeling community has identified federated approaches — where analyses are performed across distributed datasets but raw data never leaves local institutions — as the paradigm most likely to resolve this tension [45]. Critically, however, making federated approaches work for agent-based epidemiological models requires something that conventional ABMs cannot provide: the ability to propagate gradient signals back through the simulation to distributed data clients. Without this, the federated learning loop cannot close. The differentiable substep architecture of `flame` is precisely what makes this possible.

4.5.2 Federated Calibration with `flame`

The spatial infectious disease case study in Section 4.4.2 established the centralized calibration pipeline: a CalibNN ingests heterogeneous data sources and predicts simulation parameters

(θ_T, θ_P) , which are used to run the **flame** simulation, and gradients of the loss flow back through the simulation to update CalibNN. This pipeline is visualized in Figure 4.6.

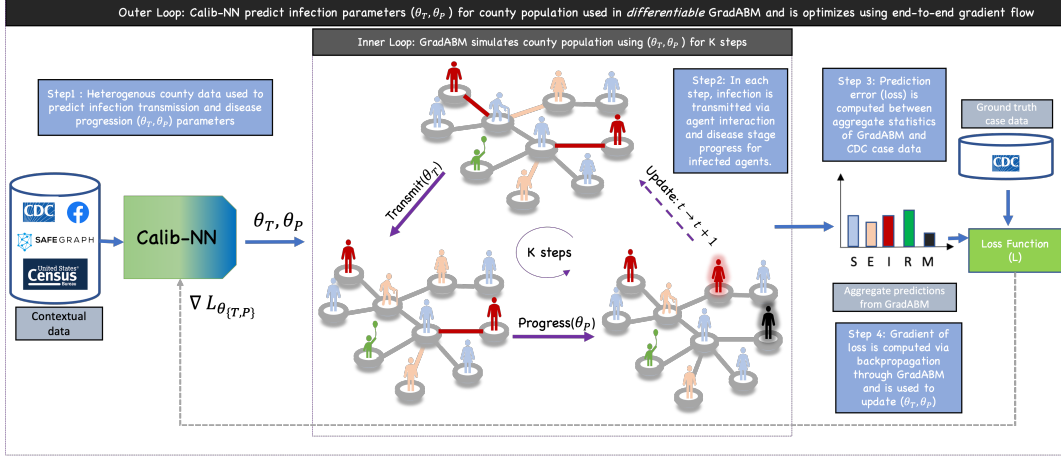


Figure 4.6: Centralized calibration of a differentiable ABM. Heterogeneous macro-level data (CDC, census, mobility, behavioral surveys) is input to a calibration network (CalibNN) which predicts epidemiological parameters (θ_T, θ_P) . These parameters drive the differentiable **flame** simulation, and gradients of the loss flow back through the simulation to update CalibNN. In practice, the contextual data is siloed across institutions due to privacy and regulatory constraints. Adapted from [40].

The key architectural insight for federated calibration is to split this pipeline across institutions. The calibration network C is decomposed into a client model C_a and a server model C_b . Each institution runs C_a locally on its sensitive data, generating only embeddings that are transmitted to the server. The server runs C_b and the **flame** simulation, and propagates gradients back to each client to update their local model. The full distributed protocol is visualized in Figure 4.7 and proceeds as follows:

1. **Local embedding generation.** Each data client i holds its own siloed dataset d_i . Each client generates embeddings $E_i = C_a(d_i; \phi_a)$ from its local data and transmits only these embeddings to the server. Raw data never leaves the institution.
2. **Server-side parameter prediction.** The server merges the client embeddings $E = [E_1, E_2, \dots, E_K]$ and generates simulation parameters $\theta = C_b(E; \phi_b)$.
3. **Simulation execution.** The server executes the **flame** simulation with parameters θ to generate aggregate predictions $\hat{y} = F(X_0; \theta)$ and computes the loss $\mathcal{L}(y, \hat{y})$.
4. **Gradient propagation.** Gradients $\nabla_{\phi_a} \mathcal{L}$ flow back through the **flame** simulation to each client, updating their local model ϕ_a . This step is only possible because **flame** ensures end-to-end differentiability through all simulation substeps, as established formally in Section 4.3.3. A conventional non-differentiable ABM cannot close this loop — without gradients, clients have no learning signal and the federated protocol collapses.

5. **Periodic synchronization.** The local models ϕ_a across clients are periodically synchronized by the server using federated averaging [72], ensuring consistent convergence across non-IID data distributions.

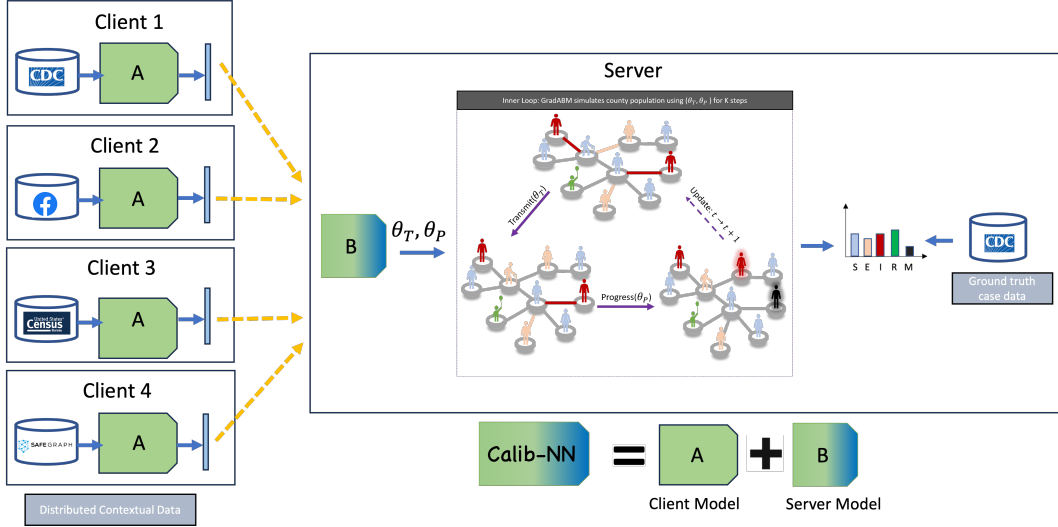


Figure 4.7: Distributed calibration of a differentiable ABM with **flame**. The calibration network is split between data clients (C_a) and a central server (C_b). Each institution generates embeddings from its siloed local data and transmits only embeddings to the server, which merges them, generates simulation parameters, executes the **flame** simulation, and propagates gradients back to clients without any raw data leaving local institutions. Adapted from [72].

4.5.3 Validation and Impact

We validate the federated calibration protocol on the same COVID-19 epidemiological setting described in Section 4.4.2 — counties across Massachusetts, calibrated against CDC ground truth case statistics — but now with the contextual data distributed across four clients rather than pooled in one place [72]: mobility signals from Google Community Surveys (Client 1), symptom surveys from Facebook (Client 2), symptom search data from Google (Client 3), and deaths and hospitalizations from the CDC and Department of Health and Human Services (Client 4).

Table 4.1 reports calibration performance across three conditions.

The distributed protocol achieves calibration accuracy statistically indistinguishable from centralized calibration using fully pooled data — ND of 1.23 versus 1.15, well within one standard deviation — while keeping all raw data decentralized across institutions. Both protocols substantially outperform calibration without any contextual data (ND of 2.79), confirming that the federated embedding mechanism successfully transmits epidemiological signal without transmitting raw data.

This result positions **flame** as the first differentiable ABM framework capable of supporting federated calibration for global outbreak response. The infectious disease modeling community

Table 4.1: Calibration performance of centralized versus distributed **flame** calibration on COVID-19 across Massachusetts counties. Lower is better for all metrics. Results from Garg and Chopra [72].

Protocol	ND	RMSE	MAE
No contextual data	2.79 ± 0.65	160.55 ± 23.59	69.16 ± 10.84
Centralized	1.15 ± 0.24	53.86 ± 14.61	28.43 ± 6.39
Distributed	1.23 ± 0.35	54.74 ± 10.61	30.21 ± 8.45

has consistently identified data centralization as the primary barrier to deploying population-scale epidemiological models across institutions [45]. The **flame** architecture resolves this barrier: institutions that cannot legally or practically share their raw outbreak data — hospitals in data-restricted jurisdictions, laboratories under biosecurity constraints, health ministries in low-resource settings — can still contribute meaningfully to calibrating global epidemiological models, improving the accuracy of forecasts and policy evaluations for all participants in the surveillance network. As the scale and frequency of zoonotic spillovers increases [45], this capability becomes not merely useful but essential.

4.6 Conclusion

This chapter introduced three capabilities that together make differentiable agent population models practical at scale. First, a formal definition that makes agent populations compatible with automatic differentiation: by decomposing each timestep into repeating substeps, each comprising observation, policy, and transition functions, the full agent population dynamics become a differentiable program through which gradients propagate via the chain rule. Second, a library of straight-through differentiable operators that extend differentiability to the mechanistic and stochastic operations that govern agent behavior, including discrete comparisons, conditional interventions, and stochastic state transitions that standard autograd frameworks cannot handle. Third, a composable pipeline that treats the full agent population as a differentiable black-box function, enabling it to be integrated with deep neural networks for calibration, sensitivity analysis, and policy optimization across heterogeneous interaction channels.

Together these capabilities instantiate the product $\prod_{j \in J}$ in the LPM coordination module as a differentiable composition: the interaction parameters Φ governing how agents influence one another across channels can be recovered from population data, and the sensitivity $\sigma_t^a = \frac{\partial s_t^a}{\partial \Phi}$ propagates faithfully through heterogeneous dynamics at population scale. The federated calibration results demonstrate that this holds even when the data required to learn Φ is distributed across institutions that cannot share raw observations.

What this chapter does not address is the cognitive side of each agent. The cognition module $\mathcal{C}(s_t^a)$ remains a fixed hand-specified function, uniform across the population. Whether richer, adaptive cognition changes collective outcomes at population scale, and whether it can be made computationally tractable for millions of agents, is the question Chapter 5 takes up next.

Chapter 5

The Limits of Agency in Agent Populations

5.1 Motivation and LPM Instantiation

Recall the Large Population Model formulation from Chapter 3. Each agent a maintains a state s_t^a and a sensitivity $\sigma_t^a = \partial s_t^a / \partial \Phi$. The state and sensitivity evolve jointly under

$$(s_{t+1}^a, \sigma_{t+1}^a) = f(s_t^a, \mathcal{C}_{\varphi_C}(s_t^a), \mathcal{K}_{\varphi_K}(s_t^a, \sigma_t^a, \mathcal{N}(a))) \quad (5.1)$$

where $\Phi = \{\phi_j\}_{j \in J}$ are the shared interaction parameters, $\mathcal{C}_{\varphi_C}$ is the cognition module parameterized by φ_C , and $\mathcal{K}_{\varphi_K}$ is the coordination module parameterized by φ_K .

The preceding chapters studied different instantiations of the coordination module $\mathcal{K}_{\varphi_K}$, covering how agents interact through multiplexed dynamics, secure message passing, and natural language. This chapter focuses on the cognition module $\mathcal{C}_{\varphi_C}$.

Chapter 4 established the infrastructure for multiplexed agent interactions at population scale. The cognition module $\mathcal{C}_{\varphi_C}$ in that framework reasons numerically: agent decisions are either hand-specified mechanistic rules or neural policies trained through gradient-based optimization on numerical population data. The sensitivity $\sigma_t^a = \frac{\partial s_t^a}{\partial \Phi}$ propagates through tensor operations, and everything the agent knows about its world is encoded in the numerical state s_t^a .

Real human behavior does not reduce to numerical reasoning alone. An agent deciding whether to isolate during a biosecurity disruption responds to how long the disruption has lasted, what their neighbors are doing, whether they received a stimulus payment, and how the situation has been framed in public discourse. These are qualitative, context-dependent signals that are not naturally expressible as gradients through a loss function. When agents reason in this richer way, their decisions propagate through interaction networks differently: behavioral adaptations cascade over space and time, reshaping collective outcomes in ways that numerical cognition cannot capture.

This chapter studies how the design of the cognition module $\mathcal{C}_{\varphi_C}$ interacts with population scale to determine collective outcomes. We ask whether language-grounded cognition changes collective outcomes relative to numerical cognition, and whether that effect depends on the scale of the population in which agents interact. We find that neither scale alone nor cognitive

richness alone is sufficient: richer cognition improves collective outcomes only when paired with population-scale interaction.

This chapter introduces LLM archetypes, an agent architecture that resolves the computational tension by grouping agents into representative types based on their state s_t^a and evaluating the cognition module once per archetype. Individual agents then sample their decisions from the resulting distribution:

$$d_t^a \sim \mathcal{C}_{\varphi_C} \left(s_t^{k(a)} \right) \quad (5.2)$$

where $k(a)$ denotes the archetype that agent a belongs to at time t , and φ_C here is realized as the LLM prompt template that maps archetype state to a decision distribution. This preserves behavioral heterogeneity while reducing the number of cognition evaluations from N to K , where $K \ll N$, allowing agents to exhibit rich, adaptive behavior at population scale.

In LPM terms, this instantiation specifies how the cognition module $\mathcal{C}_{\varphi_C}$ is realized. The coordination module $\mathcal{K}_{\varphi_K}$, the transition function f , and the interaction structure $\mathcal{N}(a)$ remain unchanged as defined in Chapter 3. Critically, φ_C in this chapter is not learned via gradient descent: the prompt template is hand-specified, and the LLM weights are frozen. The chapter’s contribution is the archetype mechanism that makes LLM cognition tractable at population scale, and the demonstration that this changes collective outcomes relative to numerical cognition.

The content of this chapter draws from the following publication:

- Chopra et al. *On the Limits of Agency in Agent-Based Models*. AAMAS 2025 [41]

5.2 Background

Agent population models have traditionally governed agent cognition through heuristic rules derived from observational data or grounded in domain theory. These rules do not distinguish between components that depend on agent behavior and those that depend on environmental dynamics. A concrete illustration is the basic reproduction number R_0 in epidemiological models: R_0 captures the expected number of cases generated by one infected individual, but does not separate whether a high case count arises from agents interacting more (behavioral) or from increased viral infectivity (environmental). Heuristic rules conflate these two sources, limiting both the realism of the model and its utility for policy design.

Large language models offer a path to decouple agent behavior from environmental dynamics. Rather than encoding behavior as a fixed function of numerical state, an LLM-based cognition module takes the agent’s context as natural language and produces a behavioral decision. For example, when modeling the probability that an agent isolates during a biosecurity disruption, an LLM can condition on demographic attributes, neighborhood case counts, pandemic duration, and stimulus payments simultaneously, producing a Bernoulli probability over the isolation decision [73]. This allows the cognition module $\mathcal{C}(s_t^a)$ to capture adaptive, context-dependent behavior that heuristic rules cannot express.

Prior work on LLM-based agent populations has demonstrated the behavioral richness this approach enables. Smallville [74] showed that 25 LLM-powered agents produce emergent

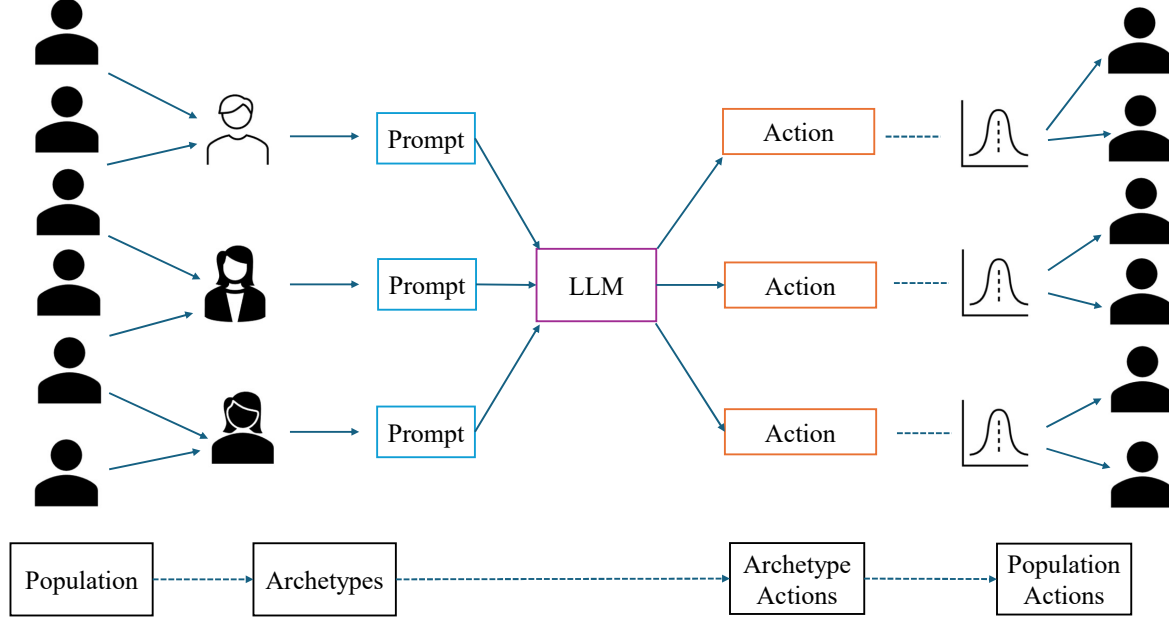


Figure 5.1: Schematic for sampling individual agent behavior using LLM archetypes. The process involves: (1) assigning individuals to representative archetypes (based on prompt template), (2) querying LLMs for archetype behaviors and estimating action distributions, and (3) sampling individual agent decisions from action distribution of representative archetype. This approach enables efficient scaling of adaptive behaviors to large agent populations.

social behaviors including relationship formation and coordinated activity. Subsequent work simulated disease spread over 100 LLM-powered agents [75], macroeconomic dynamics with 300 agents [73], and self-organization among 1000 agents in a shared environment [76]. These systems demonstrated that language-grounded cognition produces qualitatively richer collective behavior than heuristic rules. They share a common limitation: all operate at small population scale, ranging from tens to hundreds of agents, and none are designed to integrate real-world population data [77]. At population scale, querying a language model for every agent at every timestep is computationally infeasible. This is the gap this chapter addresses.

5.3 LLM Archetypes

Understanding complex systems often requires the simulation of the entire population of agents to correctly capture emergent scale-sensitive effects. For instance, while the agency or intelligence of an individual ant may be quite limited, the simulation of the entire colony captures coordination processes wherein ants use themselves as bridges for other ants to use. In these large systems, however, it is infeasible to query 5.3 for each agent, time-step, and specific action. This problem can be overcome by recognizing that the number of different behaviors is typically much smaller than the number of agents. In other words, we only need to query the LLM to inform the behaviour over each unique set of agents’ characteristics. For instance, if we consider that the behavior is purely informed by age and gender, we only need to consider one LLM query per different combination of age and sex. We refer to each of these unique combinations as **archetypes**.

For each possible agent action α , we can estimate its probability p_α in Equation (5.3) using Monte-Carlo,

$$\begin{aligned} p_\alpha(\mathbf{s}_i(t), e(t), \boldsymbol{\theta}) &= \mathbb{E}[\ell(\cdot | \mathbf{s}_i(t), e(t), \boldsymbol{\theta})] \\ &\approx \frac{1}{M} \sum_{j=1}^M \xi_j \quad \text{with } \xi_j \sim \ell(\cdot | \mathbf{s}_i(t), e(t), \boldsymbol{\theta}). \end{aligned} \quad (5.3)$$

By estimating $p_\alpha(k)$ for each archetype k , we can simulate the action of its agent by sampling the action from the archetype to which it belongs. Let K be the number of agent archetypes and A be the number of LLM-queryable actions; we can then simulate the behaviour of all agents with $K \times A$ queries, which will be typically much smaller than the number of agents N , allowing us to scale the simulation to millions of agents. This is shown in Figure 5.1.

5.4 Experimental Setup

We consider a large-scale model of New York City simulating 8.4 million individuals during COVID-19. The pandemic exemplifies the intricate interplay between individual behavior, policy interventions, and environmental factors that our approach aims to capture. Disease spread initially triggered fluctuating mobility patterns and multiple infection waves [78]. Government lockdowns, while controlling spread, caused severe economic consequences, including unprecedented unemployment [79]. Stimulus programs, introduced to mitigate economic hardship and encourage compliance to health measure [80], had unintended effects on labor markets and resource allocation [81]. As the pandemic progressed, "pandemic fatigue" emerged, further complicating public health compliance and economic recovery [82]. This feedback loop between health outcomes, economic conditions, and human behavior is ideal to study the trade-off between simulation scale and individual agency.

Environment: The agent states have static (age, gender, income, occupation) and dynamic (disease, employment status) attributes. We use 2022 American community survey (ACS) for demographic and household characteristics, the Bureau of Labor Statistics for employment data, and the center for disease control (CDC) reports for data on disease dynamics, consistent with prior work [59,67,69]. Agent attributes are specified at census-resolution with demographic and income information discretized into bins. Interactions occur

over household, workplace, and mobility networks, with recreational and workplace mobility parameterized using Google Mobility trends. Our simulations focus on the dynamics of disease spread and labor market. For disease spread, we consider a standard epidemiological model [18,69,83] wherein infection spreads through contact and the probability of agent i getting infected at step t is:

$$p_i(t) = 1 - \exp \left(-\frac{\beta S_i}{n_i} \sum_{j \in \mathcal{N}(i)} I_j(t) \right), \quad (5.4)$$

where $\mathcal{N}(i)$ is the set of neighbors of agent i , S_i the susceptibility of agent i , I_j the infection status of each neighbour, $n_i = \#\mathcal{N}(i)$ the total number of neighbors, and β a structural parameter of the ABM called the effective contact rate. The neighbourhood $\mathcal{N}(i)$ is given by contact networks constructed via household and mobility data in the US census.

For labor market, we consider a standard econometric model [73] which relates participation behavior of individual agents with aggregate unemployment rate at time t ($\mu_{w,t}$) as:

$$\mu_{w,t} = \sum_{j \in \mathcal{N}} \gamma_0 W_j(t) + \gamma_1 C_t \quad (5.5)$$

where $W_j(t)$ is the willingness to work for agent j at time t and C_t is the history of unemployment claim rates in the region, obtained from census data; and γ_0 and γ_1 are the structural parameters.

The epidemiological model forecast cases, while the economic model forecast unemployment rates. The models are coupled through a feedback loop: case numbers affect agents' willingness to work, which influences labor-force participation rates and workplace interaction networks, which in turn affect disease transmission. We implement this environment using the AgentTorch framework [84] which enables differentiate through these stochastic dynamics and scales to large populations (8.4 million agents). The parameters ($\beta, \gamma_0, \gamma_1$) are calibrated to real-world data for cases and unemployment rates, using a standard protocol for differentiable ABMs [60,83,84] (visualized in figure 5.3).

Behavior: We use LLMs to model isolation and employment behavior of individual agents. Our prompt includes agent demographics, disease dynamics, information about extrinsic interventions (stimulus payment) and intrinsic behavior adaptation (duration of pandemic to capture effect of "pandemic fatigue"). The user prompt template, motivated by [73], is given below:

User Prompt

You are a {gender} of age {age}, living in the {location} region and receiving a monthly income of {income}.

The number of new cases in your neighborhood is {cases}, which is a {change}% change from the previous month. It has been {duration} months since the start of the pandemic.

This month, you have received a stimulus payment of {payment} to support your living expenses.

Given these factors, do you choose to isolate at home? (isolation behavior)

Given these factors, do you choose to work? (employment behavior)

"There isn't enough information" and "It is unclear" are not acceptable answers. Give a "Yes" or "No" answer, followed by a period. Give one sentence explaining your choice.

The input prompt receives case numbers and pandemic duration from the past simulation trajectory, instead of ground-truth data, and outputs agents' willingness to work $W_j(t)$ which is further used in simulation. We conduct such auto-regressive prompting for two reasons: i) when simulating for prospective interventions, ground-truth data is not available and hence prompt needs to be specified entirely using simulation, ii) when simulation is un-calibrated, the model peaks may not align well with real-world data. In such case, using ground-truth data is unsuitable for capturing the adaptability of behavior (especially when incorporating time-varying information like infections). Since prompt at step t depends upon simulated trajectory at step $t - 1$, LLMs need to be sampled online during the simulation. As behavior cannot be sampled offline, the trade-off between simulation scale and agent behavior becomes particularly critical. Finally, the stimulus eligibility, timing and amounts are based on the policy implemented in NYC at the time. Specifically, December 2020 - March 2021 overlaps with the second stimulus check which provided adults \$ 600 and additional \$ 600 for every child [85]. We use GPT-3.5 for our experiments.

5.5 Validating Archetypes

The section benchmarks the capacity of Archetypes to prompt behavior consistent with measurable population-wide observations. We use LLM archetypes to prompt individual-level willingness to work for 8.4 million synthetic agents (consider responses only for eligible working adults), focusing on two time-periods: December 2020 to March 2021 (coinciding with the second stimulus round) and March to May 2022 (post-Omicron wave). We test three scenarios with increasing contextual information: (Prompt 1) we only provide demographic attributes of the agent, (Prompt 2) we add information about disease dynamics, (Prompt 3) we further include information about access to stimulus payments. For each scenarios, we initialize 3 (M in equation 5.3) queries per archetypes representing different combinations of unique prompts over the considered time frame. Given the census-resolution (demographic and income attributes binned for privacy) and prompt design (disease dynamics and stimulus information shared by all agents), this approach requires only 400 LLM queries to sample

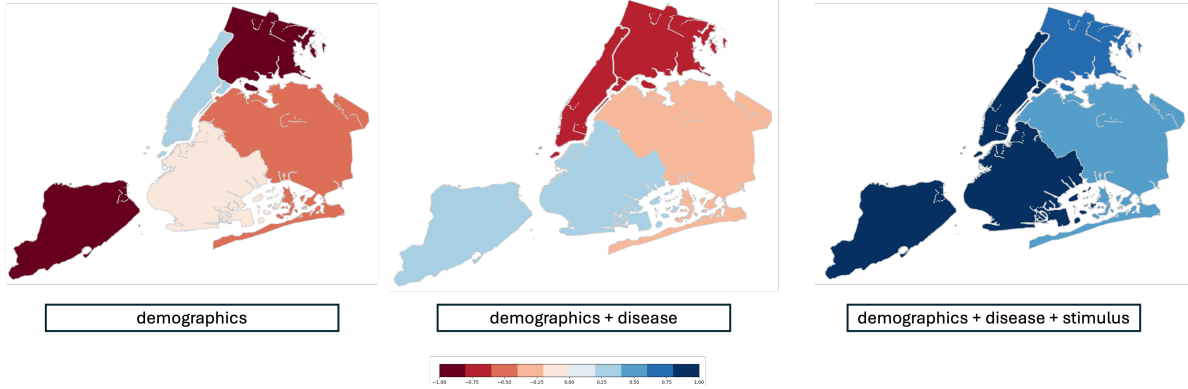


Figure 5.2: **Prompting agents via LLM archetypes:** Correlation between population-wide employment behavior predicted by LLM archetypes and observed data for 8.4 million NYC agents. Prompt 1 (left) corresponds to scenario where LLMs only see demographic attributes. Prompt 2 (middle) and Prompt 3 (right) add further contextual information regarding disease dynamics and stimulus payments. Increased correlation with additional contextual information highlights the ability of LLMs to capture behaviour trends across demographics and geography.

one decision (weekly) for each individual in the population. This is a significant reduction compared to millions of queries required in the conventional paradigm.

Following Equation (5.3), we obtain the probability of each archetype k performing action α ("will you work?") for each week, $p_\alpha(k, t)$. By sampling from the induced Bernoulli distributions, we generate a time-series of work attendance. We aggregate these individual-level actions to calculate the change in labor force participation rate across New York boroughs and compute correlation of this time series with observed data from the US Bureau of Labor Statistics¹. Results are averaged across 5 independent runs for robustness.

Figure 5.2 shows increasing correlation between behavior generated by LLM-archetypes and census data as we add more contextual information to the prompt. This demonstrates the method's ability to incorporate nuanced factors - like evolving environments and incentives - into agent decision making. Notably, archetypes capture positive time-varying correlations in census-level behaviors for 3 of 5 boroughs (roughly 5 million people across income and demographic), which is an encouraging result.

In the second experiment (March to May 2022), we test the ability of LLM-archetypes to simulate adaptive behavior over time, particularly "pandemic fatigue". This period represents the scenario post-Omicron wave, when stimulus and unemployment payments had also declined considerably. We modify the prompt to highlight the time duration since the start of the pandemic and the lack of financial incentives, testing whether our representative (and individual) agents are sensitive to these changes. We then repeat the population sampling procedure described in the previous experiment to obtain the time-series and examine cross-correlation with real-world data on participation rates.

This analysis demonstrates the potential of LLM archetypes to capture complex, adaptive behaviors in large-scale populations while maintaining computational efficiency. Despite

¹<https://www.bls.gov/charts/employment-situation/civilian-labor-force-participation-rate.htm>

high-variance individual responses, archetypes successfully capture positive time-varying correlations with census-level behaviors, an encouraging outcome for large-scale ABMs. To mitigate biased LLM responses, we estimate archetype distributions via multiple LLM generations (M queries per archetype), as motivated by [86]. While these results are promising, we acknowledge ongoing challenges. First, real-world behaviors can be significantly more complex than what our prompt can capture and second, data contamination in LLMs remains an open challenge with no formal technique to design prompts for LLM queries. Despite these limitations, the computational efficiency of archetypes - requiring only 400 queries for 8.4 million agents - compared to millions in the conventional approach represents a significant advance. This efficiency, combined with the ability to capture adaptive behaviors, makes LLM archetypes a promising tool for large-scale ABMs. We demonstrate the viability of this approach for large-scale simulations in Section 5.6.1, presenting performance benchmarking results of online LLM sampling coupled with a highly complex environment.

5.6 Results

5.6.1 Behavior Agency v.s. Simulation Scale

This section investigates the impact of incorporating adaptive agent behavior within ABMs and analyzes the trade-off between agency and simulation scale when calibrating to real-world data. We simulate dynamics of disease spread and labor market in New York City from December 2020 to April 2021 and, compare three kinds of agent architectures: LLM-as-agent, Heuristics and Archetypes. LLM-as-agent instantiates a unique LLM query per individual agent, Archetype instantiates M LLM queries per representative agent and heuristic agents use hand-crafted behaviors shared across all agents. Specifically, heuristic agents use Equation (5.4) and Equation (5.5) as they are. For archetype and LLM-as-agent, these equations are modified to incorporate an action α determined by the LLM output and obtain agent decision to isolate (I_j) and work (W_j). In terms of I_j , this is defined as:

$$I_j^{\text{LLM}} := I_j(t)(1 - \ell_\alpha(\mathbf{s}_j(t), e(t))), \quad (5.6)$$

where ℓ_α is the LLM output for the action, and

$$I_j^{\text{archetype}}(t) := I_j(t)(1 - A_j(t)), \quad (5.7)$$

where $A_j(t)$ is sampled from Bernoulli($p_j(\alpha)$) with $p_j(\alpha)$ is estimated using the LLM (see Section 5.3). Similarly, $W_j(t)$ is also modified for LLM-as-agent and archetypes.

When simulating, we sample agent decisions at each step to execute dynamics and, aggregate infection and employment states after N steps. The aggregated outputs are used to calibrate structural parameters $\theta = (\beta, \gamma_0, \gamma_1)$ to historical time-series of infections and unemployment rates, using the protocol visualized in Figure 5.3. LLMs-as-agents query behavior at individual level and hence using it to simulate 8.4 million agents is computational infeasible. This enforces a trade-off between simulation scale and individual agency. For fair comparison, we fix the prompt budget to 300 LLM queries per step and compare the following configuration: a) Heuristic agents simulate 8.4 million agents using hand-crafted

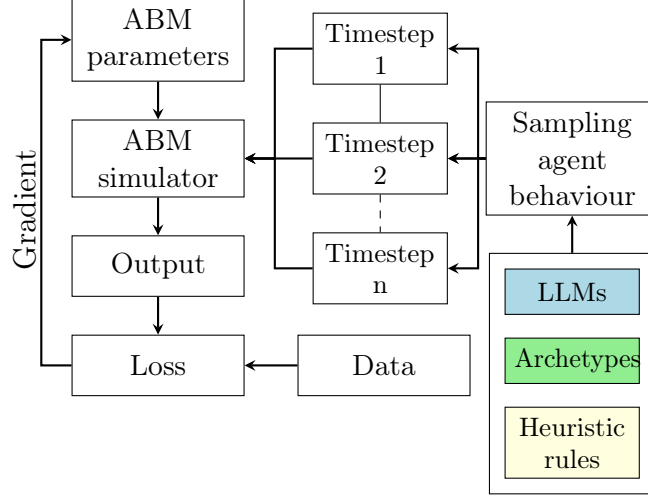


Figure 5.3: Calibration protocol for the three types of agent behaviours considered. This involves simulating ABM by sampling agent behavior at each step, comparing outputs to real-world data, and adjusting parameters through gradient-based optimization.

behaviors, b) Archetypes simulate 8.4 million agents with representative-level LLM queries and c) LLM-as-agents simulate a smaller population of 300 agents with individual-level LLM queries. Output of LLM-as-agents are scaled to the full population to compare with historical data and evaluate performance. To evaluate each calibrated model, we simulate a future time-series of 16 weeks for infection data (measured weekly) and 80 weeks for employment rates (measured monthly) and report forecasting errors.

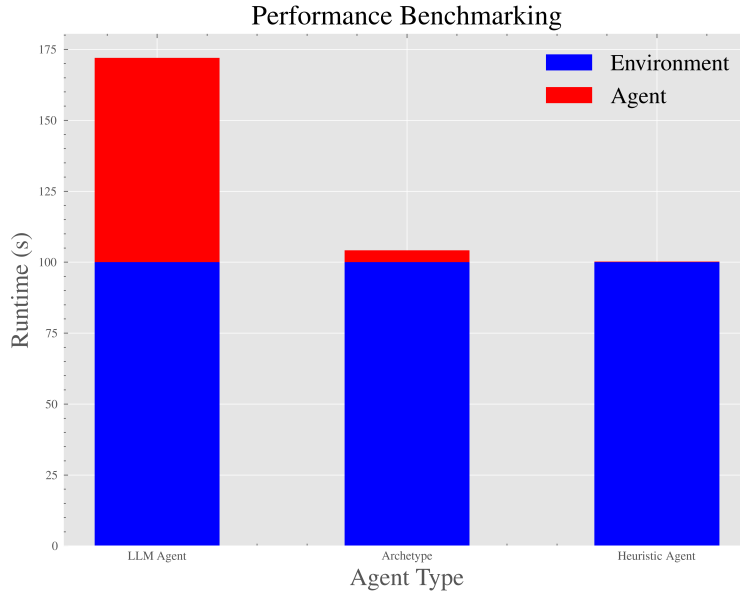


Figure 5.4: Runtime benchmarks for the environment and agent. Archetypes introduce much lower runtime overhead, enabling the simulation to scale to larger population size

Analysis: Results presented in [Table 5.1](#) show that archetype-based model achieves the

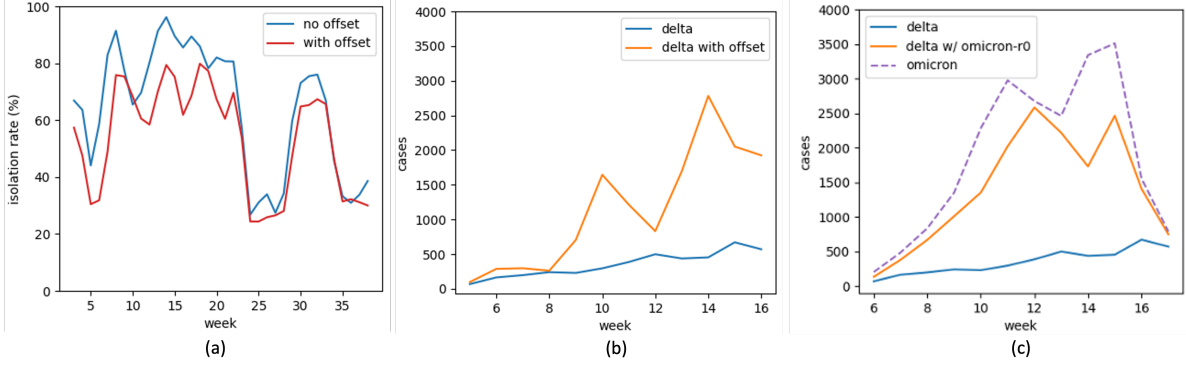


Figure 5.5: LLM archetypes help explore the interplay between behavior adaptation and environment dynamics in shaping epidemic outcomes. (left) Introducing pandemic fatigue ("the offset") to the prompt reduces relative rates of isolation behavior in the population. (middle) This decrease in isolation behavior translates to increased disease transmission in the population. (right) Comparing the original delta wave (in blue), delta wave with "omicron-like" transmissibility (in orange) and the omicron wave (shown in dashed purple to indicate this emerges at a later time). The omicron wave achieves a higher peak than both the original and "omicron-like" delta wave due to coupled impact of viral transmission and time-induced pandemic fatigue.

best performance, highlighting both the need for adaptive and expressive agents and the requirement of simulating the entire scale of the system. Further, heuristic agents simulated at population scale outperform LLM agents constrained to small population samples which shows that computational scaling is crucial. The benefit of LLM-archetypes over heuristic agents shows the benefit of capturing behavioral adaptations can be extremely useful at the right simulation scale. Figure 5.4 shows that Archetypes achieve this superior performance while consuming 95% less run-time compared to LLM-as-agents and marginally more than heuristic agents, which is encouraging for practical utility.

Agent	Error Rate	Unemployment ($\downarrow$)	Infection ($\downarrow$)
Archetype		24.59 $\pm$ 1.5	95.17 $\pm$ 20.23
Heuristic		41.05 $\pm$ 0.1	2914.73 $\pm$ 300.25
LLM-as-agent		56.98 $\pm$ 2.5	4311.70 $\pm$ 674.14

Table 5.1: Benchmark results showing the mean-square errors for each of the considered agent architectures. Archetype achieves lower test error when forecasting both infections and unemployment rates as they capture adaptive agent behaviors *without* compromising simulation scale.

5.6.2 Exploring the Lucas's Critique

The Lucas Critique [87] posits that historical data alone can never predict the outcome of new policy, since behavior may adapt while outcomes are realized. By balancing individual agency and simulation scale, LLM archetypes allow us to analyze the relative impact of

behavior adaptation and environment dynamics in shaping real world outcomes. We explore the interplay of pandemic fatigue and variant transmissibility on severity of disease outbreaks.

We consider transmission of two variants of COVID-19 - Delta ($\beta = [2.5 - 4.0]$, April 2021) and Omicron ($\beta = [5.5 - 8.0]$, November 2021) - which emerged at different stages of the pandemic. While Omicron was roughly 2-3 times more transmissible than Delta, it produced 5-20 times the case intensity [88]. We hypothesize this is due to the coupled dynamics of time-induced fatigue and increased transmissibility. Using our model calibrated to the Delta wave, we conduct two counterfactual simulations:

Q1: What if we had the delta wave later?: To simulate time-induced behavior change ("pandemic fatigue"), we update the user prompt with an artificial offset: *"it has been number of {weeks + OFFSET} since start of the epidemic"*, where OFFSET is 30 weeks.

Q2: What if we had the omicron wave earlier?: We update β in the simulation trace to mimic the Omicron variant while keeping the same behavioral dynamics.

Figure 5.5 illustrates the results of our counterfactual analysis, helping decouple the impact of behavior change and viral transmissibility on disease waves during the pandemic. First, we observe that time-induced fatigue can alter behavior of individuals with agents demonstrating lower willingness to isolate when prompted using the OFFSET(Figure 5.5(a)). This behavior change can result in a more severe outbreak even with viral dynamics do not change (Figure 5.5(b)). Second, an early onset of "omicron-like variant" would have been more destructive due to higher transmissibility, but not as severe as the actual omicron wave. The actual Omicron wave was exacerbated by the coupled influence of behavior change (additional "pandemic fatigue" due to extended duration). Such analyses can inform policy decisions during a pandemic helping appropriately allocate resources to both clinical and behavioral interventions.

We note that archetypes enable such analysis since they provide the ability to query adaptations in individual behavior, via expressive natural language, and also measure the cascading impact of individual decisions at a population scale. These analyses are challenging with other agent architectures due to trade-offs between simulation scale (compromised when using LLMs for each agent) and individual agency (lost when using heuristic agents). We note that for small populations with high-personalized interventions, LLMs-as-agent can be viable architecture (as explored in works like [31,74]). Archetypes are useful when analyzing outcomes in large populations with demographic-resolved interventions as often required for policy making.

5.7 Discussion

This section addresses implementation considerations and limitations. We analyze sensitivity of Archetype-specific design choice when querying LLMs and introduces an API to generalize utility of our contributions to diverse scenarios. We also discuss some promising future capabilities and summarize limitations of the work.

LLM Consistency: LLM archetypes are sensitive to the quality and consistency of agent behaviors, which can vary with the choice of model and number of queries per archetype (M in equation 5.3). We repeat the experiment in section 5.3 (using Prompt 3) and analyze sensitivity of individual decisions to model choice (GPT-3.5 vs GPT-4o) and number

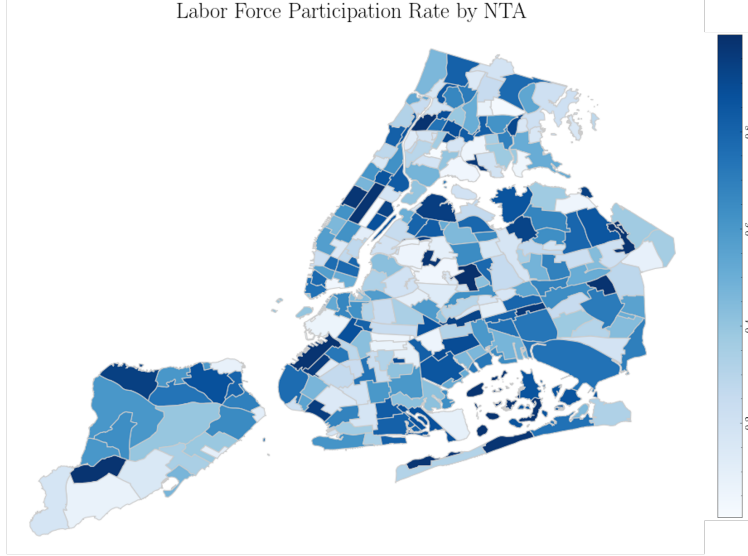


Figure 5.6: Zip-code level employment behavior for 8.4 million agents in NYC estimated using a model calibrated to coarse borough-level data. LLM Archetypes help overcome limitations of historical data for policy design.

of instances per archetype ($M=1, 3, 6$). Results are shown in [Figure 5.7](#). First, for lower M , using the superior GPT-4o model improves performance. Our algorithm is agnostic to the choice of LLMs and we anticipate that our results will become progressively better as LLMs mature, making ABMs more reliable. Second, for larger M - when archetype distributions are aggregated over multiple queries - performance significantly improves for GPT-3.5. We hypothesize that this mitigates biased LLM responses and helps capture realistic variability in individual behaviors. For future work, we plan to extensively benchmark different models (both open and closed-source) and design formal guidelines to specify archetype prompts. Finally, we also note that the choice of LLM is not the only factor affecting effectiveness. As demonstrated in [Section 5.6.1](#), population scale is also critical for ABMs where heuristic agents can outperform LLM agents. This highlights the complex interplay between agent sophistication and simulation scale in determining overall ABM utility.

Archetype API: We extend the AgentTorch framework [\[84\]](#) to integrate an 'Archetype' API, allowing the use of LLMs to prompt agent behavior in large-scale simulations. This API supports both offline and online LLMs, facilitating wider adoption of LLM archetypes in ABMs. We present a code snippet in [Figure 5.8](#).

High-resolution Analysis: LLM archetypes enable measuring granular individual behavior and integrating it with population-scale simulations. Using our model calibrated to borough-level data, we measure the impact of stimulus payments on employment behavior at a granular zip-code level. In future, such analyses can help overcome limitations of historical data in policy design ([Figure 5.6](#)).

Limitations: While our work demonstrates the utility of LLM archetypes in large-scale ABMs, several areas for improvement remain. First, ensuring the robustness and fairness of LLM-driven agents remains an open challenge, as LLMs can produce inconsistent or biased outputs, potentially leading to unrealistic agent behaviors. Future work should focus on

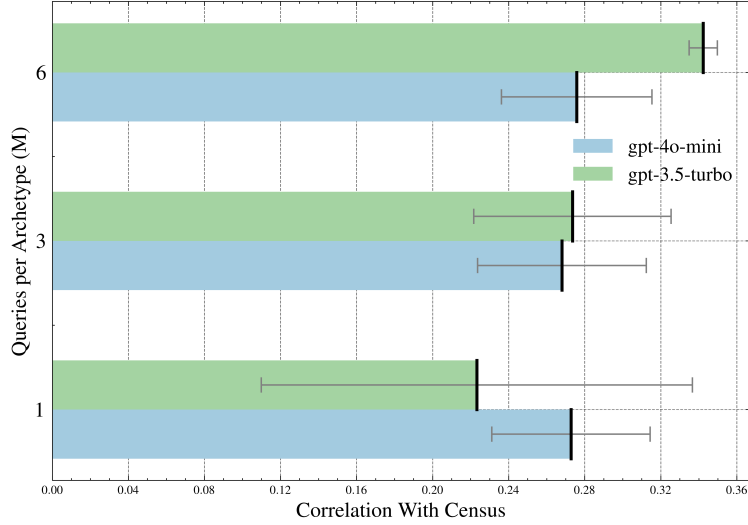


Figure 5.7: Sensitivity analysis to LLM model choice (GPT-3.5, GPT-4o) and number of queries per archetype (M=1, 3, 6)

```

from agent_torch.core import Archetype, Behavior
from agent_torch.populations import NYC

# Create an object of Archetype class
# n_arch estimates a predictive posterior over outcomes
archetype = Archetype(n_arch=7)

# Create an object of Behavior class
work_behavior = Behavior(archetype=archetype.llm(prompt),
                        region=NYC)

will_work = work_behavior.sample()

```

Figure 5.8: We extend the AgentTorch framework to generalize use of LLM Archetypes.

developing methods to detect and mitigate these biases. Second, while LLM archetypes aid scalability, they may not always capture the desired heterogeneity of individual agents, necessitating the development of more sophisticated archetype selection and interpolation methods. Third, the current implementation focuses on relatively simple agent actions, limiting the complexity of decision-making processes that can be modeled. Extending the action space and implementing multi-scale archetypes could address this limitation. Fourth, verifying the accuracy of individual agent behaviors generated by LLMs remains challenging, calling for the development of formal verification methods and benchmarks. We currently measure performance via comparisons with mesoscopic census data and generalization of macroscopic ABM predictions. Despite these limitations, we believe this work represents a significant step forward in agent-based modeling and opens new possibilities to understand and address societal challenges.

5.8 Conclusion

This chapter addressed the cognition module $\mathcal{C}(s_t^a)$ in the LPM framework, asking whether language-grounded reasoning changes collective outcomes and whether it can be made tractable at population scale. The answer to both questions is yes, but with an important qualification: richer cognition improves collective outcomes only when paired with population-scale interaction. At 8.4 million agents, LLM archetypes outperform both heuristic agents at the same scale and LLM-as-agent architectures constrained to small populations. Neither scale alone nor cognitive richness alone is sufficient.

LLM archetypes achieve this by grouping agents into representative types and evaluating $\mathcal{C}$ once per archetype, reducing cognition evaluations from N to $K \ll N$ while preserving behavioral heterogeneity through probabilistic sampling. The result is an agent population where individual behavior is adaptive and context-dependent, grounded in demographic attributes, environmental dynamics, and policy interventions expressed in natural language. The counterfactual analyses on pandemic fatigue and variant transmissibility demonstrate that this behavioral realism translates into qualitatively different collective outcomes that heuristic agents cannot reproduce.

What this chapter leaves open is the coordination side. Agents in the archetype framework reason in natural language internally through $\mathcal{C}(s_t^a)$, but their coordination signals in $\mathcal{K}_a(t)$ remain numerical. Neighbors exchange numerical messages and aggregate them through tensor operations. Whether agents can coordinate through natural language as well, and whether sensitivity signals expressed as natural language improve collective outcomes beyond what numerical coordination achieves, is the question Chapter 6 takes up next.

Chapter 6

Learning to Coordinate in Natural Language

6.1 Motivation and LPM Instantiation

Recall the Large Population Model formulation from Chapter 3. Each agent a maintains a state s_t^a and a sensitivity $\sigma_t^a = \partial s_t^a / \partial \Phi$. The state and sensitivity evolve jointly under

$$(s_{t+1}^a, \sigma_{t+1}^a) = f(s_t^a, \mathcal{C}_{\varphi_C}(s_t^a), \mathcal{K}_{\varphi_K}(s_t^a, \sigma_t^a, \mathcal{N}(a))) \quad (6.1)$$

where $\Phi = \{\phi_j\}_{j \in J}$ are the shared interaction parameters, $\mathcal{C}_{\varphi_C}$ is the cognition module parameterized by φ_C , and $\mathcal{K}_{\varphi_K}$ is the coordination module parameterized by φ_K . In this chapter, $\mathcal{K}_{\varphi_K}$ takes the multiplexed form

$$\mathcal{K}_{\varphi_K}(s_t^a, \sigma_t^a, \mathcal{N}(a)) = \prod_{j \in J} \bigoplus_{n \in \mathcal{N}(a)} M_{an}^j(s_t^n, d_t^n, \phi_j) \quad (6.2)$$

where $\bigoplus$ aggregates messages across neighbors within a channel.

In many practical settings, agents reason and communicate in natural language. Supply chain managers describe inventory decisions in prose. Public health officials report compliance behaviors through structured text. As language models become the substrate for autonomous agents, natural language becomes the default medium of interaction: structured reasoning exchanged across decentralized networks rather than numerical signals over a shared compute graph.

The challenge is coordination, not representation. Agents may correctly diagnose local conditions and still fail to act together at the right moment. Sharing decisions and reasoning traces alone is insufficient: as the Fishbanks result in Chapter 1 showed, agents can individually understand the situation and still produce collective failure. What must be shared is not only the decision d_t^a , but how that decision would change under variations in shared conditions: the sensitivity σ_t^a , expressed here as structured natural language rather than a numerical gradient.

This chapter instantiates M_{an}^j and $\bigoplus$ in the LPM coordination module as natural language representations and aggregation. Each agent n emits a message:

$$M_{an}^j(s_t^n, d_t^n, \phi_j) = (d_t^n, \sigma_t^{n,j})_{\text{lang}} \quad (6.3)$$

where $\sigma_t^{n,j}$ is a structured natural language description of how agent n 's decision d_t^n would change under variations in ϕ_j . The aggregation operator $\oplus$ synthesizes these sensitivity messages across $\mathcal{N}(a)$ into coordination variables that guide agent a 's next decision.

In LPM terms, this instantiation changes two components of $\mathcal{K}_{\varphi_K}$: the representation of M_{an}^j , from numerical signals to structured natural language, and the aggregation operator $\oplus$, from numerical aggregation to language-based synthesis. Everything else is preserved. The neighborhood structure $\mathcal{N}(a)$, the transition function f , and the cognition module $\mathcal{C}_{\varphi_C}$ remain as defined in Chapter 3. Unlike Chapters 4 and 7 where $\varphi_K = \emptyset$, here the aggregation operator is itself learned: φ_K are the parameters of a small language model fine-tuned via supervised learning to synthesize sensitivity messages into coordination signals. The cognition module $\mathcal{C}_{\varphi_C}$ is a frontier language model used with a fixed prompt template; φ_C is not learned.

Our research focuses on designing algorithms for sensitivity sharing in natural language and studying whether populations of agents coordinate better through this mechanism than through direct action and reasoning sharing alone. We validate this instantiation across varying network topologies and agent incentive structures, and against human baselines on Fishbanks and Beer Game, where over thirty years of human performance data provide a meaningful benchmark. A key test is whether structured coordination across agent populations scales where unstructured communication fails, including whether a less capable model coordinating via REP outperforms a more capable model communicating without structured sensitivity sharing.

The content of this chapter draws from the following publication:

- Chopra et al. *The Ripple Effect Protocol: Coordinating Populations of AI Agents via Sensitivity Sharing*. arXiv 2025 (In Review) [42]

6.2 Background

6.2.1 Protocol Stack for LLM Agents

Recent work has produced a set of communication protocols that enable LLM agents to interoperate across platforms. The Model Context Protocol (MCP) standardizes how agents expose tools and share resources. Cross-agent messaging frameworks such as Google's Agent-to-Agent (A2A) [89], IBM's ACP, the Agent Network Protocol (ANP) [90], and research systems like Agora [91] support discovery, authenticated capability exchange, and task hand-offs. For performance-critical settings, SLIM (Secure Low-Latency Interactive Messaging) provides a gRPC-based transport with multicast and publish-subscribe patterns [92]. These protocols form the messaging layer of the emerging Agentic Web, analogous to TCP/IP for networking. They ensure reliable message delivery, routing, and security, but stop short of coordination. While agents can exchange decisions or reasoning traces, there is no mechanism to aggregate information or ensure collective convergence. This creates a gap between communication and true multi-agent coordination.

6.2.2 Coordination in Multi-agent LLM Systems

Prior work on multi-agent LLMs has focused on centralized orchestration, where frameworks such as AutoGen [93], MetaGPT [94], CrewAI [95], and Swarm [96] enforce structure through shared prompts, memory, and control. These systems provide role definitions, task hand-offs, structured workflows, and stopping criteria. These assumptions work well within a single organization but do not generalize to open networks where agents are independently owned.

For decentralized settings, teams often reuse the same communication protocols introduced above (e.g., A2A, ACP, ANP) and rely entirely on the LLMs themselves to negotiate through free-form reasoning and message exchange. However, without explicit coordination mechanisms, this approach is fragile: agents fall into infinite loops, oscillating handoffs, or fail to detect completion [97–99]. For example, two agents negotiating a task may repeatedly hand it back and forth, each waiting for confirmation from the other, creating infinite loops that consume resources without ever reaching completion. These issues worsen as the number of agents grows, highlighting that message exchange alone cannot ensure progress or stability.

Classical distributed mechanisms – such as consensus protocols [100,101], voting rules [102,103], and auction mechanisms [104] provide such structure in traditional systems but assume static preferences and shared objectives. They cannot capture the dynamic, textual, and partially aligned reasoning of LLM agents operating in domains with diverse incentives. Recent studies on LLM voting and census-level preference aggregation [105,106] explore some of these mechanisms in controlled, centralized settings, but real-time decentralized coordination remains an open challenge. This gap motivates the need for protocols that operate above message exchange to provide true coordination capabilities. REP addresses this challenge by introducing a dedicated coordination layer for the Agentic Web, enabling agents to move from merely communicating to collectively aligning their actions in open, decentralized networks.

6.2.3 Sensitivity Sharing in Distributed Systems

Sensitivity sharing has long been used to enable coordination without central control. In distributed optimization, nodes exchange gradients or dual variables rather than final solutions, as in ADMM and gradient tracking algorithms [107,108]. In decentralized simulations, agents similarly exchange lightweight signals that summarize local state instead of sharing full datasets. This enables global calibration while preserving privacy—for example, contact tracing protocols that share infection-state updates without revealing individual health data [43,109]. These approaches assume numerical signals and shared global objectives. In contrast, LLM-based agents reason in natural language, are independently developed, and often operate with partially aligned or competing incentives. Existing sensitivity sharing techniques cannot capture this qualitative reasoning or scale to open, heterogeneous networks. REP builds on this lineage by leveraging textual sensitivity [110] as a coordination primitive, providing the missing layer above current messaging protocols and enabling large populations of LLM agents to coordinate without centralized control.

6.3 Ripple Effect Protocol

REP introduces a new coordination layer for multi-agent systems, enabling agents to move beyond merely exchanging messages to aligning their actions in open, decentralized networks. At its core, REP defines a set of shared *coordination variables* and a process for *sensitivity sharing*, where agents communicate lightweight signals indicating how their decisions would change under different conditions. These sensitivities are aggregated across local neighborhoods using either gradient-style updates or LLM-based synthesis, producing a compact coordination state that guides future decisions. This structure allows diverse LLM agents to coordinate without centralized control, providing scalability and stability missing from current systems.

REP cleanly decouples internal agent cognition from the external coordination mechanism:

- **Agents think:** Each agent uses its internal reasoning (LLM or rule-based) to decide on domain actions and generate sensitivities—signals describing how its decision would change under different environmental conditions.
- **Protocols coordinate:** REP manages the exchange and aggregation of these sensitivities, updating a shared set of *coordination variables* that influence future decisions.

6.3.1 Workflow Overview

REP operates over a network $G = (V, E)$, where each node $i \in V$ represents an agent and edges E indicate direct communication links between neighbors $N_i = \{j \in V \mid (i, j) \in E\}$. Time proceeds in discrete **rounds** $t = 1, 2, \dots$, with each round consisting of four steps:

1. **Receive Messages:** Each agent i collects messages $M_i^t = \{(d_j^t, s_j^t) : j \in N_i\}$ containing neighbor decisions d_j^t and sensitivities s_j^t .
2. **Generate Decision & Sensitivity:** The agent applies its policy π_i to the current coordination variables θ_i^t and private constraints c_i to produce:

$$(d_i^t, s_i^t) = \pi_i(\theta_i^t, c_i)$$

where d_i^t is the domain action (e.g., order quantity, vote, fleet deployment) and s_i^t is a signal describing how the decision would change if conditions shifted.

3. **Aggregate Neighbor Sensitivities:** REP combines neighboring sensitivities to update local coordination variables:

$$\theta_i^{t+1} = \theta_i^t + \text{Agg}_{j \in N_i}(s_j^t)$$

Sensitivities may be numeric or textual, with REP applying either standard optimization rules or LLM-based synthesis as needed.

4. **Consensus (Optional):** In domains requiring group agreement, REP applies deterministic rules such as coordinate-wise median to converge on shared proposals.

Example: In a supply chain, an agent’s decision might be to order 120 units. Its sensitivity could read: “*If demand increases by 10%, increase order by 15 units; if upstream capacity improves, decrease by 5 units.*” These sensitivities are aggregated by neighbors, producing updated coordination variables like `TARGET INVENTORY` and `ORDER ADJUSTMENT FACTOR`, which then shape the next round’s decisions.

6.3.2 Coordination and Aggregation

At each round t , every agent i maintains a local coordination state θ_i^t that influences its next-round decision. Using its internal policy $\pi_i(\theta_i^t, c_i)$, the agent produces two outputs: a domain action d_i^t and a *sensitivity message* s_i^t describing how d_i^t would change under counterfactual conditions. These sensitivities are exchanged among neighbors $\mathcal{N}_i$ and combined to update the agent’s local state.

REP frames this process as a generalized gradient step:

$$\theta_i^{t+1} = \theta_i^t - \eta \cdot g_i^t,$$

where η is a step size and g_i^t is a “gradient-like” signal representing the aggregated influence of neighboring agents:

$$g_i^t = \text{Agg}_\phi(\{s_j^t : j \in \mathcal{N}_i\}, \theta_i^t).$$

The operator Agg_ϕ is parameterized by a synthesis function ϕ , which determines how sensitivities are integrated. REP is modality-agnostic: in some domains, ϕ is a numeric rule such as weighted averaging or distributed gradient descent; in others, ϕ is a language model that synthesizes free-form reasoning into structured, low-dimensional updates. This approach is related to recent methods such as TextGrad [110], which treat natural language as a medium for computing gradient-like updates, but in REP these signals are used to coordinate *populations of agents* rather than optimize a single model.

This unified formulation allows REP to seamlessly handle both quantitative and qualitative coordination signals without changing the overall update rule. For example, s_j^t might encode either production adjustments in a supply chain or price/time flexibility in group scheduling.

In many domains, these local updates alone are sufficient. However, some tasks require the entire population to agree on a shared proposal, such as selecting a common meeting time or price. In these cases, REP adds a second, global consensus step after local aggregation:

$$\bar{\theta}^{t+1} = \text{Median}(\{\theta_i^{t+1} : i \in \mathcal{V}\}),$$

where $\mathcal{V}$ is the set of all agents. This coordinate-wise median prevents extreme local outliers from destabilizing the collective decision while retaining diversity of preferences. The resulting two-level process enables REP to span a broad spectrum of coordination problems, from local alignment in sequential supply chains and resource management to explicit consensus formation in distributed decision-making.

6.3.3 Implementation Architecture

REP is implemented as a lightweight coordination layer that wraps existing agents without modifying their internal policies. A `REPClient` manages message exchange, sensitivity

aggregation, and updates to coordination variables. The design is intentionally modular: the transport backend, aggregation rule, and consensus mechanism can each be configured independently. This makes REP portable across domains and independent of any single communication protocol or decision rule. For example, we use SLIM for multicast messaging and coordinate-wise median for consensus, but other systems or rules can be substituted without changing the protocol logic.

Configuration Interface. Each agent is initialized by specifying which implementations to use for these three components:

```
rep_client = rep.configure(
    agent=llm_agent,
    transport="slim",           # any messaging backend
    updater="textual_grad",    # or "numerical_grad"
    consensus="median_coordinate" # or other rules
)
```

Protocol Execution. During each round, agents run the same loop in parallel: receiving messages, generating local decisions and sensitivities, broadcasting them to neighbors, and performing local and (optionally) global aggregation:

```
# Protocol execution
neighbor_msg, neighbor_sensitivity = rep_client.receive()
rep_client.sync(neighbor_sensitivity) # aggregate neighbor sensitivities
decision, sensitivity = rep_client.decide()
rep_client.send(decision=decision, sensitivity=sensitivity)
```

When textual aggregation is enabled, domain-specific prompt templates guide the LLM in synthesizing free-form reasoning into structured updates. This allows the same core protocol to support diverse coordination problems—from supply chain optimization to distributed consensus—without changing agent logic or protocol implementation.

6.4 Experimental Setup

We evaluate REP across three domains that systematically span different network structures, incentive patterns, and coordination challenges. The Beer Game models sequential coordination in linear supply chains (4–8 agents) with aligned incentives but delayed information flows that create bullwhip effects. Movie Coordination captures consensus formation in sparse social networks (5–20 agents), where heterogeneous preferences must be reconciled despite limited connectivity. Fishbanks examines resource allocation in fully connected networks (5–8 agents) with directly competing incentives, where individual profit maximization conflicts with collective sustainability. Together, these environments test REP across a spectrum of coordination conditions: information cascades versus multi-hop influence versus broadcast interactions; aligned, heterogeneous, and conflicting incentives; and network topologies ranging from linear to sparse to dense.

Table 6.1 summarizes the core elements of each domain, including the decisions made by agents, the coordination variables introduced by REP, and the structural properties of each environment.

Table 6.1: Agent decision variables, coordination variables, and challenges for each domain.

Domain	Agent Decisions	Coord. Variables	Network	Challenge
Beer Game	Order quantities per round	Target inventory levels	Linear chain	Information cascades and bull-whip effect
Fishbank	Fleet deployment across regions	Sustainable quotas	Fully connected	Mixed-motive allocation and over-exploitation
Movie Scheduling	Participation (Y/N) or local preference votes	Preferred price/-time thresholds	Sparse small-world network	Consensus formation with heterogeneous preferences

Our baseline represents the current state of practice in open LLM-agent networks. We use an Agent-to-Agent (A2A) approach where agents share their final decisions and free-form reasoning traces. This mirrors the message exchange patterns supported by protocols such as A2A and ACP, and the coordination style seen in multi-agent orchestration frameworks like MetaGPT, CrewAI, and OpenAI Swarm today. For example, in the Beer Game, an upstream agent might message, “Ordering 12 units because inventory is low.” While this communication enables neighbors to understand intent, it cannot be systematically aggregated to update shared coordination variables. Both REP and the baseline use identical transport, consensus rules, LLM model, and network topologies, ensuring that performance differences are solely due to REP’s introduction of structured sensitivity sharing.

All experiments use Claude-3-Haiku-20240307 (temperature = 0.1, max_tokens = 300) to ensure consistent reasoning power across conditions. Each simulation runs for 20 rounds, with 3–5 trials per configuration to account for stochasticity. Coordination accuracy is reported using domain-specific metrics, detailed in the sections describing each experiment. Final results are aggregated across trials with mean and standard deviation reporting. To validate communication efficiency, we scale the number of agents from 10 to 200 and measure wall-clock coordination time. This test demonstrates that REP’s structured design, combined with SLIM’s low-latency multicast messaging, supports efficient coordination for population scale.

6.5 Results

6.5.1 Supply Chain Coordination

The Beer Game [6] is a canonical benchmark for studying coordination failures in sequential supply chains. It consists of four stages—retailer, wholesaler, distributor, and manufacturer—arranged in a linear chain. In each round, every agent observes only its local inventory and orders from its immediate neighbor, deciding how many units to order upstream. Fixed lead times delay feedback, meaning agents cannot immediately observe the downstream consequences of their actions. This limited visibility causes small fluctuations in customer demand to amplify as they propagate upstream, producing the well-known bullwhip effect [111].

In the baseline Agent-to-Agent (A2A) approach, agents share both order quantities and free-form reasoning traces describing the rationale behind those decisions. While this helps neighbors understand why an action was taken, it does not capture how the agent’s behavior would change under different future conditions. For instance, when a retailer places an unusually large order and explains it as “inventory low due to a local demand spike,” upstream agents still lack visibility into whether this action represents a temporary fluctuation or a longer-term shift. Without explicit modeling of decision flexibility, these free-form messages cannot be aggregated into a stable shared understanding, leaving the system vulnerable to information cascades and runaway amplification.

REP introduces structured sensitivity messages, where each agent communicates how its decision would adapt under key environmental changes. For example, “If demand increases by 10%, increase order by 15 units. If upstream capacity improves, reduce order by 5 units. Current spike seems temporary.” These sensitivities are combined across neighboring agents into low-dimensional coordination variables, such as `TARGET_INVENTORY` and `ORDER_ADJUSTMENT_FACTOR`, which guide subsequent decisions. By circulating structured decision flexibility rather than only final actions and free-form text, agents gain predictive context that allows them to distinguish transient noise from structural change, reducing instability.

Insight 1: REP mitigates bullwhip effects. Under identical LLM agents and network structures, REP reduces total supply chain cost by 41.8%, from \$7,300 (A2A) to \$4,251, and stabilizes demand shocks within 3–4 rounds, compared to 10+ rounds under A2A (Fig. 6.1(a-b)). REP’s structured signals prevent the repeated over- and under-reactions that drive oscillations in traditional systems, leading to rapid convergence after initial disturbances.

Insight 2: Textual sensitivities outperform numerical gradients REP supports both numerical aggregation—where sensitivities are treated as structured derivatives—and textual aggregation, where natural language reasoning is synthesized into compact updates. Textual sensitivities yield lower total cost (4,251 vs 4,680, a 9.2% improvement; Fig. 6.1)(c) by capturing nuanced causal relationships, such as supplier behavior or distinguishing short-term volatility from systemic shifts, that numerical signals cannot express.

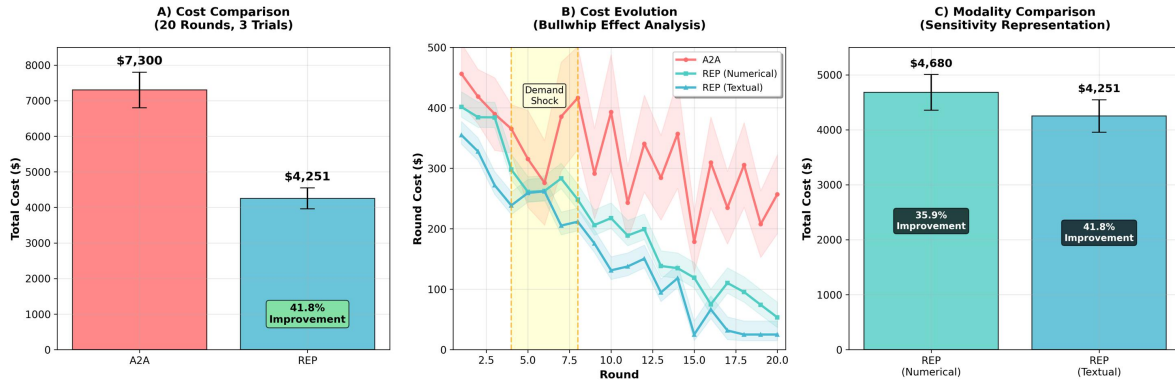


Figure 6.1: Supply chain coordination outcomes over 20 rounds with a demand shock introduced at round 4, where customer demand doubles. (A) Baseline A2A shows persistent bullwhip oscillations as agents repeatedly over- and under-react to incomplete information. REP reduces total supply chain cost by 41.8%, from \$7,300 (A2A) to \$4,251 (B) REP rapidly stabilizes within 3–4 rounds, preventing runaway amplification. (C) REP textual aggregation outperforms numerical aggregation, achieving lower total costs by capturing richer causal reasoning.

6.5.2 Resource Allocation in Tragedy of the Commons

Resource allocation under sustainability constraints exemplifies the tragedy of the commons, where individually rational decisions lead to collectively irrational outcomes [112,113]. The Fishbanks simulation models this dynamic through 12 fishing companies operating in shared waters [5]. Each company aims to maximize profits by deploying its fleet to different fishing zones, but uncoordinated strategies lead to overfishing and eventual resource collapse, harming all participants.

In the baseline Agent-to-Agent (A2A) approach, companies share their deployment decisions and reasoning—such as the number of boats sent to each zone—based on local observations of catch rates and market prices. Without additional structure, agents cannot determine whether others are pursuing short-term opportunistic gains or making long-term sustainability commitments. This information isolation drives competitive escalation: each company acts defensively, deploying aggressively to avoid being left behind, which accelerates resource depletion.

REP extends this setting by allowing companies to communicate textual sensitivities about resource conditions and coordination willingness alongside deployment decisions. For example, “Fish population shows recovery signs based on catch composition. Fleet maintenance costs justify a conservative approach if others also coordinate.”. These sensitivities capture how decisions would shift under different environmental or strategic conditions, enabling agents to infer whether others are willing to restrain harvesting for collective benefit. Aggregating these signals across companies provides a shared picture of ecosystem health and alignment, supporting voluntary, decentralized coordination without a central controller.

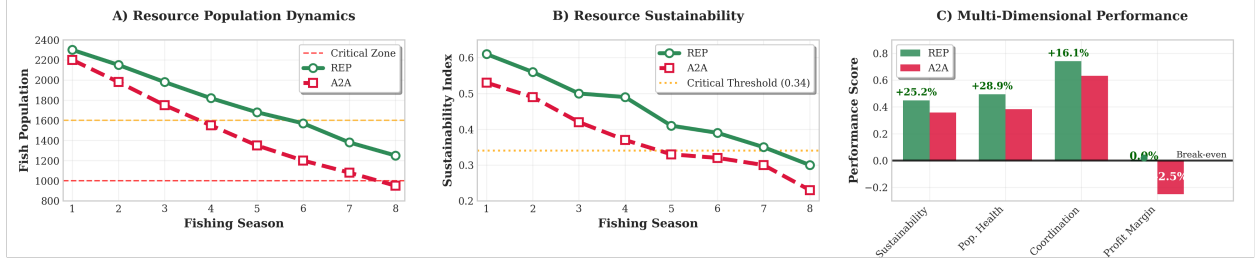


Figure 6.2: Fishbanks simulation with 12 LLM agents over 8 seasons. A demand for fish remains constant while overharvesting leads to eventual depletion. (A) Fish population dynamics showing delayed resource collapse under REP compared to rapid decline in the A2A baseline. (B) Sustainability index with a critical viability threshold at 0.35; REP maintains the resource above this threshold 2 seasons longer than A2A. (C) Comparison of final outcomes across normalized metrics and profit margins. REP improves sustainability (+25.2%), population health (+28.9%), and coordination (+16.1%) while avoiding the financial losses (-2.5%) experienced by A2A agents. Results averaged over five trials.

Insight 3: Coordination under competing incentives. Across 8-season experiments with 12 LLM agents, REP achieves 25.2% sustainability improvement and 28.9% better population health while preventing the financial losses experienced by A2A agents (-2.5%). This demonstrates REP’s capability to coordinate in tragedy-of-commons scenarios—the most challenging multi-agent setting where individual and collective objectives directly conflict. Unlike traditional approaches that result in resource depletion harming all participants, REP enables win-win outcomes through voluntary coordination. This is visualized in figure 6.2

Insight 4: Transparency enables conditional cooperation. The shared sensitivities include both ecosystem assessments and signals of willingness to cooperate. Companies can condition their behavior on others’ stated intentions—e.g., “We can maintain a conservative fleet size if others do the same”. This transparency allows trust to emerge without explicit enforcement, shifting agents from pure profit-maximization toward coordinated stewardship of the shared resource.

6.5.3 Preference Aggregation

Group decision-making in sparse social networks presents a fundamental coordination challenge: agents must converge on a shared outcome despite heterogeneous preferences and limited connectivity. Each agent represents an individual with distinct budget and scheduling constraints, communicating only with immediate neighbors. As a result, preferences must propagate indirectly through the network.

In the baseline Agent-to-Agent (A2A) setting, agents exchange their final participation choices and free-form reasoning. Without structured aggregation, this leads to information isolation: small clusters of neighbors may agree locally, but there is no mechanism to align the full network. This produces fragmented, unstable outcomes, where local proposals conflict and the group fails to converge on a global consensus.

REP introduces a structured preference-sharing process that unfolds in two stages during each round. In the first stage, every agent communicates both its current participation decision and a set of preference sensitivities that capture how its utility would change under adjustments to the shared coordination variables. These sensitivities can take numerical form, such as $\partial U / \partial \text{price} = -0.8$ to indicate high price sensitivity, or textual form, such as “*Price matters more than timing — strongly budget constrained.*” Messages from neighboring agents are then aggregated for the two global coordination variables, **TIME** and **PRICE**, with local updates nudging these variables toward values that reflect the collective preference landscape. In the second stage, agents vote again on the updated proposal, and REP applies a median consensus rule to produce the next shared state. This consensus mechanism provides stability by preventing extreme local outliers from destabilizing the group decision. Through repeated rounds, sensitivities ripple across multiple network hops, allowing the population to converge even when many agents are not directly connected.

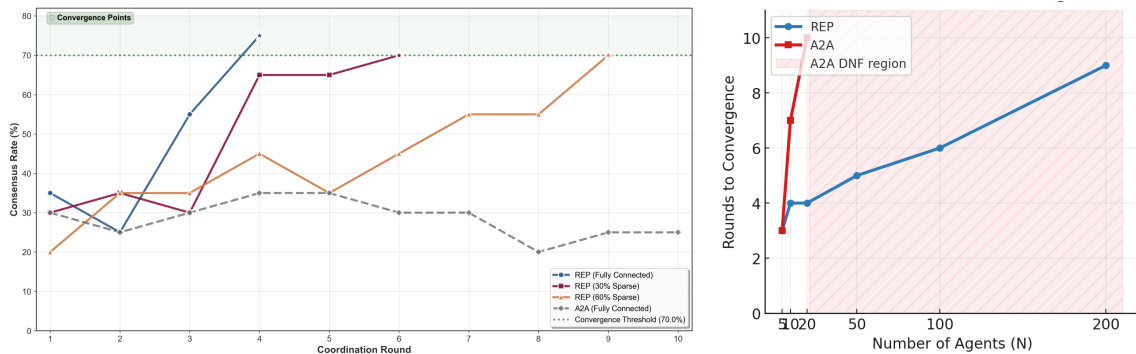


Figure 6.3: REP performance under network sparsity and population scaling. (Left) REP maintains effective coordination even in sparse social networks (30% and 60% sparse), with only gradual increases in convergence rounds (4 $\rightarrow$ 6 $\rightarrow$ 9) while sustaining 70–75% consensus. Stars mark convergence points where the 70% threshold is reached. In contrast, A2A fails to achieve meaningful coordination even when fully connected, showing that sensitivity sharing is essential for global alignment. (Right) As population size grows from 5 to 200 agents, REP continues to converge reliably (3–15 rounds), while A2A requires 7–10 rounds at small sizes and fails entirely beyond 20 agents (shaded DNF region). Together, these results demonstrate REP’s robustness to both sparse connectivity and large-scale populations.

Insight 5: Coordination effectiveness across network sparsity. We evaluate REP on a 20-agent network under three connectivity regimes: fully connected, 30% sparse, and 60% sparse. **REP** achieves **70–75% consensus** across all conditions (Fig. 6.3(left)), with only gradual increases in convergence time: Round 4 $\rightarrow$ Round 6 $\rightarrow$ Round 9 as connectivity decreases. This resilience arises because sensitivity signals propagate indirectly, allowing agents to align even when some connections are missing. In contrast, **A2A** fails to reach meaningful agreement, plateauing at **35% maximum consensus** even in fully connected networks. Without structured aggregation, each agent is overwhelmed by conflicting inputs, leading to cognitive overload and preventing the system from integrating distributed reasoning

into a stable global outcome.

Insight 6: Scalability to larger populations To evaluate REP’s scalability, we increase the network size from 5 to 200 agents at fixed sparsity. Fig. 6.3(right) projects the number of rounds required to reach convergence as population size grows. REP maintains stable performance: convergence occurs in just 3–9 rounds across the entire range, with only modest growth as the network scales. In contrast, A2A requires 7–10 rounds even for small populations (≤ 20 agents) and fails to converge entirely beyond this point, as indicated by the shaded DNF region. Because per-round message cost grows near-linearly with N under SLIM multicast, fewer rounds directly translate to lower total communication. Wall-clock profiling further shows that communication is negligible: at 200 agents, sensitivity sharing is 3% of runtime, with the rest dominated by LLM inference (38%) and wait time (59%). Thus, REP’s scalability is promising for designing an agentic web.

6.6 Conclusion

This chapter instantiated the message function M_{an}^j and aggregation operator $\oplus$ in the LPM coordination module as natural language representations and synthesis. The Ripple Effect Protocol enables populations of agents to share not only decisions but structured sensitivity signals describing how those decisions would change under varying conditions. These signals ripple through local neighborhoods, allowing agents to align collectively without centralized control.

The results across three coordination domains demonstrate that sensitivity sharing in natural language produces substantial gains over decision-only communication. In supply chain coordination, REP reduces total cost by 41.8% and stabilizes demand shocks within 3 to 4 rounds compared to 10 or more rounds under the A2A baseline. In resource allocation under competing incentives, REP achieves 25.2% sustainability improvement and 28.9% better population health while avoiding the financial losses experienced by uncoordinated agents. In preference aggregation, REP maintains 70 to 75% consensus across sparse networks where A2A fails to achieve meaningful agreement. The key finding is that structured coordination across agent populations scales where unstructured communication fails: a less capable model coordinating via REP outperforms a more capable model communicating without structured sensitivity sharing.

What this chapter leaves open is a different constraint on coordination: privacy. The agents in REP share their state s_t^a and sensitivity $\sigma_t^{n,j}$ freely with neighbors. In many real-world settings this is not possible. An agent’s state may encode sensitive information: health status, location, financial position, or organizational strategy. A neighbor may be unwilling or legally unable to reveal which other agents it interacts with. The interaction graph $\mathcal{N}(a)$ itself may be sensitive. Whether agents can still coordinate and learn shared dynamics Φ from population data when individual states and neighborhood identities cannot be revealed in plaintext is the question Chapter 7 takes up next.

Chapter 7

Learning to Coordinate Under Information Asymmetry

7.1 Motivation and LPM Instantiation

Recall the Large Population Model formulation from Chapter 3. Each agent a maintains a state s_t^a and a sensitivity $\sigma_t^a = \partial s_t^a / \partial \Phi$. The state and sensitivity evolve jointly under

$$(s_{t+1}^a, \sigma_{t+1}^a) = f(s_t^a, \mathcal{C}_{\varphi_C}(s_t^a), \mathcal{K}_{\varphi_K}(s_t^a, \sigma_t^a, \mathcal{N}(a))) \quad (7.1)$$

where $\Phi = \{\phi_j\}_{j \in J}$ are the shared interaction parameters, $\mathcal{C}_{\varphi_C}$ is the cognition module parameterized by φ_C , and $\mathcal{K}_{\varphi_K}$ is the coordination module parameterized by φ_K . In this chapter, $\mathcal{K}_{\varphi_K}$ takes the multiplexed form

$$\mathcal{K}_{\varphi_K}(s_t^a, \sigma_t^a, \mathcal{N}(a)) = \prod_{j \in J} \bigoplus_{n \in \mathcal{N}(a)} M_{an}^j(s_t^n, d_t^n, \phi_j) \quad (7.2)$$

where $\bigoplus$ aggregates messages across neighbors within a channel.

In many practical settings, agents cannot share their state, interaction messages, or even the identity of their interaction partners due to privacy, regulatory, or competitive constraints. In contact tracing, an agent may become infected without observing which neighbor caused the exposure. In auction markets, agents influence outcomes without revealing bids or strategies. In multi-institutional biosecurity surveillance, organizations must coordinate without exposing individual records or interaction patterns. In each case, both the interaction messages M_{an}^j and, in some settings, the interaction graph $\mathcal{N}(a)$ are sensitive.

Can agents still coordinate with their neighbors, sending messages, sharing sensitivities, and learning shared dynamics Φ from data, when individual interaction messages cannot be observed and interaction may occur without revealing explicit identities?

This chapter introduces an algorithm for agents to coordinate with neighbors without directly exchanging individual messages or neighborhood identities in plaintext. We instantiate $\bigoplus$ as a secure aggregation operator that preserves the coordination and learning objectives of the LPM while hiding individual contributions. The neighborhood structure $\mathcal{N}(a)$, the message function M_{an}^j , the cognition module $\mathcal{C}_{\varphi_C}$, and the transition function f are all

preserved from Chapter 3. This instantiation changes only the aggregation operator $\oplus$, from open message passing to secure aggregation.

Critically, the learning objective is unchanged. The goal remains fitting Φ to observed population data. The sensitivity σ_t^a survives the privacy constraint: agents can still coordinate, learn from each other, and calibrate shared dynamics to real data without observing individual messages or neighborhood identities.

In LPM terms, this instantiation specifies how the aggregation operator $\oplus$ within $\mathcal{K}_{\varphi_K}$ is realized as a secure aggregation primitive. As in Chapter 4, $\varphi_K = \emptyset$: the coordination rule itself remains structural rather than learned. What is learned is Φ , recovered from population data via secure gradient flow through the differentiable product. The cognition module $\mathcal{C}_{\varphi_C}$ is hand-specified in the case studies considered. The threat model is honest-but-curious: agents follow the protocol but may attempt to infer information from the messages they observe.

We validate this instantiation on biosecurity case studies, testing whether execution, calibration, and analysis are preserved under secure aggregation, and through real-world deployment across a global collaboration of 20 institutions, demonstrating that private agent coordination is practically deployable at scale.

The content of this chapter draws from the following publications:

- Chopra et al. *Private Agent-Based Modeling*. AAMAS 2024 [43]
- Khurana, M.*, Tsui, J.*, Gutierrez, B.*, Chopra, A., (40 others) et al. Kraemer, M. *Global Approaches to Infectious Disease Surveillance and Modelling*. Nature Medicine 2026 [45]

7.2 Background

This chapter addresses two interconnected challenges: the analytical value of granular agent population data, and the privacy constraints that prevent that data from being centralized. The first subsection establishes why granular analysis of agent populations matters. The second surveys existing approaches to data privacy and identifies the gap that motivates this chapter’s contribution.

7.2.1 Analysis of Agent Populations

One of the core strengths of agent population models is their granularity, which enables research questions that are beyond reach for coarser methodologies. Epidemiological models can analyze observed disparities in biosecurity disruption infections among demographic groups [114–116], plan effective vaccination rollout plans [117], or assess the role of latent transmission for a given network structure [118]. All these studies require examination of the distribution of state values of agents and their responsiveness to changes in the model’s structural parameters. Specifically, we are interested in collecting summary statistics over the agent population, $\xi_k = h_k(\{\mathbf{z}_i(t) \mid i \in A\})$, and their sensitivity $\nabla_{\Phi} \xi_k$. For instance, in an epidemiological model, we might investigate the distribution of infections among different age groups and assess how these infections react to variations in the reproduction number

R_0 . Sensitivity analysis is computationally demanding due to the high cost of running large agent populations and the high dimensionality of the parameter space. Differentiable agent populations mitigate this bottleneck through direct gradient computation [116].

The challenge is that producing these granular analyses requires access to individual agent states: age, health status, location, household composition. This is precisely the data that privacy constraints prevent from being centralized. The analytical capability that makes agent population models valuable is also the capability that creates the greatest privacy risk.

7.2.2 Data Privacy for Agent Modeling

Recent improvements in agent population scalability have been matched by an increased risk to individual privacy [119], which has already manifested in data leaks [120–122]. Statistical privacy methods such as K-anonymity and differential privacy [123,124] protect individual information at an aggregate level and have been used to release US census data [125] and mobility trace data [126]. Although these methods hide identifiable information in aggregate statistics, they have limited capability when applied at the level of individuals. State propagation in agent populations requires accumulating individual agent states, while policy evaluation requires access to individual mobility data. Statistical hiding methods provide poor privacy-utility trade-offs in this context.

Cryptographic protocols offer stronger privacy guarantees at the individual level. Secure multi-party computation has been explored for federated learning [127] and training of distributed graph neural networks [128], but these approaches are constrained by the high computation and communication cost of operating on deep neural networks, and the intractability of secure backpropagation through non-linear mechanisms. The simplicity of mechanistic agent population models makes MPC tractable in fully decentralized scenarios where DNN-based approaches require hybrid trusted execution environments. Prior work introduced privacy-preserving contact tracing through Private Information Retrieval [129] and encrypted individual states [130], but neither supports privacy-preserving calibration or sensitivity analysis of agent population models. This chapter introduces the first framework that enables simulation, calibration, and analysis of agent populations while preserving individual agent privacy at every step.

7.3 Characterizing Privacy

First, we formalize a threat model to setup constraints for a privacy-preserving solution. Then, we provide background on secure multi-party computation and describe the GMW protocol which we use to design algorithms for privacy-preserving simulation, calibration and analysis of agent-based models.

7.3.1 Threat Model

We assume an honest-but-curious (a.k.a. semi-honest) attacker [131] which aims to learn private information about participating agents without altering the protocol. This private

information is included in an agent’s state $\mathbf{z}_j(t)$, interaction trace $\{\mathcal{N}_i(t) \mid \forall t\}$, and neighborhood messages $\{M_{ij}(t) \mid i \in A, j \in \mathcal{N}(i)\}$. For instance, in epidemiological models, this can correspond to the health and demographic traits and mobility patterns of individual agents. Such an attacker can manifest itself as the coordinating server that wants to surveil agents using the mobility trace or a (sub-group) of adversarial agents, which may be incentivized to steal the personal health information of agent cohorts. In the context of agent-based modeling, this information can be leaked during message passing over per-step neighborhoods and during the collection of summary statistics over the population. The goal of this work is to alleviate such challenges and design a privacy-preserving mechanism that can compute functions over agents’ states without revealing private information.

7.3.2 Secure Multi-party Computation

Secure multi-party computation enables a set of agents to interact and compute a joint function of their private inputs while revealing nothing but the output [132]. MPC protocols are coordinated with a server (MPC server) and are designed to protect against behavior of adversarial participants. These participants, either an agent or the server, aim to learn private information (of other entities) or cause the computation result to be incorrect. The idea was first introduced by Yao for the two-party case [133] and generalized to multiparty settings by Goldreich, Micali and Wigderson (GMW) [134]. Among other properties, GMW protocols guarantee (1) *privacy*: so that no entity can learn anything more than its prescribed output and, (2) *correctness*: so that agents receive the correct output. For instance, in an epidemiological ABM, this would ensure both that the personal disease status of agents is not leaked and that agents receive the correct transmission probability as in a centralized simulation.

7.3.3 The GMW protocol

The GMW protocol uses additive secret sharing to communicate (or aggregate) private inputs across the participant entities. The key insight is to divide a secret input into multiple shares in such a way that the secret can be reconstructed only when a sufficient number of shares are combined together. The scheme supports diverse aggregation queries such as secure addition or secure multiplication [135] of the secrets held by the participating agents. Here we focus on the addition case and we assume that all participating agents are required to compute the secret, usually denoted by $t = N$, but the same methodology can be extended to multiplication and composite queries (see, e.g., [132]).

Consider N agents holding private values s_i . We want to compute the sum $\sum_i s_i$ without any agent j acquiring knowledge about $s_{\{k \neq j\}}$. To setup the protocol, the agents agree on an integer $n > \sum_i s_i$ defining the finite group $\mathbb{Z}_n$ on which all computations will be carried ¹. Each agent i then samples $N - 1$ random numbers, $r_{ij} \sim \mathcal{U}\{0, n - 1\}$, such that the input is

¹The choice to perform finite group arithmetics is so that no information about the secret can be gained by holding $< N$ shares.

divided into N shares, s_{ij} defined by

$$s_i = \sum_{j=1}^N s_{ij} \pmod{n} = \sum_{j=1}^{N-1} r_{ij} + \left(s_i - \sum_{j=1}^{N-1} r_{ij} \right) \pmod{n}. \quad (7.3)$$

Each agent then sends each share of their secret to each corresponding agent; agent i sends s_{i1} share to agent 1, s_{i2} share to agent 2, etc. Locally, each agent performs the sum

$$\sigma_k = \sum_{i=1}^N s_{ik} \pmod{n}. \quad (7.4)$$

Finally, all values σ_k are shared so that the reconstructed sum, $S = \sum_k \sigma_k \pmod{n}$, can be computed, corresponding to the sum of the agent inputs s_i by construction. Typically, this reconstruction may be conducted by a central MPC server or a trusted agent. We summarize the protocol in [Algorithm 1](#), and we provide an illustrating example below.

Additive secret sharing example

Consider $N = 3$ agents — Alice, Bob, and Carol — holding private values $s_A = 2$, $s_B = 3$, and $s_C = 5$. They wish to compute the sum of these values without disclosing their individual inputs. They agree on an integer $n = 11$, defining a finite group $\mathbb{Z}_n$. First, the agents generate 3 shares each by sampling 2 random numbers from $\mathbb{Z}_n$. For instance, Alice generates random numbers 7 and 5, so that

$$s_A = s_{AA} + s_{AB} + s_{AC} = 7 + 5 + 1 \pmod{11} = 2, \quad (7.5)$$

and similarly for Bob and Carol with $s_B = 2 + 0 + 1 \pmod{11}$, and $s_C = 3 + 1 + 1 \pmod{11}$. Second, the agents communicate with each other to keep one of the shares and send the other two to the other two agents and perform the sum of the received shares. For example, Alice receives s_{BA} from Bob and s_{CA} from Carol and computes

$$\sigma_A = s_{AA} + s_{BA} + s_{CA} = 7 + 2 + 3 \pmod{11} = 1 \pmod{11}, \quad (7.6)$$

and similarly for Bob and Carol with $\sigma_B = 5 + 0 + 1 \pmod{11} = 6 \pmod{11}$ and $\sigma_C = 1 + 1 + 1 \pmod{11} = 3 \pmod{11}$. Finally, the secret can be reconstructed by doing $S = \sigma_A + \sigma_B + \sigma_C = 10 \pmod{11}$ as expected.

In the following section, we apply the GMW protocol to generalize the above insight to share information containing agent's private information to other agents or a central MPC server, providing protocols for the computation of agent updates, and gradients in a secure way, enabling privacy-preserving simulation, calibration, and analysis of agent population models.

7.4 Private Agent-based Population Models

With the MPC background introduced in the preceding section, we formulate protocols to conduct the simulation, calibration, and analysis of ABMs in a privacy-preserving way.

Algorithm 1: SECURESUM

Data: Agents $\{1, \dots, N\}$ with secret inputs $s_1, \dots, s_n$, integer $n > \sum_i s_i$.

Result: The sum of all shares $S = s_1 + \dots + s_n$.

1 Splitting secret into shares and distributing:

2 Each party i generates N shares $s_{i1}, \dots, s_{iN} \in \mathbb{Z}_n$ which sum up to s_i .

3 Each party i distributes all their shares $s_{i1}, \dots, s_{iN} \in \mathbb{Z}_n$ to $1, \dots, N$, including themselves.

4 Secure Computation (Addition):

5 To add the inputs securely, parties simply add their respective shares

$$\sigma_i = s_{1i} + \dots + s_{Ni} \mod n.$$

6 Reconstruction:

7 To reveal the final result of the computation, parties collaborate by summing their shares:

$$8 \ S = (\sigma_1 + \sigma_2 + \dots + \sigma_n) \mod n.$$

7.4.1 Secure Simulation

In order to update the state of an agent during a simulation, an agent requires an aggregation over the agent's neighbors, revealing their private information. As described in the previous section, MPC is well-equipped to perform this kind of calculation without revealing the individual parties' data. Without loss of generality, we present our protocols for the case where the aggregation function $\oplus$ is a summation Σ , so that we can make use of the SECURESUM protocol introduced in Algorithm 1. Furthermore, as long as the agent's update function f is differentiable respect to the structural parameters θ , which is indeed the case for many ABMs [40], each agent can store $\nabla_{\theta} f$ for use during the calibration step. With all this in mind, we present in Algorithm 2, a privacy-preserving protocol for updating agent states.

Algorithm 2: SECUREAGENTUPDATE

Data: Agent i with state $\mathbf{z}_i(t)$, Neighboring agent's messages $\{M_{ij}(t) \mid j \in \mathcal{N}(i)\}$, Integer n , State update rule f , ABM parameters θ

Result: New state $\mathbf{z}_i(t+1)$

1 Agent i calls the SECURESUM protocol with neighbors $\{j \mid j \in \mathcal{N}(i)\}$ and integer n to obtain the sum $M_i(t) = \sum_{j \in \mathcal{N}(i)} M_{ij}(t)$.

2 Agent i updates its state $\mathbf{z}_i(t+1) = f(\mathbf{z}_i(t), M_i(t), \theta)$ and stores the gradient $\nabla_{\theta} f$.

It is worth noting that, in contrast to general applications of the GMW protocol, only the agent who starts the protocol receives the result of the computation since there is no need for the neighboring agents to have access to that information.

Next, we introduce the SECURESIMULATION protocol in Algorithm 3, where, in addition to performing agent updates, we collect a time series of aggregate statistics over the agent's population and its gradient with respect to the ABM structural parameters θ .

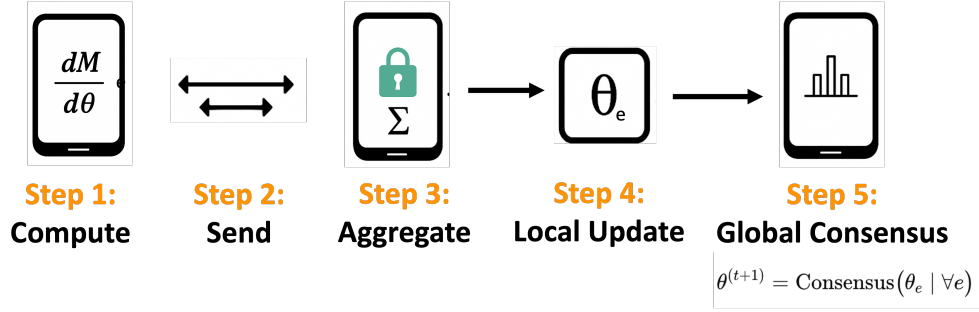


Figure 7.1: Privacy-Preserving Agentic Coordination. The protocol operates in five steps: (1) Each agent computes local sensitivities $\partial M / \partial \theta$ based on private health and mobility data; (2) Agents exchange encrypted sensitivity messages with neighbors; (3) Secure multi-party computation aggregates sensitivities without revealing individual data; (4) Each agent updates local coordination parameters θ_e based on aggregated signals; (5) Global consensus mechanisms align population-level responses. This workflow enables millions of contact tracing devices to coordinate testing, isolation, and mobility decisions while preserving individual privacy through distributed gradient computation.

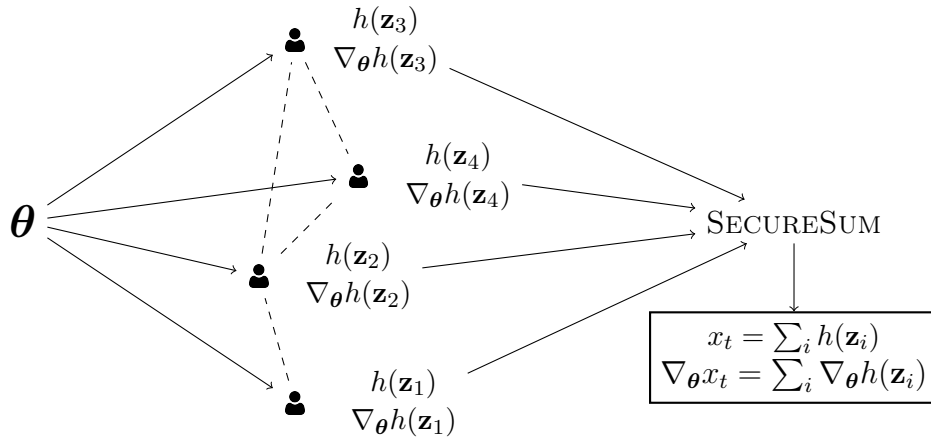


Figure 7.2: Diagram illustrating the SECURESIMULATION protocol for ABM parameters θ .

Algorithm 3: SECURESIMULATION

Data: MPC server C , Agents $\{1, \dots, N\}$ with states $\{\mathbf{z}_1, \dots, \mathbf{z}_N\}$, ABM parameters θ , State update rule f , Number of time-steps T

Result: Aggregate statistics $\mathbf{x} = x_1, \dots, x_T$ and gradients $\nabla_{\theta}\mathbf{x}$.

- 1 C generates a large enough prime number P and the requested statistics collecting function h ; and sends them to all agents along ABM parameters θ .
- 2 **for** $t = 1, \dots, T$ **do**
- 3 **for** $i = 1, \dots, N$ **do**
- 4 Agent i calls the SECUREAGENTUPDATE protocol (Algorithm 2) to compute $\mathbf{z}_i(t+1)$.
- 5 Agent i gathers its information of interest $h(\mathbf{z}_i(t+1))$ and gradient $\nabla_{\theta}h(\mathbf{z}_i(t+1))$.
- 6 C calls the SECURESUM protocol with all agents to collect the aggregate statistics x_t and their gradients $\nabla_{\theta}x_t$.
- 7 C returns the accumulated $\mathbf{x}$ and $\nabla_{\theta}\mathbf{x}$.

7.4.2 Secure Calibration

During the calibration of an ABM, the modeler (central MPC server) requires the ability to evaluate the ABM at different values of θ , and, in the case of gradient-assisted calibration, the gradient of the outputs with respect to θ . The retrieval of these quantities is enabled by the SECURESIMULATION protocol and so classical calibration algorithms [136,137] can be seamlessly applied to conduct a privacy-preserving calibration. Here, as outlined in Subsection 5.6.1, we focus on conducting GVI with a deep neural network trained to approximate the generalized posterior. This neural network seats on the central MPC server and it is trained using the collected gradients. To this end, what remains to be detailed is the transition from the gradient of aggregate statistics, $\nabla_{\theta}\mathbf{x}$, to the loss gradient required for GVI. This process is detailed in Algorithm 4.

7.4.3 Secure Analysis

As outlined in Subsection 7.2.1, we require a secure protocol to retrieve summary statistics over the agent's population, $\xi_k = h_k(\{\mathbf{z}_i(t) \mid i \in A\})$, and their sensitivity, $\nabla_{\theta}\xi_k$. We note that we have already addressed this issue in the SECURESIMULATION protocol since retrieving the time-series $\mathbf{x}$ is a particular case of this more general problem. In fact, we formulate a more general approach wherein a summary statistic ξ over the population states can be obtained by aggregating over the entire population the outputs of an indicator function $\mathbb{1}_{\Omega}(\mathbf{z})$ acting on the agent's state,

$$\xi = \bigoplus_{i \in A} \mathbb{1}_{\Omega}(\mathbf{z}_i), \quad (7.7)$$

where Ω denotes the set of characteristics we want to aggregate on, and $\oplus$ denotes a kind of aggregation, usually a sum. For instance, Ω may correspond to the property of being infected, in which case taking $\oplus$ to be a sum would return the number of infected agents.

Algorithm 4: SECURECALIBRATION

Data: MPC server C , Agents $\{1, \dots, N\}$ with states $\{\mathbf{z}_1, \dots, \mathbf{z}_N\}$, State update rule f , Prior $\pi(\boldsymbol{\theta})$, Observed time-series $\mathbf{y}$, Loss function $\ell(\mathbf{x}, \mathbf{y})$, Number of epochs N_e , Number Monte-Carlo samples N , Number of time-steps T , Learning rate η

Result: Trained candidate posterior q^*

```
1  $C$  initializes candidate posterior  $q_\phi$ .
2 for  $1, \dots, N_e$  do
3   for  $1, \dots, N$  do
4      $C$  samples ABM parameters  $\boldsymbol{\theta} \sim q_\phi(\boldsymbol{\theta})$ .
5      $C$  executes SECURESIMULATION protocol (Algorithm 3) to obtain  $\mathbf{x}$  and  $\nabla_{\boldsymbol{\theta}} \mathbf{x}$ .
6      $C$  uses the chain rule to compute
        
$$\nabla_{\boldsymbol{\theta}} \ell(\mathbf{x}(\boldsymbol{\theta}), \mathbf{y}) = \nabla_{\mathbf{x}} \ell(\mathbf{x}, \mathbf{y}) \cdot \nabla_{\boldsymbol{\theta}} \mathbf{x}$$

        
$$\nabla_{\phi} \ell(\mathbf{x}(\boldsymbol{\theta}), \mathbf{y}) = \nabla_{\boldsymbol{\theta}} \ell(\mathbf{x}(\boldsymbol{\theta}), \mathbf{y}) \cdot \nabla_{\phi} \boldsymbol{\theta}$$

        using the reparameterization trick.
7    $C$  accumulates gradient to estimate
        
$$\nabla_{\phi} \mathbb{E}_{q_{\phi}} [\ell(\mathbf{x}(\boldsymbol{\theta}), \mathbf{y})] \approx \frac{1}{N} \sum_{i=1}^N \nabla_{\phi} \ell(x(\theta), y).$$

8    $C$  computes divergence  $\text{KL}(q_{\phi} \parallel \pi(\boldsymbol{\theta}))$  and derivative  $\nabla_{\phi} \text{KL}(q_{\phi} \parallel \pi(\boldsymbol{\theta}))$ .
9    $C$  updates  $\phi \rightarrow \phi - \eta \nabla_{\phi} \mathcal{L}(\phi)$ , where
        
$$\mathcal{L}(\phi) = \mathbb{E}_{q_{\phi}} [\ell(\mathbf{x}(\boldsymbol{\theta}), \mathbf{y})] + \text{KL}(q_{\phi} \parallel \pi(\boldsymbol{\theta})).$$

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The algorithm is formally described in Algorithm 5, which holds the same security guarantees as the one exposed in Subsection 7.4.1.

Likewise, sensitivity analysis can be performed securely using Algorithm 6, where each agent computes the sensitivity of their state change to the model parameters, and the central agent retrieves the aggregation by employing the SECURESUM protocol.

7.5 Case Study: Privacy-preserving Epidemiology

In this section, we illustrate how our methodology may be deployed in practice by describing a decentralized privacy-preserving agent-based SIR model, which supports simulation, calibration, and analysis.

The model follows a standard parameterization where agents' interactions are specified through a contact graph $\mathcal{G}$, which in this case is only locally defined by each agent having access to their neighbors. Each agent has 3 possible states: 0 (Susceptible), 1 (Infected), and 2 (Recovered). We initialize the simulation by infecting a fraction I_0 of agents sampled

Algorithm 5: SECURESUMMARYSTATISTIC

Data: MPC server C , Agents $\{1, \dots, N\}$ with states $\{\mathbf{z}_1, \dots, \mathbf{z}_N\}$, Indicator function $\mathbb{1}_\Omega(\mathbf{z})$.

Result: Aggregate quantity $\bigoplus_{i \in A} \mathbb{1}_\Omega(\mathbf{z}_i)$.

- 1 C generates prime number P and sends them to all agents alongside indicator function $\mathbb{1}_\Omega(\mathbf{z})$.
 - 2 Each agent i computes result of $\mathbb{1}_\Omega(\mathbf{z}_i)$.
 - 3 C executes SECURESUM protocol to securely retrieve $\bigoplus_i^N \mathbb{1}_\Omega(\mathbf{z}_i)$.
-

Algorithm 6: SECURESENSITIVITYANALYSIS

Data: MPC server C , Agents $\{1, \dots, N\}$ with states $\{\mathbf{z}_1, \dots, \mathbf{z}_N\}$, Indicator function $\mathbb{1}_\Omega(\mathbf{z})$, State update rule f

Result: Sensitivity $\nabla_\theta \xi$.

- 1 C generates prime number P and sends them to all agents alongside indicator function $\mathbb{1}_\Omega(\mathbf{z})$.
- 2 Each agent i executes SECUREAGENTUPDATE with parameters θ to obtain $\nabla_\theta f(\mathbf{z}_i, M_i, \theta)$ so that

$$\nabla_\theta \xi_i = \nabla_\theta f(\mathbf{z}_i, M_i, \theta) \cdot \mathbb{1}_\Omega(\mathbf{z})$$

- 3 C executes SECURESUM protocol to securely retrieve $\nabla_\theta \xi = \bigoplus_i^N \nabla_\theta \xi_i$.
-

uniformly from the population, while the remaining agents are considered susceptible. At each time-step, agent i updates its state as follows:

$$M_{ij}(t) = I_j(t) \tag{7.8}$$

where $I_j(t)$ is the infected status of the neighbor (0 or 1), so that

$$\begin{aligned} z_i(t+1) = & \mathbb{1}_{\{z_i=0\}} \cdot \text{Bernoulli} \left(p_{\text{inf}}^{(i)}(t) \right) + \\ & \mathbb{1}_{\{z_i=1\}} \cdot \left(1 + \text{Bernoulli} \left(p_{\text{rec}}^{(i)} \right) \right) + \\ & \mathbb{1}_{\{z_i=2\}} \cdot 2 \end{aligned} \tag{7.9}$$

with

$$p_{\text{inf}}^{(i)}(t) = 1 - \exp \left(-\frac{\beta S_i \Delta t}{n_i} \sum_{j \in \mathcal{N}(i)} I_j(t) \right), \tag{7.10}$$

where $\mathcal{N}(i)$ is the set of neighbors of agent i , S_i is the susceptibility of agent i , $n_i = \#\mathcal{N}(i)$ is the total number of neighbors, Δt is the duration of the time-step, and β is a structural parameter of the ABM called the effective contact rate. Infected agents can recover at each time step with recovery rate γ , so that

$$p_{\text{rec}}^{(i)} = 1 - \exp(-\gamma \Delta t). \tag{7.11}$$

For the case of a complete graph, the model reduces to the standard ODE-based SIR model with $R_0 = \beta/\gamma$ as the basic reproduction number. The model is run for n_t time steps.

To ground the example on real data, we consider the contact graph of the city of Oxford, extracted from the June ABM model [36] to determine the neighborhood of each agent, $\mathcal{N}(i)$. This contact graph includes agents' interactions in households, companies, and schools, and it is based on English census data. The choice of parameter values for the experiment is given in Table 7.1.

Parameter	Value
β	0.5 day^{-1}
γ	0.1 day^{-1}
I_0	0.01
Δt	1 day
n_t	60
$\mathcal{G}$	Oxford

Table 7.1: Parameter values for the agent-based SIR model.

7.5.1 Private Policy Assessment

We first consider the application of the SECURESIMULATION protocol (Algorithm 3). Let us pose a situation where a policymaker wants to study the efficacy of mask-wearing at different compliance levels using agent-based simulation. We introduce a slight modification to Equation (7.10) to incorporate a reduction in the infection probability due to mask-wearing with certain compliance α ,

$$p_{\text{inf}}^{(i)}(t) = 1 - \exp \left(-\frac{\beta S_i \Delta t}{n_i} \sum_{j \in \mathcal{N}(i)} I_j(t)(1 - c_j) \right), \quad (7.12)$$

where $c_j \sim \text{Bernoulli}(\alpha)$, so that $\alpha_i = 1$ corresponds to full compliance where there is no transmission. Note that we are assuming complete protection against infection when wearing a mask. We proceed to execute 3 simulations for 3 different values of α . At each simulation, α is sent to the agents, where they locally compute their own compliance to the measure. The SecureSimulation protocol is then used to run the simulation and retrieve the aggregate statistic of interest, $\mathbf{x}$, which in this case is the number of infections over time. The results are shown in Figure 7.3, where we observe that little transmission occurs when compliance is above $\gtrsim 75\%$. Thus, we observe that within our methodology, the policymaker could still have access to the same level of insight as a traditional ABM, all while protecting the individual agent's privacy.

7.5.2 Private Calibration

Next, we pose a situation where we want to calibrate our ABM with structural parameters $\phi = (\beta, \gamma)$ to observed ground-truth data. For simplicity, we present the calibration of the β parameter given an observed curve of infections ($\mathbf{y}$), obtained by running the ABM model with the baseline parameters in Table 7.1.

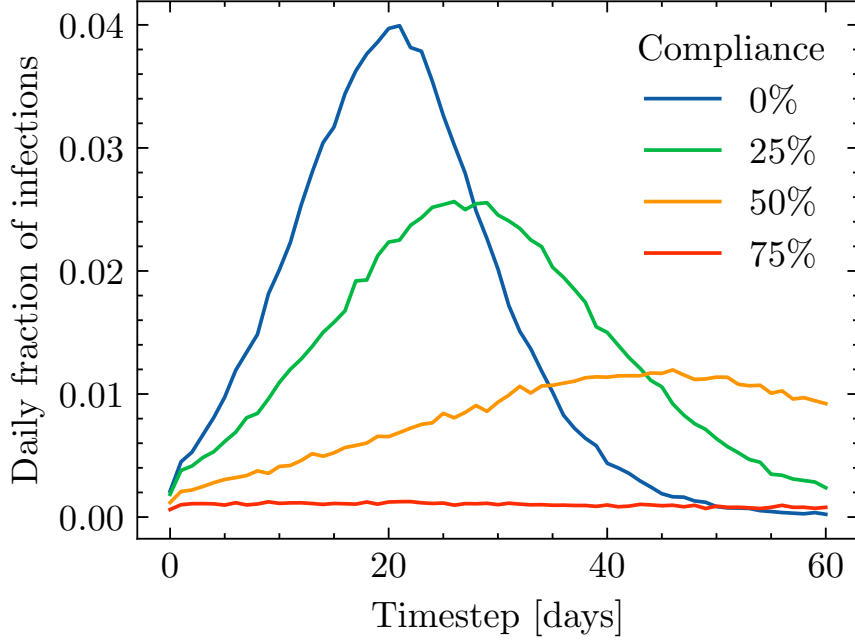


Figure 7.3: Infection curves for different levels of compliance: 0% (blue), 25% (green), 50% (orange), 75% (red). The number of infections has been normalized to the number of agents N . These plots are generated without releasing the infection status or compliance decision of any individual agent.

The first step is to compute the gradient $\nabla_{\phi} \mathbf{x}$, where $\mathbf{x}$ is the number of daily infections and $\phi = \beta$. We note that this gradient can be approximated by the gradient of the average number of new infections with respect to β ,

$$\frac{\partial x_t}{\partial \beta} \approx \frac{\partial \mathbb{E}[\Delta I(t)]}{\partial \beta} = \sum_{i=1}^N \chi_i(t) \exp(-\chi_i(t)/\beta), \quad (7.13)$$

where

$$\chi_i(t) = \exp \left(-\frac{\beta S_i \Delta_t}{n_i} \sum_{j \in \mathcal{N}(i)} I_j(t) \right). \quad (7.14)$$

A central agent can safely retrieve the gradient by performing the SECURESUM protocol across all agents as described in [Algorithm 3](#) and [Algorithm 4](#). We thus conduct GVI by considering $\mathcal{Q}$ to be a masked-autoregressive normalizing flow [138] and assume the prior is a normal distribution with $\mu = 0.7$ and $\sigma = 0.5$. [Figure 7.4](#) (left) shows the trained normalizing flow, which correctly assigns high probability mass to the ground-truth value. To further evaluate the goodness of the fit, we plot simulated runs from ABM parameters sampled from the trained flow in [Figure 7.4](#) (right), where we compare it to runs simulated from prior samples.

This experiment highlights how privacy-preserving ABMs can be integrated into differential and probabilistic programming pipelines. This opens the door to integrating ABM insight

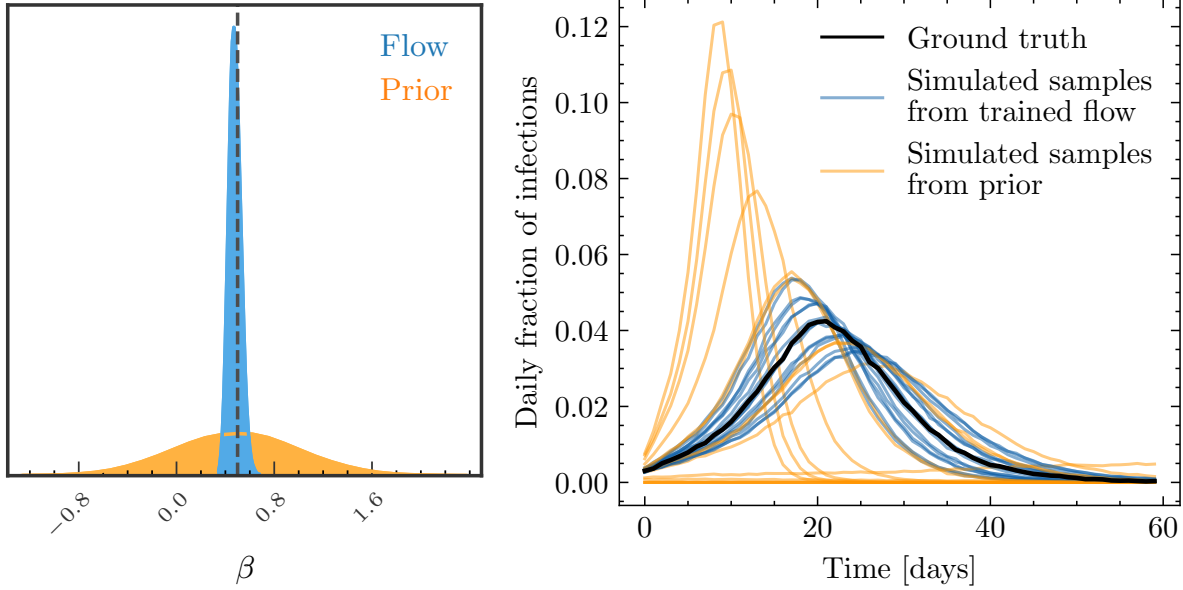


Figure 7.4: Left: Probability density plot for the trained normalizing flow (blue) against the prior distribution (orange). Ground-truth value is marked as a dashed black line. Right: Results from simulating β samples from the trained flow (blue) and prior (orange) compared to the ground-truth data (black). The number of infections has been normalized to the number of agents N .

into more complex ML pipelines leveraging heterogeneous data streams to boost the model’s insight capabilities.

7.5.3 Private Demographic Analysis

In this section, we apply the SECURESUMMARYSTATISTIC protocol to analyze our calibrated ABM. In particular, we study the distribution of infections by age, ethnicity, and geographical location (ZIP code). This analysis may be relevant to, for instance, understanding the causes of asymmetric distribution of infections among different demographic groups [114–116].

For this exercise, we import a synthetic population of Oxford from the June model, as we did for the contact graph, so that we have access to a population with realistic age, sex, and ethnicity distributions and geographical location. Note that the ethnicity categorization follows the English census [139].

We can construct histograms that highlight the distribution of infection among different ethnic groups by considering the relevant indicator functions so that we can apply the SECURESUMMARYSTATISTIC protocol. For instance, for the case of age distribution, we have

$$\mathbb{1}_{\Omega}(\mathbf{z}) = \mathbb{1}_{\{(z_{\text{age}} \in a_k) \wedge (z_{\text{inf}}=1)\}}(\mathbf{z}), \quad (7.15)$$

where z_{age} and z_{inf} are the age and infected status of the agent, and a_k are each of the histogram age bins. The number of counts in the particular age bin a_k can then be obtained upon executing SECURESUM over the entire population following Algorithm 5. Figure 7.5 shows the age and ethnicity histogram of infections for the calibrated simulation obtained in

Subsection 7.5.2, where the ethnicity histogram has been computed analogously to age one by considering the indicator function

$$\mathbb{1}_{\Omega}(\mathbf{z}) = \mathbb{1}_{\{(z_{\text{ethnicity}}=e_i) \wedge (z_{\text{inf}}=1)\}}(\mathbf{z}), \quad (7.16)$$

for each ethnicity category e_i . The results are in agreement with the particular demographics of the city of Oxford, dominated by students in the 20-30 age bin with a significant presence of non-White groups.

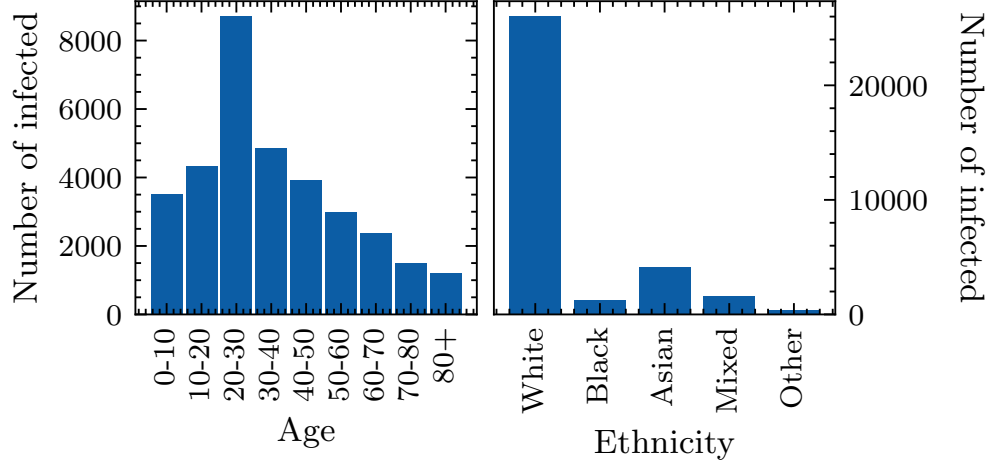


Figure 7.5: Age (left) and ethnicity (right) histogram of the infected population. These statistics are computed without leaking the infection or demographic properties of any agent.

Finally, a similar analysis can be done at the geographical level, where infections can be collected by ZIP code sector by executing the SECURESUMMARYSTATISTIC protocol with $\mathbb{1}_{\Omega}(\mathbf{z}) = \mathbb{1}_{\{z_{\text{ZIPcode}}=r_i\}}$ for each ZIP code sector r_i . In Figure 7.6, we show the obtained distribution of infections across the city of Oxford.

Summarising, we have illustrated how a detailed analysis of agent properties can be performed in a privacy-preserving way by following Subsection 7.4.3.

7.5.4 What the Oxford Case Study Demonstrates

The three Oxford experiments correspond directly to the capabilities required for real-time contact tracing and outbreak response.

The private policy assessment demonstrated that a policymaker can evaluate the effect of a masking mandate at different compliance levels and observe that transmission drops sharply above approximately 75% compliance (Figure 7.3), without any agent revealing their individual infection status or compliance decision. In a real-time monitoring context, this translates directly to the ability to evaluate intervention efficacy as an outbreak evolves, using live individual-level data that never leaves personal devices.

The private calibration demonstrated that structural parameters $\phi = (\beta, \gamma)$ can be calibrated to observed data using gradients that flow through the SECURESUM protocol, with the trained normalizing flow correctly concentrating probability mass on the ground-truth

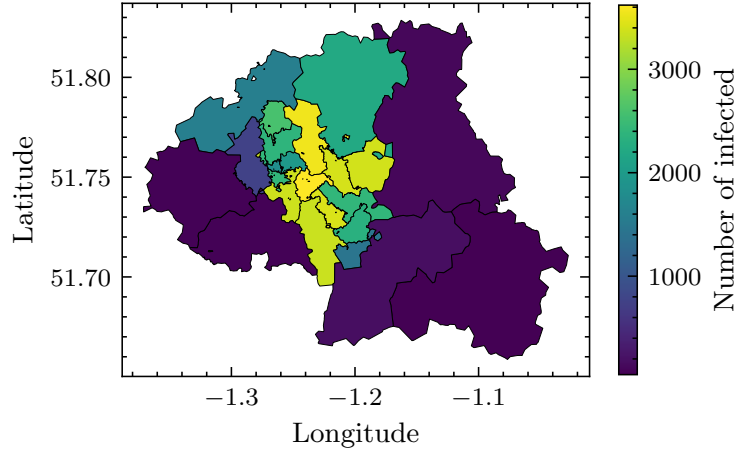


Figure 7.6: Geographical distribution of infections by ZIP code sector within the city of Oxford. These statistics are computed without leaking the infection status or geo-location of any individual agent.

value (Figure 7.4). In the contact tracing setting, this translates to continuously updating the epidemiological model as new individual exposure data accumulates on devices, enabling a living calibrated model of the outbreak that improves in real time without centralizing any data.

The private demographic analysis demonstrated that infection distributions by age, ethnicity, and geographical location can be retrieved using the SECURESUMMARYSTATISTIC protocol (Figures 7.5 and 7.6), without leaking any individual’s demographic properties or infection status. This translates to the ability to identify which communities are most at risk in real time, enabling targeted situational awareness without requiring individuals to disclose their demographics to any central authority.

7.6 Application: Secure Real-Time Disease Monitoring

The contributions in this section are based on the following publication:

Khurana, M.P., Tsui, J.L.-H., Gutierrez, B., Chopra, A., et al. “Global Approaches to Infectious Disease Surveillance and Modelling.” *Nature Medicine*, 2026. [45]

The author contributed the private agent-based modeling components described in this section, enabling privacy-preserving real-time disease monitoring directly on personal devices without compromising individual data privacy.

The federated calibration approach introduced in Chapter 4 addressed the case where sensitive outbreak data is siloed across a small number of institutions: hospitals, laboratories, and health ministries, each capable of running a local calibration model and sharing embeddings with a central server. Infectious disease surveillance presents a harder and more prevalent version of this problem. The most granular and real-time data about disease spread is not held by institutions at all. It lives on the personal devices of millions of individuals: in

their mobility traces, their contact logs, their symptom reports, their exposure histories. No institution has the authority or the infrastructure to centralize this data, and centralizing it would constitute a severe privacy violation even if it were technically feasible. The question is whether live population-scale agent population modeling and real-time monitoring can run directly on personal devices, learning from individual-level data without that data ever leaving the device. Private agent populations answer this question.

7.6.1 From Institutions to Individuals

Digital contact tracing emerged during the biosecurity disruption of recent years as a potentially powerful tool for real-time outbreak management. Contact tracing apps collect mobility and contact data directly on personal devices, precisely where the most granular individual-level epidemiological data resides. When an individual is exposed to an infected contact, their device knows it. When their symptoms begin, their device can record it. When their behavior changes in response to a public health intervention, their mobility trace captures it. This data stream, aggregated across millions of individuals, would constitute the most informative real-time signal available for calibrating epidemiological models and guiding outbreak response.

Yet current contact tracing implementations cannot use this data for population-level epidemiological analytics. As Khurana et al. identify, current implementations do not support federated analytics such as exposure management, intervention evaluation, or risk minimisation [45]. A contact tracing app can tell an individual whether they were exposed. It cannot tell a policymaker what the current reproduction number is, how effective a masking mandate would be, or which demographic groups are most at risk, without centralizing the raw contact data, which would be both a privacy violation and a security liability.

This gap is not merely a software engineering problem. It reflects a fundamental tension in epidemiological modeling: the data that would most improve models is the data that is most sensitive, most personal, and most legally constrained. Resolving this tension requires a framework that can run agent population modeling and calibration where the data already lives, directly on personal devices, without requiring that data to move anywhere.

7.6.2 Private Agent Models as Real-Time Monitoring Infrastructure

The private agent population framework introduced in this chapter resolves this tension directly. The SECURESIMULATION, SECURECALIBRATION, and SECUREANALYSIS protocols are designed precisely for this setting: each agent runs its own local state update, shares only cryptographically protected aggregate statistics via SECURESUM, and the central MPC server never observes any individual agent’s state.

Applied to contact tracing, this architecture enables a qualitatively new mode of real-time disease monitoring. As described in Khurana et al., private agent population models can resolve this challenge by running agent population modeling and calibration directly on personal devices, where individual contact tracing information resides. This approach allows contact tracing apps to provide proactive, personalized guidance, for example asking “How can I reduce my exposure risk?”, without compromising data privacy [45].

Concretely, the private agent population architecture maps onto the contact tracing setting as follows. Each personal device is an agent: it holds the individual’s contact history, mobility trace, symptom reports, and vaccination status as its private state $\mathbf{z}_i(t)$, which never leaves the device. Using the SECUREAGENTUPDATE protocol, each device updates its own infection probability based on secure aggregate signals from neighboring devices, without any device revealing its individual infection status. Using SECURESIMULATION, a central coordination server retrieves aggregate statistics, including the number of infections, the reproduction number, and the demographic distribution of new cases, without observing any individual agent’s state. Using SECURECALIBRATION, structural parameters such as the effective contact rate β and recovery rate γ can be calibrated to real-time population-level observations using gradients that flow through secure aggregation without exposing individual states. Because each device runs its own local agent population model, it can answer individualized counterfactual questions: if an individual reduces their contact rate by 50%, what is their personal exposure risk? This is the proactive personalized guidance use case described in Khurana et al. [45], a capability that no centralized system can provide without a privacy violation.

7.7 Conclusion

This chapter instantiated $\oplus$ in the LPM coordination module as a secure aggregation operator, enabling agents to coordinate, calibrate shared dynamics Φ , and conduct granular population analysis without revealing individual states, interaction messages, or neighborhood identities. The three protocols, SECURESIMULATION, SECURECALIBRATION, and SECUREANALYSIS, preserve the full analytical capability of agent population models under cryptographic privacy guarantees. The Oxford case study demonstrated that mask-wearing policy evaluation, gradient-assisted calibration to ground-truth data, and demographic analysis of infections by age, ethnicity, and geography can all be executed without any individual agent’s private state leaving their local device.

The application to real-time disease monitoring extends this result to the most demanding deployment context: population-scale surveillance directly on personal devices. Private agent populations occupy a distinct position in the landscape of federated infectious disease surveillance. Where the federated calibration approach of Chapter 4 operates at the level of institutions, enabling hospitals, laboratories, and health ministries to collaborate on model calibration without sharing raw data, private agent populations operate at the level of individuals, enabling the most granular possible data to inform epidemiological models without that data ever leaving the personal device on which it was generated. Together they form a complementary architecture across two scales: institution-level privacy through federated calibration, and individual-level privacy through secure multi-party computation.

Several directions remain open. The current protocols operate on pairwise interactions and would benefit from extension to higher-order networks that capture group dynamics and complex contagion. Practical deployment at mobile scale requires minimizing communication overhead and supporting asynchronous message passing across heterogeneous devices. Combining secure aggregation with federated institutional learning, for example privately integrating genomic surveillance data or insurance claims into epidemiological models, would

further close the gap between what these models can learn and what data exists to inform them.

The four algorithm chapters of Part II have now addressed each component of the LPM coordination module under a different constraint: multiplexed numerical dynamics, language-grounded cognition, natural language coordination, and private aggregation. Together they establish that Large Population Models can represent, learn from, and deploy across realistic population settings. Part III applies this framework at national scale to the two domains that motivated the thesis: biological disruption and labor market disruption.

Part III

Applications: Mitigating Risks to Human Systems

Chapter 8

Biological Disruption at National Scale

8.1 Motivation and LPM Application

The preceding chapters developed four algorithms for building Large Population Models: multiplexed differentiable interactions, language-grounded cognition, natural language coordination, and private secure aggregation. This chapter deploys that framework at national scale on one of the most consequential biosecurity risks facing the United States: the spread of highly pathogenic avian influenza H5N1 across livestock, wild bird, and human populations.

Biological disruptions are the canonical testbed for population-scale modeling. Zoonotic spillover does not emerge from any single pathogen or individual host. It emerges from the interaction networks of species, populations, and behaviors through which pathogens propagate across species boundaries. Understanding spillover risk requires modeling heterogeneous agents: farms, wild bird populations, livestock workers, and the general human population, interacting across overlapping geographic and contact networks. The data required to calibrate such models is distributed across institutions, arrives at different spatial and temporal resolutions, and is subject to privacy constraints that prevent centralization. This is precisely the setting the LPM framework was designed for.

This chapter presents FIELD: a national-scale digital similar of the contiguous United States that represents livestock production, wild bird populations, and human demographics as interacting agent populations. Applied to H5N1 risk assessment, FIELD produces the first commodity-specific spatiotemporal risk maps for zoonotic spillover across the full contiguous United States, validated against historical incidence data and phylogenetic analysis.

The content of this chapter draws from the following publication:

- Adiga*, Chopra*, Wilson* et al. *A High-Resolution, US-scale Digital Similar of Interacting Livestock, Wild Birds, and Human Ecosystems with Applications to Multi-host Epidemic Spread*. Proceedings of the National Academy of Sciences 2026 [38]

8.2 Introduction

Highly pathogenic avian influenza (HPAI) poses a serious global threat to health, environment and food security. In the Americas alone, the unprecedented spread of H5N1 virus

clade 2.3.4.4b has led to severe loss of wildlife [140–143]. In the US, the incidence among wild birds continues to be widespread. Large-scale outbreaks in poultry and dairy cattle threaten food production [144–147] and health [148]. There have also been several instances of zoonotic transmissions [149] through exposure to poultry and cattle, which poses a serious pandemic risk [150]. Recent works analyzing this phenomenon in poultry and cattle [145,147,151,152] underscore the urgent need for a modeling platform to help understand, prepare, and respond to the spread of HPAI [153], accounting for the various agents that shape, and are affected by, this phenomenon [154,155].

Several national-scale, realistic in silico representations of populations, socioeconomic activities, and built infrastructures have been developed at fine spatiotemporal resolutions to study complex phenomena such as epidemiology, emergency response and food security [145,156–160]. Here, we refer to such synthetic datasets as *digital similars*. They have statistical similarity to real data, but differ from “digital twins” [161–164], which are intended as precise “living” replicas of the real-world systems they represent [165,166]. These realistic datasets constructed from diverse datasets and modeling are used for risk assessment and simulation modeling, as evidenced by studies conducted during the COVID-19 pandemic to analyze the dynamics of infectious diseases [20,167–171].

The first large-scale high-resolution digital similars of socio-technical systems were developed more than two decades ago [159]. Subsequently, many products have been proposed to address problems in multiple domains, ranging from simple gridded distributions of populations with demographic attributes, to definitions of activities and interactions with built infrastructures [171,172]. Some of these digital similars focus on modeling the spatial distribution of livestock [145,151,156,157,173,174]. Of these previous efforts, those on US-scale data sets, including recent ones in the context of HPAI, focus on a single livestock type. Key data challenges stem from the need to explore and fuse diverse, often sparse, data sets, which are misaligned in format and spatial resolution. Methodological challenges in this context include the choice of appropriate objectives, assumptions and constraints in the algorithmic formulations in order to achieve realistic representations that are statistically consistent with the parent data sets (for example, composition and distribution of livestock farms).

8.2.1 Summary of our contributions

This work presents a framework for high-resolution multi-layered spatiotemporal representation of the contiguous US, henceforth referred to as FIELD (Framework for In silico representation of Environment-Livestock-Demographics interface), that captures (i) the distribution of livestock populations and operations for multiple types (like cattle or poultry) and commodities within each type (like beef or dairy cows, chickens, turkeys, etc.); (ii) associated food processing center locations, capacities, and functions; (iii) spatiotemporally-varying wild bird abundances for multiple species affected by H5N1; and (iv) human populations with demographic features and attributes consistent with agricultural employment, as illustrated in Figure 8.1.

We leverage diverse open datasets (listed in Table 8.1), and address data gaps by using a combination of statistical tools and mathematical programming. Mapping livestock populations to farms and assigning them to grid cells are cast as optimization problems and solved using integer linear programs. We perform rigorous data quality checks with reference

to the source data sets, and verification and validation studies using independent data sets. Our work builds on earlier important work (e.g., [145, 151, 156]) and extends it to multiple livestock types and commodities on a national scale. The dataset is openly available through an interactive web dashboard, Digital Twin for Transboundary OneHealth (DiTTO) [175].

We develop a host-specific, county-level incidence likelihood prediction model that estimates risk of farm-level outbreaks in a future time window. The model incorporates spillover risk from wild birds and conditional risk based on ongoing outbreaks, to infer the effects of the unobserved pathways of transmission that increase the risk of infection from nearby farms. The model quantifies the increased risk of infection due to clustering of livestock operations thus assisting in prioritization for real-time surveillance and data collection. The model also brings to light the differences in the dynamics of H5N1 spread in different livestock types. We utilize phylogenetic analysis to benchmark the risk model against the evolutionary history of H5N1. By segregating the novel genotype B3.13 (associated with dairy cattle) from wild-type lineages, we evaluate how well our risk metric captures distinct transmission regimes. While the risk model successfully identifies traditional wild bird-poultry interfaces in non-B3.13 lineages, it also adapts to capture the novel bovine-centric interfaces driving the B3.13 outbreak. We identify geographical hotspots where dense human populations and intensive livestock operations intersect, creating significant potential for large-scale outbreaks.

8.3 Results

8.3.1 The Digital Similar

FIELD provides a unified gridded representation of livestock production and processing operations, the human population, and wild bird populations in the contiguous US. Figure 8.1 provides a layered view along with a summary of population sizes. Table 8.1 provides an overview of the data sources used to construct FIELD. Formally, $\text{FIELD}(V, \mathcal{L}, \mathcal{P}, \mathcal{B}, \mathcal{H})$ is defined over a grid V overlaid on the study region. Each grid cell $v \in V$ has attributes that capture the details of each of these components. In the current setting, we use a 5×5 square-arcminute grid. Descriptions of the components $\mathcal{L}$, $\mathcal{P}$, $\mathcal{B}$, and $\mathcal{H}$ are provided below.

Livestock $\mathcal{L}(\Theta_{\mathcal{L}}, \{\Gamma_{\theta} \mid \theta \in \Theta_{\mathcal{L}}\}, \{\mathcal{F}_{\theta} \mid \theta \in \Theta_{\mathcal{L}}\})$. We develop a novel, extensible approach to construct the livestock layers from agricultural census and grid-level estimates of livestock populations, which comprise four types of animals: $\Theta_{\mathcal{L}} = \{\text{cattle}, \text{poultry}, \text{hogs}, \text{sheep}\}$. For each type $\theta \in \Theta_{\mathcal{L}}$, Γ_{θ} denotes the set of different “commodities” of animals. For example, $\Gamma_{\text{cattle}} = \{\text{beef}, \text{dairy}, \text{other}\}$. The full list of commodities is provided in the Supporting Information (SI). For each type of livestock, the population is partitioned into farms¹. The collection of farms for each livestock type θ is denoted by $\mathcal{F}_{\theta}$. For each farm $f \in \mathcal{F}_{\theta}$, the population of each commodity γ , denoted by $H_{f\gamma}$, is specified. (We use H for head counts). Also specified is the grid cell v to which this farm is assigned. Figure 8.3(a) shows the sequence of steps performed to generate livestock representations from census and gridded data using statistical and optimization techniques.

¹All livestock production operations will be referred to as farms.

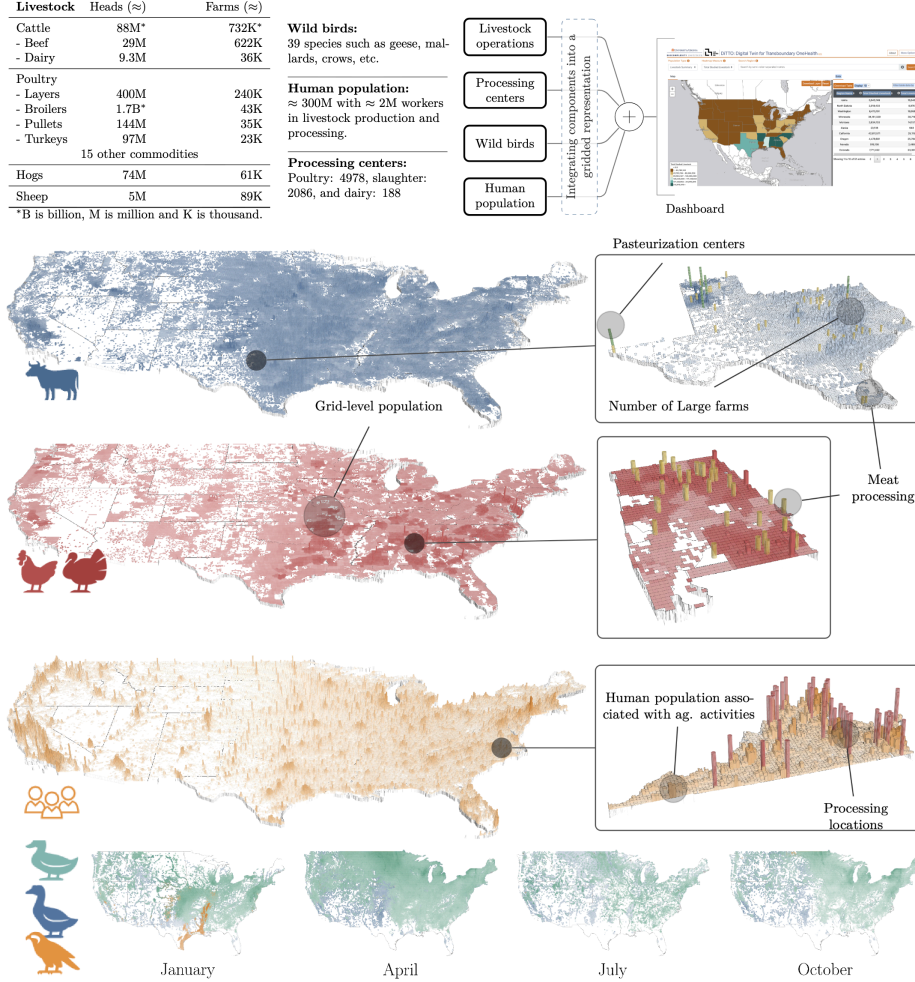


Figure 8.1: **Overview of the digital similar FIELD.** A schematic of the system highlighting the various components and the dashboard through which the data is exposed is provided at the top. Two of the four livestock layers are shown. We have zoomed in on major production regions for the respective livestock. Both population density and counts of farms are depicted. Also shown are livestock and dairy processing centers. For the human population, agricultural workers are highlighted. The spatiotemporal distribution of three wild bird populations is shown in the bottom layer. We provide an interactive visualization dashboard, Digital Twin for Transboundary OneHealth (DiTTO) [175] to make the data available.

Processing centers $\mathcal{P}$. This layer captures livestock-associated food processing centers such as meat processing, dairy processing, and poultry processing units. Each processing unit $p \in \mathcal{P}$ contains attributes such as the location of the unit, type of processing, and the size estimate.

Wild birds $\mathcal{B}(\Theta_{\mathcal{B}}, A(\cdot))$. Spatiotemporal distribution of 39 species of birds were identified as significant vectors of avian influenza as derived from EBIRD (year 2022) and H5N1 incidence data from 2022–2025 (see Table 8.1 for the data sources). Let $\Theta_{\mathcal{B}}$ denote the set of different

species. The abundance of a species $\theta \in \Theta_B$ in grid cell $v \in V$ at time t is denoted by $A(\theta, v, t)$. The data is available at a weekly resolution.

Human population $\mathcal{H}(\Delta, \mathcal{E}, \pi(\cdot))$. This component provides a grid-level representation of the human population with emphasis on agricultural workers (workers associated with livestock and its processing). For each cell v , $\pi(v, \delta, \epsilon)$ denotes the size of the subpopulation that belongs to the demographic group $\delta \in \Delta$ defined by attributes including age group, sex, and employment in professional classes specified by $\epsilon \in \mathcal{E}$. An employment group ϵ is defined by occupation and industry attributes, where non-agricultural employees are all binned into a single group.

8.3.2 Incidence Likelihood Estimation

We consider a general modeling approach to estimate the risk of H5N1 incidence. Let $I(C, s, t)$ represent the event that at least one new farm containing commodity s in county C is infected at month t . In our model, there are two components that contribute to the likelihood of this event. The first component is the farm-level susceptibility to *spillover from wild birds* $P_{\text{birds}}(f, s, t)$, where f is a farm containing livestock commodity s due to spillover from wild birds. It is modeled as a distance-weighted function of the infected wild bird population in the vicinity of the farm. The total introduction susceptibility to a county C is the probability that at least one farm is infected in time step t : $P_{\text{birds}}(C, s, t) = 1 - \prod_{f \in C, s_f = s} (1 - P_{\text{birds}}(f, s, t))$. The second component is the *conditional susceptibility* model, which captures the increased risk of an outbreak conditioned on the presence of infection in the region. Specifically, we compute a k -month ahead forecast given the historical incidence for δ months. Formally, given the time sequence $\mathcal{I}(C, s, t, k, \delta) = \langle I(C, s, t - \delta - k), I(C, s, t - \delta - k + 1), \dots, I(C, s, t - k) \rangle$, we compute the probability of an incidence at time t , that is, $P_{\text{cond}}(C, s, t, k, \delta) = \Pr(I(C, s, t) \mid \mathcal{I}(C, s, t, k, \delta))$. To estimate $P_{\text{cond}}(\cdot)$, we employ a Hidden Markov Modeling (HMM) approach. We aggregate the two types of risk to make a comprehensive k -months ahead forecast: $P_{\text{comp}}(C, s, t, k, \delta) = 1 - (1 - P_{\text{birds}}(C, s, t))(1 - P_{\text{cond}}(C, s, t, k, \delta))$. Details on model definitions and calibration are in the SI. At each time step, the conditional susceptibility model uses the past sequence of incidences to update its state and, subsequently, estimates the likelihood of incidence for future time steps.

Strong performance validates risk assessment framework. We evaluated host-specific risk models against ground-truth H5N1 incidence data, starting from the end of the training period and continuing through December 2025. In the case of dairy cattle, we combined the susceptibility of all counties in the Central Valley of California as a single region for evaluation to be consistent with the reporting format. In Figures 8.2(a) and (b), the top row shows a recall-based comparison that quantifies the following: if a county C reported a case in month t , what was its estimated risk profile at that time? For the model that only accounts for spillover from wild birds, P_{birds} (Figure 8.2(a)), around 60% of the instances correspond to High or Very high risk categories for turkeys and layers, while three of the four regions in the case of dairy cattle also fall in the High or Very high risk categories. In the context of poultry, this observation aligns with previous findings [145,151] that employ similar risk estimation

approaches. For dairy cattle, while farm-to-farm transmission likely dominates over wild bird spillover, we observe that several production-intensive regions align with migratory bird corridors, warranting continued surveillance for potential future emergence.

The inclusion of the conditional susceptibility model in the comprehensive risk model (P_{comp}) highlights the distinct commodity-specific dynamics (Figure 8.2(b)). For poultry, we observe minimal change in recall performance, suggesting that outbreaks are sporadic and short-lived (1 month-ahead forecasts are much better than longer-term forecasts), while farm-to-farm spread is still likely (as discussed later). For dairy cattle, though, we observe a marked improvement in the recall performance, showing that the chance of an outbreak is strongly influenced by the presence of infection in the region.

There are three caveats to this analysis. First, aggregating counties in California’s Central Valley creates potential bias in susceptibility estimates; however, the region’s high density of dairy farms ensures that risk estimates and profiles remain meaningful. Second, some counties show Very low risk profiles, primarily due to the fact that there are no large farms (as per our thresholds) in those counties, highlighting data challenges. There are very few such instances in each commodity. Finally, delays in reporting of time-stamped farm-level incidences can affect the accuracy of the risk estimates. Timely collection and dissemination of this data at the highest possible spatial resolution is critical.

Commodity-specific conditional risk amplification from regional infection presence.

A key distinction between the unconditional risk and comprehensive risk models lies in the absolute risk values (bottom rows of Figures 8.2(a) and (b)). When there is no incidence for several months, both models yield identical risk estimates. However, among present instances, the comprehensive model assigns substantially higher risk probabilities than the unconditional model. This indicates that while baseline spillover risk from wild birds is generally low, the presence of infection in a county significantly elevates risk across livestock commodities by at least an order of magnitude (as learned from the data). Our estimates of P_{birds} , the likelihood of spillover from wild birds, is based on setting one spillover event per case cluster (see Methods for details). Even setting it to a higher value does not explain the observed risk amplification due to this mode of transmission. In Figure 8.2(c), we plot the time series of susceptibility scores for counties with outbreaks. The plots show how the susceptibility spikes triggered by events. For cattle, P_{comp} is much higher than for poultry, but sharply drops off in the absence of new cases, indicating that farm-to-farm spread is a key for this commodity. For poultry, we see that the susceptibility remains elevated for longer periods, perhaps indicating a higher level of incidence in wild birds. A previous study based on phylogenetic analysis reveals that for most of the detections in poultry (based on a 3-month sampling window in 2022), the transmission was likely due to spillover from wild birds, and not from farm to farm spread [176], supporting our findings. However, the outbreak clusters are of shorter duration, which could be attributed to interventions such as culling or early marketing [177]. An analysis of the HMM models is presented in the SI.

Characterizing commodity-specific risk using phylogenetics. We validated the correlating comprehensive risk scores with phylogenetically inferred cross-species links. We divide the analysis of 17,325 genomes into two groups – B3.13 genotype (bovine interface)

and the non-B3.13 genotypes (wild bird interface). For non-B3.13 lineages ($N = 12,096$ sequenced genomes), interactions are dominated by the wild bird reservoir. The most frequent cross-species interaction was between *Avian-Wild* (row) and *Chicken* (column), $N = 166$, with minimal cattle involvement. The risk scores from the comprehensive risk model (Fig. 8.2(d)) for poultry commodities were significant predictors of occurrence (e.g., *Avian-Wild* $\leftrightarrow$ *Layers*, $r = 0.15$, $p = 0.0$). This suggests that the comprehensive risk model effectively captures the ecological interface where wild birds introduce virus into poultry operations. On the other hand, as expected, the B3.13 genotype ($N = 5,229$) exhibits interactions heavily shifted toward cattle. Despite this shift in host dynamics, the comprehensive risk model remains robust (Fig. 8.2(e)), with the dairy risk layer serving as a significant predictor of B3.13 events involving cattle. For the B3.13 genotype we observe large numbers of cross species links between *Cattle* and *Chicken* ($N = 59$), *Cattle* and *Turkey* ($N = 23$), and *Cattle* and *Wild Birds* ($N = 23$).

High-risk regions yet to report H5N1 cases. The risk map in Figure 8.2(f) identifies counties at high risk according to the comprehensive model, despite having no reported H5N1 cases. In December 2025, Wisconsin reported an outbreak [178]. Our risk maps show a high potential for a multi-farm case cluster in the state. Notably, several key milk-producing counties in Pennsylvania, Minnesota, and Ohio fall into the Very High risk category as well. Similarly, major turkey-producing regions in South Carolina and egg-laying operations across many eastern states are identified as facing elevated risk. This is covered in the SI.

8.3.3 Validation of Farm Locations

We evaluated the construction process for the livestock layers by comparing the constructed layers with the parent datasets. The algorithms for generating farms and assigning them to grid cells (See Figure 8.3(a)) are also evaluated. These analyses are available in the SI. Here, we focus on the validation of the resulting synthetic representation using independent datasets. Concentrated Agricultural Feed Operations (CAFOs) are large animal feeding operations that are a potential hazard to the environment and health. CAFOs are regulated by the Environmental Protection Agency (EPA), and some state agencies provide location information, among other attributes. We obtained such data for cattle, hogs, and chickens from CAFOMAPS [179]. For each county–livestock instance for which such data is present, we selected large farms from our farm assignment based on livestock-specific thresholds informed by CAFO size specifications provided by various states [180]. We computed the Haversine distance of each CAFO location to the centroid of each grid cell that contains a large farm. We construct a weighted complete bipartite graph $G(A, B)$ for each county–livestock instance. Here, A corresponds to farm locations from our assignments; each farm is assumed to be located at the centroid of the grid cell to which it belongs. The set B corresponds to CAFO locations. For each $u \in A$ and $v \in B$, the weight on the edge (u, v) is the inverse of the distance between the two locations. We compute a maximum weighted perfect matching² of this bipartite graph to match each CAFO location to a farm in FIELD. The main objective is

²For a complete bipartite graph $G(A, B, E)$, where $|A| = m$ and $|B| = n$, a perfect matching consists of $\min\{m, n\}$ edges.

to map as many CAFO locations as possible. It is possible that the number of farms is greater than the number of CAFO locations, as not all locations are listed in CAFOMAPS. The results of the matching are analyzed in Figure 8.3(b). We considered two sets of thresholds, the second set corresponding to larger farms compared to the first. A large percentage of CAFO locations were matched in the case of cattle ($> 95\%$) and chickens ($> 83\%$), while in the case of hogs we observe only 50% match. A closer examination of Census of Agriculture (AGCENSUS) data reveals the reason for the low number of matches for hogs: the number of farms specified by the AGCENSUS dataset for the relevant size categories is less than the number of CAFO locations specified in CAFOMAPS. Among the matched locations, we observe that 90% of the CAFO locations are at most 10 miles from the grid centroid of the corresponding farm from FIELD, which places it in the same grid cell or a neighboring one.

8.3.4 Wild Bird Abundance and H5N1 Incidence

We compare the relative abundance of the chosen bird species, as captured in FIELD, with the occurrence of H5N1 cases at the state level. We recall that we chose to include all species with reference to H5N1 incidence data from 2022–2024. This exercise establishes the relevance of the included bird population to the colocation-based risk analysis. For a given state, we partition the wild bird species into two categories: (i) top- k species based on their abundance and (ii) the remaining set of species. Next, we compute the number of case reports that fall in the top- k species category. This count is divided by the total number of case reports in the state. Bird species are grouped into 15 categories (e.g., ‘Duck’, ‘Goose’, ‘Eagle’) based on species similarity to provide more robust comparisons. We employ the **top-k recall** metric to quantify the relationship between bird abundance and H5N1 cases. It is the proportion of species with H5N1 cases that are among the k most abundant species in a state within the target period and administrative region.

$$\text{Top-k Recall} = \frac{\frac{\text{\#Hosts with cases among top } k \text{ abundant hosts}}{\text{\#Hosts with cases}}}{\text{\#Hosts with cases}}.$$

The results presented in Figure 8.3(c) summarize the Top-k recall results for all states and different parts of the year. We observe that, for all quarters, the larger the bird population, the greater the likelihood of observed H5N1 cases. Secondly, across the eight reporting states for which the plot is shown, the top 10 ($k \geq 10$) species cover almost all the reported H5N1 cases, so the recall value is close to 1. More plots are shown in the SI for multiple US states with details about specific bird types. These results suggest a robust correlation between bird abundance and H5N1 case occurrences.

8.3.5 Livestock Worker Population

Here, we compare the livestock worker population in FIELD with counts obtained from the Quarterly Census of Employment and Wages data corresponding to the year 2023 (Table 8.1, BLS). Details of how livestock workers were identified in each dataset are present in the SI. The plot in Figure 8.3(d) shows the difference between our data and the BLS data in counts of livestock worker population by county and state, respectively, for 2769 counties

that are common to both the datasets. We note that, for more than 96% of the counties, the synthetic population counts exceed that of BLS. This is expected, as BLS excludes a significant population of farm workers by only counting workers covered under unemployment insurance. For the remaining counties, the difference is usually very small compared to many of the remaining instances where the counts in FIELD far exceed BLS. It is possible that a significant portion of the farm worker population in these counties did not participate in the census.

Using data from various states and USDA (see Table 8.1), we estimated the labor per farm as a function of the number of heads for dairy and poultry operations (details in Methods and the SI) using an optimization algorithm which minimizes the deviation between assigned workers and target values. Figure 8.3(e) shows the results of this analysis as a function of the maximum distance traveled to work. For both poultry and dairy, we observe that the number of assigned workers is generally less than the estimated target values. This could be due to various reasons such as under-representation of workers in our synthetic population, an inability to capture workers visiting multiple farms, or part-time labor. Based on this assignment, we model the mixing of livestock workers and the general population to assess risk to the latter population from potential exposure through interactions with livestock workers. Figure 8.3(f) shows the spatiotemporal mixing for dairy farmworkers and the general population as a function of a distance threshold M and mixing parameter β . The figure maps the areas where there is high mixing between workers and the general population, indicating potential hotspots for zoonotic transmission. The methodology and more results are presented in the SI for various combinations of parameters.

8.4 Discussion

Our work presents a comprehensive methodology to construct synthetic spatiotemporal datasets of food systems, human populations, and wildlife. Several models have been proposed for subcounty-level disaggregation of livestock populations [156,157,173,181]. Using a random forest model, Gilbert et al. [173] develop a global distribution of populations of multiple livestock types. Burdett et al. [156] develop the FLAPS model to simulate populations and locations of individual farms for swine using the AGCENSUS dataset and a microsimulation model; this work has been used to analyze the spread of porcine deltacoronavirus in the US [181]. In very recent work, Prosser et al. [145] address estimation of transmission risk at the wild waterfowl–domestic poultry interface. They develop a spatiotemporal model combining 10 species-level wild bird abundance models [182] with a commodity-level poultry farm model [183]. Humphreys et al. [151] use a variety of datasets, including GLW, to model waterfowl movement and interactions with poultry farms and human populations.

A risk-based monitoring and surveillance system was developed by USDA’s National Wildlife Disease Program [184] initially as in DeLiberto et al. [185]. The current system employs adaptive surveillance methodology (e.g., [186,187]) but using data streams from wild bird abundance from eBird, wild bird movement networks from bird banding data (e.g., [188]), livestock densities, environmental variables, and historical avian influenza surveillance data. The data are integrated in a dynamic occupancy modeling framework to predict hotspots of influenza avian virus in wild birds at a watershed scale across four distinct biological seasons.

This system guides annual sampling quotas for each watershed and seasonal window. The surveillance data provide early detection for new HPAI strains [189]. New surveillance data are made available rapidly [190] to provide spatio-temporal risk assessment of spillover for producers.

In the context of HPAI outbreaks, there is a need for significant extensions to these works given the requirement to simultaneously account for colocated farms of multiple livestock types, wild birds, and human populations. Evaluating potential scenarios of spillovers to other major livestock types like swine [191] or beef cattle is critical for pandemic preparedness. Our work takes the first few but significant steps towards this necessary integrated view of the agricultural ecosystem.

The risk assessment framework utilizing FIELD quantifies livestock-type-specific characteristics of H5N1 spread. The comprehensive model and subsequent phylogenetic analysis clearly highlight different pathways at play, and these pathways affecting different livestock types in distinct ways. Further host-specific investigation is required to discover and quantify their significance and better explain incidence clusters. These include movement of animals among farms, biosecurity practices, the role of livestock operations as reservoirs, humans as pathways, etc. These estimates can guide targeted surveillance and biosecurity assessments [153] by prioritizing high-risk farm locations and regions. The analysis of mixing between livestock workers and the general population bring to light the existence of large numbers of hotspots that could be potential sites for zoonotic spillover. These are conservative estimates as they do not account for workers not captured by the census or the possibility of risk due to exposure in processing centers. Future work could focus on validating these findings using other data streams like waste water surveillance and strengthening surveillance in such areas for early detection and mitigation.

While this dataset offers valuable insights, it faces challenges stemming from its parent sources. Specifically, the synthetic GLW dataset and AGCENSUS are misaligned in time. Our analysis of farms with known locations also shows that some assigned operations remain unmatched, confirming spatiotemporal misalignment. Unlike previous works [145,156], we do not provide coordinate-level assignments. While this potentially limits fine-grained analysis of farm-to-farm movement, such precision requires additional operation-level data and rigorous validation. Emerging works [192–194] on mapping concentrated animal feeding operations (CAFOs) (using remote sensing and deep learning) can greatly enhance FIELD in this regard. Furthermore, we do not model mixed-livestock farms (e.g., cattle and poultry). For certain livestock like poultry, limited information on farm size leads to heavy-tailed population distributions. Future exploration of AGCENSUS cross-tabulation data could address these gaps. For wild birds, EBIRD abundance measures are subject to observational biases and underestimation. Finally, mapping agricultural workers remains difficult due to variations in farm size and automation [195], as evidenced by our comparison with BLS data and subsequent risk assessment. The risk assessment model performance remains contingent on the quality and completeness of underlying data, which may be subject to reporting biases and delays. For more accurate risk assessments, understanding and quantifying the significance of various pathways of introduction and spread is required. Recent work [152] addresses interstate spread via cattle movement models based on historical data [196]. While incorporating such data would enhance introduction risk estimates, accurately assessing per-county susceptibility requires additional data on intra-county movements of both animals

and humans, and how they are being shaped by the spread. As H5N1 variants emerge, dynamics may shift, necessitating continuous recalibration. Exploring the role of wild birds in cross-species transmission is another critical area for future work [197].

While the choice of agents and design decisions were influenced by the focus application—the spread of HPAI-like diseases—the utility of such a digital twin extends well beyond this domain. Its modular architecture enables leveraging subsets of layers depending on application requirements. This dataset can model the spread of other pathogens, such as West Nile virus, Salmonella, or cholera, which involve interactions between livestock, infrastructure, and human populations [198–201]. Such systems have value beyond infectious diseases in domains including food safety, agricultural economics, environmental impact, pollution, disaster response, biosecurity, and supply chain optimization [156,173,179,202]. For biodiversity and conservation, incorporating models on bird movement can help identify critical habitats in relation to livestock operations [203]. Spatially explicit synthetic datasets are extensively developed for non-epidemiological applications [204–208]. FIELD provides a critical foundation for understanding our complex agricultural ecosystem essential to pandemic preparedness, food security, and environmental sustainability.

8.5 Method

8.5.1 Datasets

Table 8.1 summarizes all data used. We used publicly available datasets, which can be categorized into three types: (i) census, (ii) synthetic realistic datasets derived from models and data samples, and (iii) real location-level datasets. From a spatial unit perspective, some data (e.g., AGCENSUS and H5N1 incidence data) are specified at various administrative levels (county or state), some data (e.g., GLW and EBIRD) are specified at the grid level, while exact locations are provided for the rest. Some of these datasets have been used for the construction of FIELD, while the rest have been used for subsequent analysis. More details about each dataset are provided in the relevant sections.

8.5.2 Livestock

Two data sources were used to construct the livestock layers: Census of Agriculture (AGCENSUS) and Gridded Livestock of the World (GLW). An overview of the methodology is shown in Figure 8.3(a). Additional details, including data organization, gaps, and other challenges, are described in the SI. We use a combination of integer linear programs (ILPs) and iterative proportional fitting (IPF), the latter following Burdett et al. [156,237,238]. We use the ILP FILLGAPS (described in the SI) to fill in missing data for the following types of counts: state-total, state-by-farm-size, and county-total. It takes as input all the known counts and the bounds on the unknown counts, and distributes the heads that are unaccounted for equitably across all entities for which the counts are missing. To fill gaps for county-by-farm-size counts, we apply IPF. At each step, the objective is to make use of all available data (in all count types).

The next step is the generation of farms. Given a livestock type and county, the objective

Table 8.1: Datasets explored to construct and validate FIELD. Throughout the paper, each dataset will be referred to by its abbreviation.

Name	Abbrev.	Source	Description
Census of Agriculture	AgCENSUS	[209–211]	Provides location-level Ag data such as number of farms, farm sizes, crop types, and fallowed status. Provides individual-level Ag data such as number of workers on a farm.
Gridded Livestock of the World	GLW	[173,212,213]	GLW4.0 provides distribution maps for several livestock types. FAO hosts this website.
eBird Status and Trends	EBIRD	[182,214]	Weekly data of relative abundance of migratory birds across geospatial regions throughout the year.
Dairy processing	Dairy plants	[215]	Large dairy processing centers and their attributes regulated by USDA Agricultural Marketing Service (AMS).
Meat, poultry, and egg processing	Meat and poultry	[216]	Listing of establishments that produce meat, poultry, and/or egg products regulated by USDA Food Safety and Inspection Service (FSIS).
US population	USPOP	[19,217]	Synthetic digital twin of the US human population.
Quarterly Census of Employment and Wages	BLS	[218]	Quarterly counts of livestock workers (NAICS code 112) from Bureau of Labor Statistics.
CAFOs in the US	CAFOMAPS	[179,180]	A map of Concentrated Animal Feeding Operations (CAFOs) in the southern United States covering nine states.
Farm labor	Labor	[219–225]	Data on labor for poultry and dairy farms.
H5N1 outbreaks	H5N1CASES	[149,190,226,227]	H5N1 bird flu detections in wild birds, livestock, and humans by state and county.
Bacterial and Viral Bioinformatics Resource Center	BV-BRC	[228]	Repository of viral genomes and metadata including H5N1 sequences from the US and worldwide.
Nextstrain H5N1 cattle outbreak North America	Nextstrain	[229–236]	Full genome analysis of the ongoing influenza A/H5N1 cattle outbreak in North America.

is to obtain a grid-level distribution of farms that is consistent with operations and their sizes as well as the grid-level distribution of livestock populations. We use a two-step procedure using optimization algorithms: (i) generating farms consistent with the county-by-farm-size counts (either provided for or estimated) and (ii) assigning farms to cells. The GENFARMS algorithm for generating farms is described in the SI. The objective function encodes several minimization criteria. They are stated in order of priority: (i) feasibility: modifies the commodity head count minimally to ensure feasibility of the assignment, (ii) equitable distribution of head counts for each commodity of livestock across farms within a category, (iii) minimize the number of commodities within a farm, and (iv) align with known county-by-farm-size counts. The FARMSTOCELLS algorithm (described in the SI) assigns a cell to each farm. The objective of this algorithm is to ensure that the head counts resulting from the assignment and GLW head counts are closely aligned.

We provide a layer for animal product processing plants with attributes such as size, type of processing (dairy, meat, egg, etc.), livestock type, etc. (see Table 8.1). While coordinate-level data was available for the location of meat processing plants, for dairy processing plants, the locations at the city level were inferred by manual inspection.

8.5.3 Wild Bird Abundance and Movement

We leveraged data from EBIRD’s Status and Trends products [214] (see Table 8.1) to construct this component. We recall that this component $\mathcal{B}(\Theta_{\mathcal{B}}, A(\cdot), G_{\mathcal{B}})$ captures spatiotemporal abundance and movement of multiple species of birds. The EBIRD data provides weekly estimates of relative bird abundance across a high-resolution grid (1.84miles $\times$ 1.84miles). A relative abundance of 1.0 for a species at a particular location and time would indicate that an average EBIRD checklist at that place and time would be expected to count one individual of that species. Higher values indicate more individuals would be expected, while lower values indicate the species would be observed less frequently or in smaller numbers. We chose bird species for which H5N1 cases were observed in the period 2022–2024. Out of 40 species,

abundance data was available for 36 species. For each of the selected species, we extract relative abundance values along with their associated geographic coordinates for all 52 weeks in the year.

8.5.4 Human Population

We develop a gridded representation of the US population with rich demographic and employment-related attributes. This data is derived from a synthetic population [19,159,160] that is developed using diverse datasets such as census data, land use data, activity patterns, building maps, etc. Each individual in the population is associated with a residential location, an occupation identified by the Standard Occupational Classification code (SOCP), and an industry, identified by the North American Industry Classification System (NAICS) code. We identified livestock workers in the synthetic population using combinations of SOCP and NAICS codes (the rest were tagged as non-livestock workers), and aggregated the population at the grid cell level as a function of occupation and demographic parameters. To assess risk to the general population from potential exposure through livestock workers from poultry and dairy farms, we model the assignment of workers from grid cells to farms using openly available data on farm labor and as a function of the distance traveled to work. An optimization algorithm WORKERFARMASSIGN minimizes the deviation between assigned workers and target values. The details are in the SI.

8.5.5 Incidence Likelihood Model

The incidence likelihood model estimates the probability of at least one new farm containing commodity s in county C being infected at time t , denoted by $P(I(C, s, t))$. We model this probability as a combination of two components. The first component is the *wild bird spillover susceptibility* model defined at the farm-level as $P_{\text{birds}}(f, s, t) = 1 - e^{-\alpha(s)I_{\text{birds}}(f, t)}$, where f is a farm containing livestock commodity s at time t , $I_{\text{birds}}(f, t)$ captures the influence of wild bird relative abundance on farm f at time t while the parameter $\alpha(s)$ scales this influence for commodity s . The parameter $\alpha(s)$ was tuned via Bayesian optimization with Gaussian processes to minimize the difference between the expected number of infected farms and the estimated number of spillover events, which, in turn, was estimated by analyzing reported cases. Our model is unlike Prosser et al. [145], where spillover probability values for different poultry types is based on values gathered from past literature.

For the second component, the county-level *conditional susceptibility* model $P_{\text{cond}}(\cdot)$, we use Hidden Markov models (HMM) to model the conditional susceptibility probability $P_{\text{cond}}(\cdot)$. Suppose $\text{HMM}(Q, O, A, E, \pi_0)$ is a model with N states $Q = \{q_0, \dots, q_N\}$. O is the observations set, a collection of binary time series per county; 1 if at least one case was reported at time step t , else 0, where each time step is a month. The transition probability matrix A defines the probability β_{ij} of transitioning from state q_i to q_j . The set E defines the emission probabilities γ_{ib} of observing $b \in B$ when the model is in state q_i . Since the objective is to model the susceptibility conditioned upon having observed an incidence, we only accounted for the time series data across all counties for which there was at least one incidence pertaining to the target commodity. The training details are in the SI. The susceptibility to a county C

at time step t with respect to a commodity s is $P_{\text{cond}}(C, s, t) = \sum_{i=1}^N \beta_i(t) \cdot \gamma_{i1}$, where β_i is the a posteriori probability of being in state q_i at time t .

8.5.6 Phylogenetic Identification of Cross-Species Links

We integrated metadata and hemagglutinin (HA) sequences from both Nextstrain and BV-BRC (Table 8.1) to construct a unified view of H5N1 evolution. We constructed two separate phylogenetic trees: one for the B3.13 genotype and one for all non-B3.13 genotypes, utilizing the Nextstrain *avian-flu* pipeline. We define a *Cross-species link* as a phylogenetically inferred ancestral connection between distinct host categories within the same geographic state. Since sparse sampling makes directionality problematic, these links are treated as bidirectional markers of interface risk. Links were identified using a depth-first search algorithm on the ancestral state-reconstructed trees. A link between hosts within a single state was recorded when a most recent common ancestor (MRCA) was found subtending two host labels and the maximum temporal branch length from the MRCA to the implicated descendants did not exceed 140 days. This temporal window was chosen to capture *regional lineage persistence* across seasons, acknowledging that it likely aggregates multiple transmission events into longer-term environmental interfaces. Events were aggregated monthly to correlate with the digital similar’s Comprehensive Risk scores (P_{comp}) and by state to accommodate the limited resolution of genomic metadata. Further details provided in the SI.

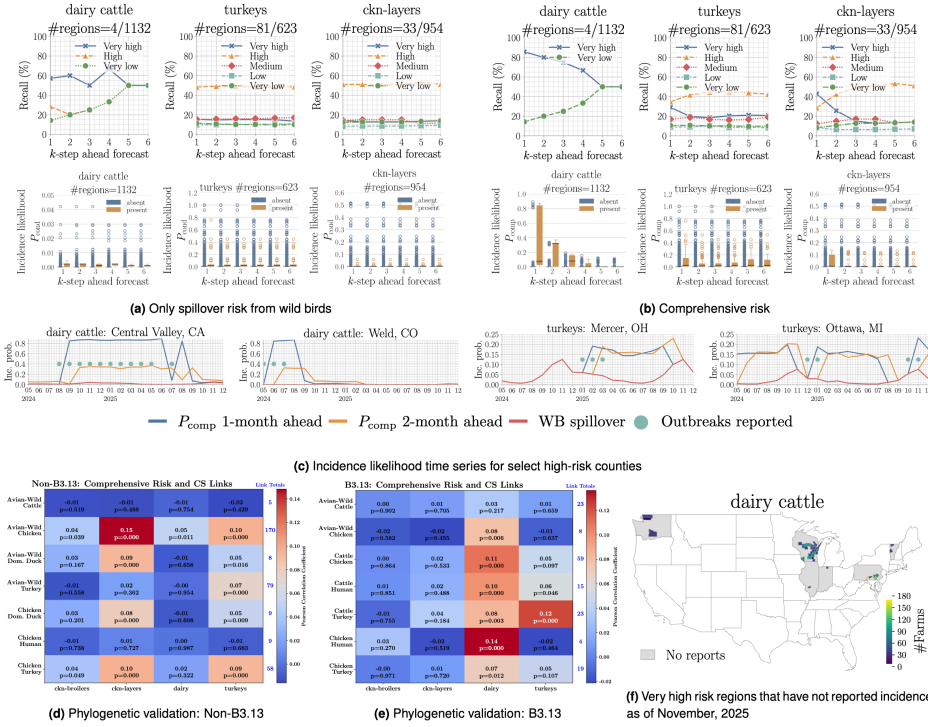


Figure 8.2: (a) Wild bird spillover susceptibility: Recall-based model performance for 1-6 month ahead forecasts: Each H5N1 instance corresponds to a unique county-month pair (denoting at least one H5N1 incidence report from that county in that month). Each county-month pair is associated with a percentile-based risk profile using relative susceptibility risk as the metric: Very high (95-100 percentile), High (75-95), Medium (50-75), Low (25-50) and Very low (0-25). For the risk computation, we only considered counties with at least one farm (poultry ≥ 1000 heads and dairy ≥ 100 heads). We added to this set counties where outbreaks are reported, but FIELD has no farms meeting the threshold constraints. In the top row, each plot provides normalized counts of the risk profiles for the positive county-time instances. The second row plots the risk scores distribution for negative and positive instances. Plots without outliers are provided in the SI. For the dairy cattle, the entire Central Valley, CA, is represented as a single region. (b) Comprehensive susceptibility model: Similar recall-based evaluation as (a). (c) For selected counties, the time series of different susceptibility scores are provided for 20 months. Markers are also plotted for each month in which at least one case was reported. We have two instances each for dairy cows and turkeys. (d) Correlation between state-month aggregated cross-species links and Comprehensive Risk Score (P_{comp}) for Non-B3.13 genotypes. Network is driven by Wild Bird-poultry interactions. (e) Correlation between state-month cross-species links and Comprehensive Risk Score (P_{comp}) for B3.13 genotypes. The model successfully identifies these interfaces, with the “dairy” risk serving as a significant predictor of B3.13 activity, likely reflecting shared high-density agricultural geography. (f) For states that have not reported incidence till date, we plot counties with “Very high” risk profile for at least three months between April 2025 and October 2025. The colors correspond to the number of large farms. A green/yellow color implies a higher number of farms and, therefore, a greater risk of a negative impact.

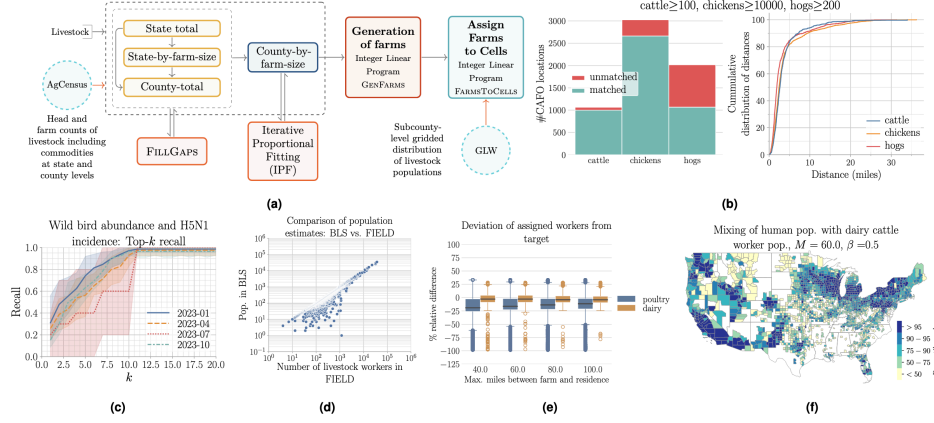


Figure 8.3: (a) A schematic of the livestock layer construction, including data processing, generation of farms and assignment of cells to farms. It takes as input AGCENSUS data that comprises head and farm counts at various administrative levels and the gridded distribution of the livestock populations from GLW. FILLGAPS is an integer program that fills gaps in the census data. GENFARMS is an integer linear program (ILP) for distributing the livestock populations to farms consistent with the census data. FARMSTOCELLS is an ILP that assigns farms to grid cells with the objective of aligning the population with GLW. Validation of components: (b) Analysis of mapping CAFO locations by livestock type to farms from FIELD. Farms were chosen based on the thresholds stated in the title. The first subplot shows how many CAFO locations were matched. The SI has an additional plot for a different set of thresholds. The second subplot provides the cumulative distribution of the distances (in miles) between matched pairs of CAFO locations and farms. An additional plot for a different set of thresholds is in the SI. (c) Analysis of H5N1 cases and bird abundance for the period of January to December of 2023. The plot summarizes the results across eight reporting states for the four quarters. The x-axis corresponds to the number of top species groups considered, while the y-axis corresponds to the count of those groups with H5N1 incidence. Also plotted is the coefficient of variation (CV). (d) Comparing livestock worker counts between FIELD and data from the Bureau of Labor Statistics (BLS) for each county. (e) Relative difference between assigned and target workers for varying maximum distance traveled to work for poultry and dairy farms. (f) Mixing of livestock workers and the general population. The figure shows the heatmap of the quantile-based binning of the county-level scores for $M = 60$ miles and $\beta = 0.5$ for dairy cattle.

Chapter 9

Labor Market Disruption at National Scale

9.1 Motivation and LPM Application

Labor market disruption driven by technology evolution shares the same analytical structure as biological disruption, operating through different interaction channels at a different scale. When technology capabilities change the demand for particular skills, the consequences propagate through skill interaction networks, occupational dependencies, and geographic labor markets in cascades that no individual-level analysis can capture. Workers, firms, and institutions are the agents; skills, occupations, and geographic proximity define the interaction topology; and the collective outcome is an emergent property of how those interactions accumulate across the full population.

Traditional workforce metrics measure employment outcomes after disruption has occurred, at aggregate levels that do not resolve skill-level interactions. They are essential for retrospective assessment but not designed to capture forward-looking technical exposure at the skill level. The structural overlap between human skills and AI capabilities is distributed across geographies and occupations that aggregate statistics cannot resolve. Quantifying it requires representing the full US workforce as a population of interacting agents and simulating how AI capabilities propagate through that population’s interaction network.

This chapter presents Project Iceberg, which deploys the LPM framework at national scale on the US labor market. It represents 151 million workers as autonomous agents executing skills across 3,000 counties, computes the Iceberg Index as the wage-weighted output of the simulated cascade, and projects that technical automation exposure under stated assumptions is approximately fivefold larger than what visible adoption metrics suggest. The Index has been formally adopted by multiple US states and is monitoring over \$1 trillion of estimated labor exposure across the economy.

The content of this chapter draws from the following publication:

- Chopra et al. *The Iceberg Index: Measuring Skills-centered Exposure in the AI Economy*. 2025 [1]

9.2 Introduction

Artificial Intelligence is reshaping America’s \$9.4 trillion labor market, creating cascading effects across industries and communities that extend beyond visible technology sectors. When AI systems automate quality control in automotive manufacturing, the consequences propagate through supplier networks, logistics operations, and local service economies. These ripple effects multiply the economic stakes of AI adoption, yet remain invisible to the systems that guide billion-dollar workforce and infrastructure decisions.

While consumer applications such as ChatGPT capture public attention - helping with homework, writing, or quick research - the larger restructuring is already underway inside firms. AI systems now generate more than a billion lines of code each day, prompting companies to restructure hiring pipelines and reduce demand for entry-level programmers [239]. These observable changes in technology occupations signal a broader reorganization of tasks that extends beyond software development.

AI capabilities now span diverse operational contexts. Financial institutions deploy AI for document processing and analytical support. Healthcare systems automate administrative tasks, enabling clinical staff to allocate more time to patient care. Logistics operations integrate AI-powered systems to optimize fulfillment while creating demand for maintenance and coordination roles. Manufacturing facilities deploy AI-driven quality control to automate inspection tasks while creating roles in system coordination and oversight. Analysis of Bureau of Labor Statistics skill taxonomies reveals that current AI systems can technically perform approximately 16 percent of classified labor tasks (see Appendix A) - yet this technical capability remains largely invisible to the workforce planning systems guiding billion-dollar investments.

Historical precedent demonstrates both the opportunities and risks of such technological shifts. The internet revolution generated substantial economic value: technology companies now represent over one-third of S&P 500 market capitalization [240] and the U.S. digital economy contributes approximately \$4.9 trillion (18%) to GDP [241]. The impact extends beyond technology firms - research indicates each high-technology job correlates with approximately five additional positions in local services, healthcare, and retail [242,243]. Early-moving regions captured durable advantages during the internet era: North Carolina’s Research Triangle matured into a global research hub [244]; Texas scaled Austin into a top tech market [245]; Tennessee and Kentucky became national logistics leaders as FedEx’s Memphis super-hub and UPS Worldport in Louisville expanded [246,247]; and Utah’s “Silicon Slopes” rose as a cloud computing center [248].

Similar dynamics will now shape AI adoption patterns. Regions that align skill development, infrastructure investment, and industry strategy early may establish competitive advantages, while delayed response could result in widening disparities. Project Iceberg provides analytical infrastructure to support evidence-based planning as AI capabilities expand across the economy.

Physical Economy (Census Visible)		AI Economy (Census Blind Spot)	
Manufacturing Jobs	Factory locations mapped	AI-mediated work	No geographic anchor
Service Employment	Business address tracked	Human-AI teams	No job category exists
Worker Residences	Household surveys capture	AI Coordination	Siloed in private analytics
Economic Activity	Geographic boundaries	Digital value creation	Platform controlled data
Physical economy = measurable economy People work in places, Census counts people and places		AI economy ≠ measurable economy AI automation tools have no state id, work through private platforms census can't see	

Figure 9.1: Traditional workforce metrics miss AI-mediated tasks. Census data captures jobs tied to geographic locations and business addresses. Human-AI collaboration—where workers and AI systems jointly perform tasks within occupations—creates new forms of labor that existing metrics don’t capture. The Iceberg Index provides a forward-looking measure of this technical automation exposure, revealing skill-based transformation that remains invisible.

9.3 Measurement Gaps in Workforce Planning

Federal and state governments are already committing billions to prepare for the AI economy. The federal AI Action Plan [249] outlines 90 policy positions and has launched a national AI Workforce Research Hub [249]. DOE has identified 16 federal sites for data center development [250].

States are responding with major initiatives:

- North Carolina has secured a \$10 billion investment to expand data center capacity [251]
- Tennessee’s Google-Kairos reactor will power data centers while anchoring Oak Ridge as a nuclear innovation hub [252]
- Utah’s Operation Gigawatt will double statewide clean-energy production over the next decade [253]
- Virginia has committed \$1.1 billion to train 32,000 AI graduates [254]
- DOE has also committed \$100 million for nuclear safety training as that workforce is projected to triple by 2050 [255]

These efforts reflect real momentum in infrastructure, energy, and talent development. Yet evidence suggests workforce change is occurring faster than planning cycles can accommodate. Payroll data covering millions of workers shows a 13% relative decline in early-career employment (ages 22–25) for AI-exposed occupations [256]. Analysis of job postings across 285,000 firms for 62 million workers reinforces the pattern: the demand for entry-level positions have subdued while the focus has shifted to hiring for experienced roles [257]. These shifts, alongside widespread restructuring in the technology sector [239], indicate that the pace of change is accelerating across the economy. States are committing billions to workforce programs while key workforce dynamics remain invisible to traditional planning tools.

9.3.1 Challenge 1: Anticipate Workforce Shifts Before They Happen

The labor market is evolving faster than current data systems can capture. AI automates some skills and augments others, creating uneven effects across industries and regions. Much of this activity occurs on digital platforms—gig marketplaces, AI copilots, freelance networks—that fall outside conventional reporting [258]. By the time these changes appear in official statistics, policymakers may already be reacting to yesterday’s disruptions, committing billions to programs that target skills already displaced. Without forward-looking capability to test strategies before implementation, states cannot distinguish investments that prepare workers from those that arrive too late.

The Invisible Economy:

AI adoption and worker behavior unfold on platforms outside official data. Feedback loops between technology maturity, firm adoption, and state interventions mean that by the time disruptions appear in surveys, policy responses may target skills already displaced or industries already transformed.

9.3.2 Challenge 2: Measure Human–AI Work, Not Just Human Work.

Existing workforce planning frameworks were designed for human-only economies. They track employment, wages, and productivity, but were not designed to measure where AI capabilities overlap with human skills before adoption reshapes occupational structure. Each major transition required new measurement: in the postwar era, output per hour captured physical productivity [259]; in the internet era, the Digital Economy Satellite Account measured the value of online services [260]. The AI era is defined by intelligence itself becoming a shared input between humans and machines. What matters is not only the number and complexity skills to execute, but how much of each skill’s value can be performed by AI and which human skills remain differentially valuable. Without a skills-centered metric, states lack a systematic way to align investments with where AI-human skill overlap is concentrated.

Every Revolution Needs a New Metric:

Industrial era → Output per hour (measured physical productivity)
Internet era → Digital economy accounts (captured online service value)
Intelligence era → Skills-centered measure (reveals AI-human skill overlap)

9.4 Project Iceberg: Sandbox for the Human-AI Workforce

Iceberg simulates the emerging human-AI workforce: 151 million American workers and the AI capabilities that increasingly act on the same skills. The chapter’s contribution is methodological — a national-scale LPM in which workers, skills, and AI tools are connected through hypergraphs, and AI capability propagates through these structures over discrete

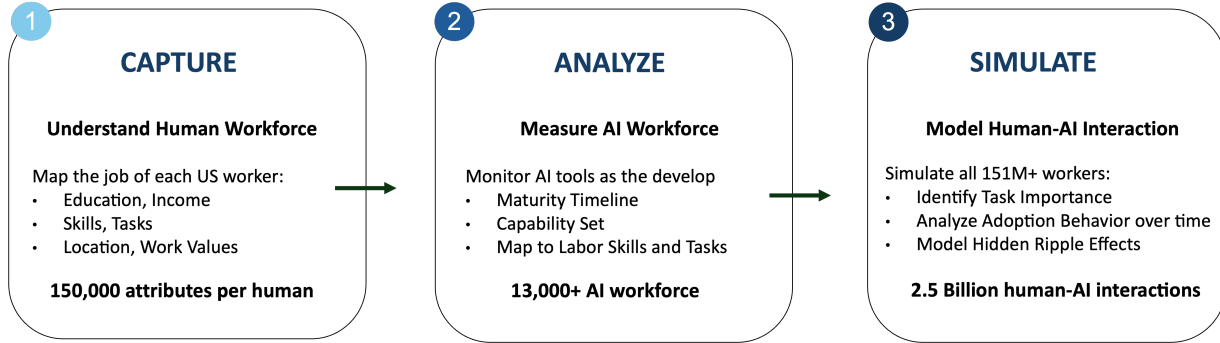


Figure 9.2: **Project Iceberg prepares states for the AI economy.** First platform to simulate human-AI workforce interactions at national scale, enabling policymakers to assess technical exposure, test workforce strategies, and target investments before committing billions to implementation.

adoption rounds. The Iceberg Index, presented in Section 9.5, is one summary statistic of the resulting dynamics. The same simulation supports other instantiations: projecting adoption timelines, evaluating reskilling interventions, or comparing alternative regulatory regimes. Built on MIT’s Large Population Models [261] and powered by Oak Ridge National Laboratory’s Frontier supercomputer [262], Iceberg turns trillions of workforce data points into scenario-planning capability.

9.4.1 The Iceberg Workforce Model

Iceberg represents both sides of the emerging labor market: what human workers currently do, and what AI systems are becoming capable of doing. By mapping these onto a shared skill taxonomy, it simulates how work evolves under different assumptions about policy intervention, adoption behavior, and technology readiness. The model is constructed in three steps (Fig 9.2):

Step 1 - Human Workforce Mapping: We construct a digital representation of the workforce: 151 million workers across 923 occupations in 3,000 counties, covering more than 32,000 distinct skills [263,264]. Each worker is modeled as an agent with attributes such as skills, tasks, and location. This enables analysis of reskilling potential and occupational similarity, which is essential for designing pathways when skills and tasks shift.

Step 2 - AI Workforce Mapping: We catalog over 13,000 AI tools - including copilots and workflow automation systems - using the same skill taxonomy. This allows direct comparison between human and AI capabilities. The alignment identifies where AI augments human work (e.g., automating hospital paperwork so nurses can spend more time with patients) and where it transforms tasks entirely (e.g., accelerating code generation so engineers shift toward oversight, testing, and integration skills).

Step 3 - Human-AI Simulation: We combine these two populations in Large Population Models (LPMs), simulating billions of interactions between workers, skills, and AI tools. The simulation projects how exposure spreads under given assumptions about technology readiness, adoption behavior, and regional variation. Policymakers can use the resulting scenarios to anticipate disruption, test reskilling strategies, and allocate resources where they

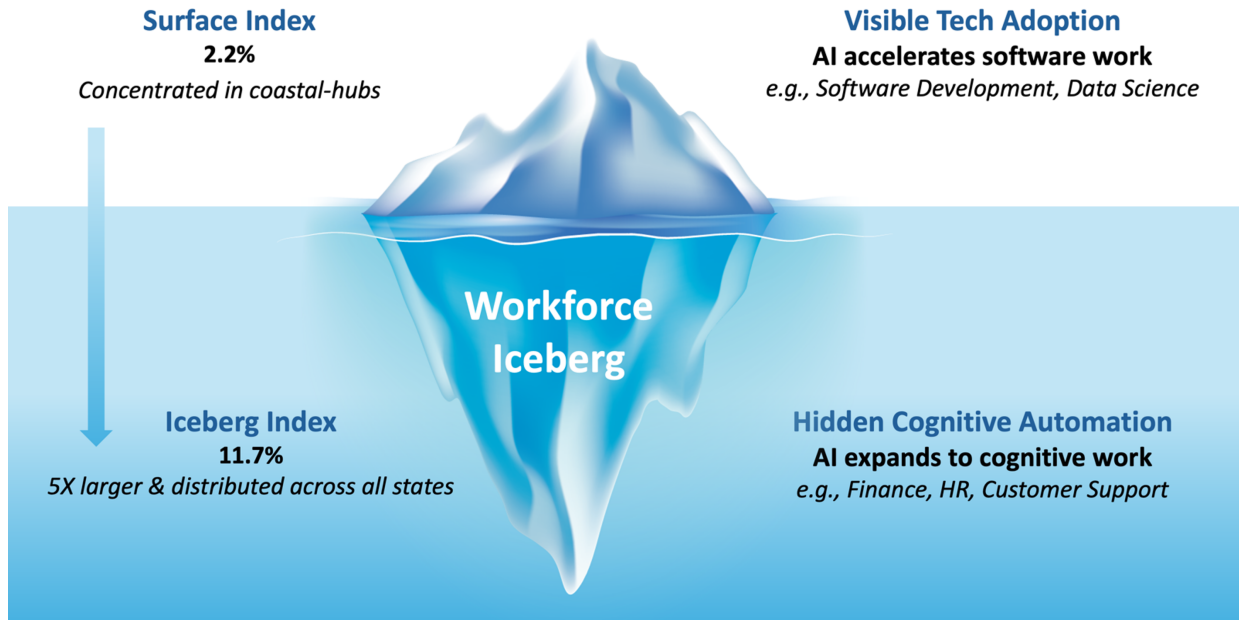


Figure 9.3: The Iceberg Index estimates workforce exposure approximately five times larger than visible tech adoption. Above the waterline: the Surface Index (2.2%) tracks current AI adoption in coastal technology hubs. Below the waterline: the Iceberg Index (11.7%) estimates technical capability spanning administrative, financial, and professional services nationwide. This larger value represents where the simulation projects exposure beyond visible tech-sector adoption, indicating where workforce preparation strategies based solely on visible signals may need to extend.

will have the most impact.

Iceberg Simulations: Scale and Infrastructure

- **Workforce Scale:** 151 million workers across 923 occupations in 3,000 counties
- **Skill Coverage:** 32,000+ distinct skills from standardized taxonomies
- **AI Systems:** 13,000+ tools across software, cognitive, and physical capabilities
- **Simulation Engine:** Large Population Models with AgentTorch [261].
- **Computing Platform:** Frontier supercomputer at Oak Ridge National Lab [265].

9.4.2 LPM Instantiation and Structural Assumptions

In LPM terms, each worker is an agent a whose state s_t^a encodes its current capability across the O*NET skill vocabulary. Workers are connected through two hypergraphs: a **skill-skill**

hypergraph H_{skill} derived from O*NET major and minor occupational groups together with skill-embedding similarity edges, and a **skill-tool hypergraph** H_{tool} linking each skill to the AI tools that can perform it. Geographic proximity at the county level provides a third interaction channel. The coordination module $\mathcal{K}_{\varphi_K}$ propagates AI capability across these structures over discrete adoption rounds, and the post-cascade exposure profile of each worker is the output from which the Iceberg Index is computed in Section 9.5.

The dynamics rest on five structural choices that are fixed by methodology rather than learned from outcome data. We state them explicitly so that downstream results can be interpreted in light of these choices.

1. **Skill vocabulary.** The simulation operates over the O*NET taxonomy of skills and tasks. Skills outside this vocabulary are not represented.
2. **Hypergraph topology.** H_{skill} combines O*NET major and minor occupational groups (yielding hyperedges over the skills shared within an occupational family) with pairwise similarity edges added when skill-embedding cosine similarity exceeds a fixed threshold. The threshold influences how exposure propagates across loosely related skills.
3. **Automatability criterion.** A skill is treated as subject to AI substitution if at least one tool in the catalog can perform a task drawn from that skill. This is a capability-availability criterion, not an adoption criterion: the simulation estimates technical exposure assuming tool availability, not realized displacement.
4. **Wage weighting.** Aggregate exposure is computed as a wage-weighted sum across occupations. This enables comparison across regions and industries on a common monetary scale, but smooths over within-occupation heterogeneity, where some tasks may face high automation potential while others remain human-intensive.
5. **Cascade dynamics.** The propagation rule is fixed: capability spreads through hyperedges with attenuation, and the simulation runs to a fixed point. Adoption timing, firm-level decisions, regulatory responses, and worker adaptation are not modeled in the present work; they are research directions for future LPM instantiations of this framework.

The Iceberg Index reported in Section 9.5 is the output of the simulation under these assumptions. Different choices would yield different Index values. The validation in Section 9.6 tests whether the simulated dynamics produce structures consistent with observed workforce patterns (career transitions and AEI adoption) rather than whether the Index predicts displacement outcomes, which are not yet observed at scale.

Project Iceberg Goal: To provide a sandbox for testing workforce strategies before implementation—enabling policymakers to estimate technical exposure under stated structural assumptions, evaluate interventions under different scenarios, and target preparation where the structural exposure is concentrated.

9.5 The Iceberg Index

The Iceberg Index is the wage-weighted summary statistic of the LPM simulation introduced in Section 9.4. It aggregates the post-cascade exposure profile of each worker into a single state-level value that estimates where human skills and AI capabilities overlap under the structural assumptions stated in Section 9.4.2. The Index is intended as forward-looking intelligence to complement traditional workforce metrics, which measure employment outcomes after disruption occurs rather than technical capability before adoption crystallizes.

The Iceberg Index: A skills-centered KPI for the AI economy.
It estimates the wage value of skills that AI systems can perform within each occupation under stated structural assumptions, indicating where human and AI capabilities overlap.

9.5.1 How the Index Is Computed

The LPM simulation in Section 9.4 produces, for each worker, a post-cascade exposure profile across the O*NET skill vocabulary. The Iceberg Index aggregates these profiles into a state-level measure along three dimensions: the skills required for each occupation, the automatability of those skills as determined by the cascade, and the value of the work in wages and employment. Formally, the Index for a given occupation weights each skill by its relative importance, post-cascade automatability score, and prevalence, producing a single exposure value between 0 and 100%. State-level and industry-level Index values are wage-weighted aggregations of the underlying occupation values.

The Index estimates the share of tasks that are *technically automatable* based on demonstrated AI capabilities. A skill is treated as automatable if at least one tool in the catalog can perform a task drawn from that skill, as specified in the structural assumptions of Section 9.4.2. This is a capability-availability criterion rather than an adoption criterion: the Index estimates technical exposure assuming tool availability, not realized displacement. Because the Index is skill-based rather than job-based, it can be consistently compared across geographies, industries, and occupation clusters, making it useful for both national and local decision-making.

9.5.2 What the Index Estimates

The Index estimates *technical exposure*—where AI capabilities and human skills overlap under the simulation’s structural assumptions—not displacement outcomes. For example, financial analysts will not disappear, but AI systems may demonstrate capability across significant portions of document-processing and routine analysis work. This reshapes how roles are structured and which skills remain in demand, without necessarily reducing headcount.

Systemic exposure, not task capability. The Index estimates how AI tool availability transforms workforce skills at scale, across millions of workers. This differs from capability benchmarks like GDPval [266] and APEX [267], which test base language model performance on isolated professional tasks. Both perspectives inform workforce planning: benchmarks

establish what models can do, while the Index estimates how tool ecosystems enable automation in practice. Future work can incorporate capability benchmark data to strengthen assessments of language model performance.

The Index does not predict job losses, adoption timelines, or net employment effects. Actual workforce impacts depend on firm adoption strategies, worker adaptation, regulatory choices, societal acceptance, and broader economic conditions. Policymakers should interpret the Index as a capability map—similar to how earthquake risk zones identify exposure without predicting when events occur—that enables scenario testing and proactive planning.

9.5.3 Two Ways to Use the Index

The Iceberg Index serves two complementary purposes. This chapter focuses on the baseline Index produced by the static cascade simulation; scenarios exploring adoption dynamics, policy interventions, and reskilling are research directions for future LPM instantiations.

Baseline assessment. The Index estimates maximum technical exposure—the share of occupational skills where AI has demonstrated capability in at least one context. This establishes problem scope without making adoption assumptions. It identifies where skill overlap is concentrated, enables geographic and industry comparisons, indicates structural patterns in workforce composition, and provides a metric for tracking how capability overlap evolves over time. *Key question: Where might AI capabilities reach if adoption follows patterns observed in technology sectors?*

Simulation input. The baseline Index feeds into scenario modeling that explores how technical capability might translate to workforce impact. Future work will test adoption assumptions (speed, sector patterns), model skill transferability across occupational contexts, evaluate policy interventions, and compare alternative futures under different technology maturity assumptions. Simulations explore: *Key question: Under specific adoption scenarios, how might baseline exposure materialize into workforce change?*

9.5.4 Applications for Policymakers

The Index enables three complementary strategic functions:

- **Status Quo Assessment:** Establish a baseline of automation potential if no action is taken, helping governments identify where adaptation needs are most concentrated. This makes the risks of inaction visible across industries and regions.
- **Strategic Opportunity Identification:** Highlight where AI adoption can advance broader goals while strengthening, rather than displacing, the workforce. For example, automating healthcare administration can free nurses for direct patient care, easing shortages while improving quality.
- **Recalibrating Investments:** Update traditional job-multiplier assumptions (e.g., one construction job creating 1.8 allied jobs) for the AI economy, showing how ripple effects change when support functions are automated or augmented. This enables more accurate projections of returns on training, infrastructure, and incentive programs.

The Index provides a measurable framework for evaluating policy choices, reframing AI as a planning tool for strategic workforce investments. This enables states to prepare for workforce change before it appears in unemployment data or GDP figures. For detailed validation methodology, see Section 9.6.1.

9.6 Interpreting the Index

In the previous section, we defined the Iceberg Index and described how it is constructed. Here we demonstrate how it can be applied to the U.S. workforce. The results quantify technical exposure—the share of occupational tasks that current AI systems can perform as mapped across 151 million workers, 923 occupations, and 3,000+ counties. In this analysis we focus on digital AI tools that can perform technical and cognitive tasks, where technology maturity and deployment patterns are observable. Physical automation through robotics is excluded here but will become increasingly relevant as capabilities mature.

Throughout this section, all numerical values are projections of the LPM simulation under the structural assumptions stated in Section 9.4.2; we use ‘projects’ and ‘estimates’ to mark statements that follow from the simulation rather than from observed displacement data. We illustrate four uses of the Index: identifying visible disruptions already occurring, uncovering larger hidden exposures beneath the surface, detecting blind spots where states risk being caught off guard, and distinguishing concentrated versus distributed industry patterns. These demonstrate how the Index can function as a policy laboratory: a tool for exploring alternative futures and guiding investment choices.

9.6.1 Validation with Real-world Data

We validate our methodology through two tests: first, whether skill-based occupational representations capture genuine labor market structure; second, whether exposure predictions align with actual AI adoption.

Skill-Based Validation: We test whether occupations our framework identifies as similar based on skill profiles exhibit similar characteristics in empirical workforce data. Using skill-based embeddings derived from O*NET occupational profiles, we calculate similarity scores between all occupation pairs and compare these against career transition networks that capture observed worker mobility patterns [268]. Career transition data reflects which occupations workers commonly move between, providing ground truth for occupational relationships independent of our framework. We identify the top percentile of similar occupation pairs according to our skill embeddings and test whether these correspond to frequent career transitions in the empirical data. Our embeddings achieve 85% recall in predicting these transition relationships (Figure 9.4(a)): 85% of commonly observed career moves involve occupations our framework identifies as highly similar based on skills. This high recall rate confirms that skill-based representations capture genuine labor market structure rather than theoretical constructs, validating our approach to measuring AI-human skill overlap using the same framework.

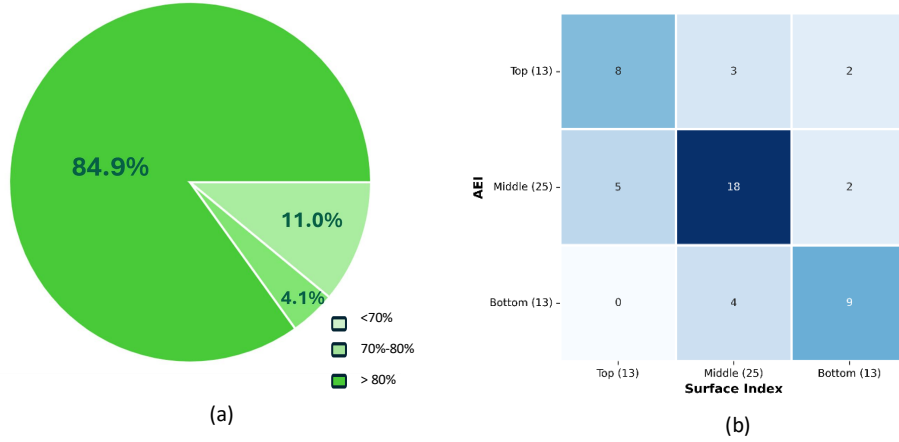


Figure 9.4: Validation of skill-based framework against empirical data. (a) Skill similarity validation: Pie chart shows prediction accuracy when matching occupation pairs by skill similarity to actual career transitions from O*NET data. For pairs our framework identifies as highly similar (>80% similarity score, dark green), 85% correspond to observed career transitions, confirming skill embeddings capture genuine labor market structure. Lower similarity thresholds (70-80%; <70%) show decreasing alignment with actual transitions. (b) Adoption validation: State-level comparison shows 69% agreement between Anthropic Economic Index (actual AI usage patterns) and Surface Index (predicted technical exposure). Strong alignment at both extremes: 8 of 13 leading states and 9 of 13 aspiring states match perfectly, confirming the Index identifies structural exposure before widespread adoption.

Adoption Validation: Building on this validated skill framework, we test whether automation exposure predictions align with actual AI usage. The Anthropic Economic Index (AEI) measures real-world AI usage from millions of Claude users nationwide, with industry analysis showing concentration in computing and technical occupations [269]. We define a Surface Index that models technical exposure in computing and technology tasks across all 50 states. We rank states by both AEI usage and Surface Index exposure. AEI groups states into four adoption tiers; we consolidate the two middle tiers into a single category, creating three balanced groups (leading, emerging, aspiring) to reduce sensitivity to boundary effects. Agreement is measured by whether states fall into the same category in both rankings. We find 69% geographic agreement, with strong consensus at extremes: 8 of 13 leading states and 9 of 13 aspiring states match perfectly. For instance, Washington, California, and Colorado consistently appear as leaders in both measures, while Wyoming, Mississippi, and Alaska align as laggards. Discrepancies reveal an interpretable pattern: our Index occasionally (18%) shows higher exposure than current usage in states like Texas and North Carolina, reflecting workforce structures with high technical vulnerability regardless of current adoption levels. However, the framework rarely underestimates—states (13%) where high AEI usage shows low Index values. This asymmetry validates the Index as a leading indicator: it identifies structural exposure before widespread adoption occurs.

Result 1: Iceberg Validated with Real-World Data

Validation against independent data from millions of AI usage interactions shows strong agreement on leading and aspiring states, indicating our approach is consistent with observed adoption patterns. Skill-based representations predict 85% of career transitions, and exposure predictions achieve 69% geographic agreement with actual usage patterns.

9.6.2 Quantifying the Tip of the Iceberg

Building on our validated methodology, we now quantify technical exposure within occupations where AI adoption is currently concentrated. In 2025, more than 100,000 job losses were linked to AI restructuring. AI systems now write over a billion lines of code each day, exceeding human developer output. We measure skill overlap within computing and technology occupations—the Surface Index—which aligns with real-world adoption patterns from millions of AI users (Section 9.6.1).

Nationally, the Surface Index stands at 2.2%, representing approximately \$211 billion in wage value across 1.9 million workers in technology occupations (Figure 9.5). This includes software engineers, data scientists, analysts, program managers, and related roles where current AI adoption is concentrated. Washington (4.2%), Virginia (3.6%), and California (3.0%) lead in exposure values, yet even in these states, direct technology tasks account for only a small share of employment. The technology sector represents more than 30% of the S&P 500's market capitalization but only around 6% of the workforce, showing that states dominating technology maintain highly diversified labor markets.

States like Mississippi and Wyoming show minimal values on this Surface Index, reflecting their limited technology sector presence and fewer occupations with skill profiles matching current AI adoption patterns. However, this represents only the visible beginning. The full Iceberg Index indicates that these same states show substantial technical exposure in administrative, financial, and professional services—indicating broader potential impact beyond current visible adoption.

Result 2: Tech Disruption Is Visible but Small

Headlines focus on tech skills, but these span occupations representing only 2% of labor market wage value. The simulation projects exposure beyond visible tech sectors that is approximately five times larger.

9.6.3 The Hidden Mass Beneath the Surface

Beyond technology occupations, AI capabilities extend to cognitive and administrative work. Tools developed for coding demonstrate technical capability in document processing, financial analysis, and routine administrative tasks - illustrating how capabilities demonstrated in technology contexts translate to other domains. Some adoption is already occurring: IBM reduced HR staff through AI automation [270], Salesforce froze hiring for non-technical

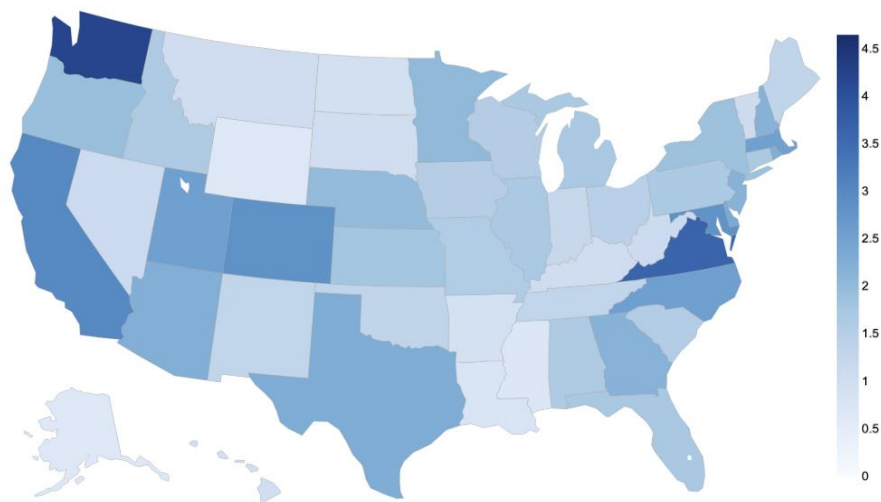


Figure 9.5: Surface Index validation: Current technology-sector exposure across US states. The average surface index of the country is 2.2, maximum is 4.2 (Washington) and minimum is 0.2 (Guam). 18 states (or territories) are above the national average and 36 are below the national average.

roles [271], and McKinsey projects that 30% of financial tasks could be automated by 2030 [272].

We apply the same skill-overlap methodology to administrative, financial, and professional service occupations beyond the technology sector. The Iceberg Index for digital AI shows values averaging 11.7%—five times larger than the 2.2% Surface Index. Unlike technology-sector exposure concentrated in coastal hubs, this broader skill overlap is geographically distributed. South Dakota, North Carolina, and Utah show higher Index values than California or Virginia.

Industrial states illustrate this pattern. Tennessee (11.6%) and Ohio (11.8%) show substantial Index values driven by administrative and coordination roles within factories and supply chains. These white-collar functions show technical exposure that maybe invisible to policymakers while states focus largely on physical automation. These patterns indicate where skill overlap extends beyond current visible adoption, though actual workforce impacts will depend on adoption decisions, quality thresholds, and organizational constraints (Figure 9.6(a)).

Result 3: White-Collar Exposure Is Nationwide

Administrative and financial tasks where AI demonstrates capability span five times more wage value than visible tech disruption—and are geographically distributed across all states, not just coastal. States like Delaware and South Dakota show higher Index values than California.

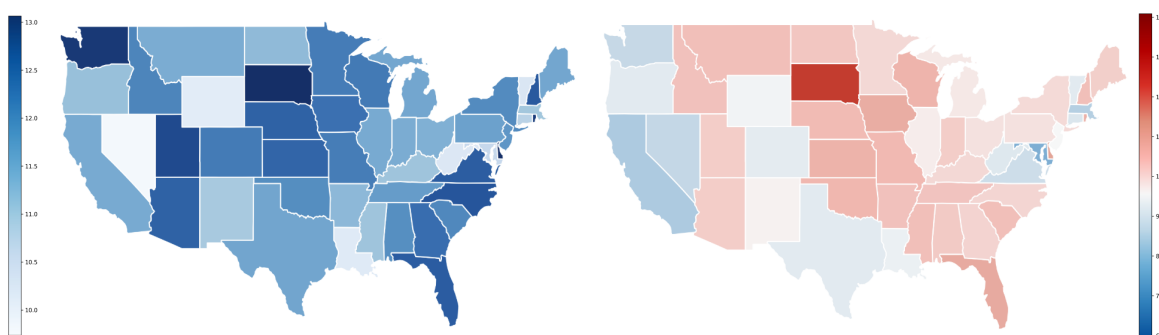


Figure 9.6: Geographic patterns of hidden automation exposure. (Left) Digital Iceberg Index shows cognitive automation spreads beyond coastal tech hubs, with unexpected leaders like Delaware and South Dakota showing higher exposure than California due to concentrated administrative and financial sectors. (Right) Automation surprise shows which states face the largest gaps between current visibility and projected transformation. Manufacturing states like Ohio and Michigan show substantial hidden white-collar exposure years before anticipated physical automation, requiring proactive preparation for administrative and coordination role changes.

9.6.4 The Blind Spot: Automation Surprise

Some states show large differences between Surface Index (technology occupations) and Iceberg Index values (including administrative, financial, and professional service occupations). These gaps indicate substantial skill overlap in cognitive work despite minimal visible technology-sector adoption, potentially creating mismatches between preparation strategies and technical capability patterns.

America’s industrial heartland shows the largest gaps. Rust Belt states such as Ohio, Michigan, and Tennessee register modest Surface Index values but substantial Iceberg Index values driven by cognitive work—financial analysis, administrative coordination, and professional services—that supports manufacturing operations. Tennessee illustrates this pattern: Surface Index of 1.3% but Iceberg Index of 11.6% indicating that administrative and service functions show up to ten times greater technical exposure than visible technology occupations. South Dakota follows a similar pattern, combining low technology-sector Index values with high administrative-sector overlap. Figure 9.6(b) maps these gaps across states.

By contrast, technology-intensive states such as California and Washington show high values on both Surface and Iceberg Indices, resulting in smaller gaps. These states may recognize preparation needs earlier because skill overlap is already visible in their dominant technology sectors. The gaps emerge because current workforce planning often focuses on visible technology-sector adoption, while cognitive and administrative work has received less attention in preparation strategies. As a result, states with small technology sectors underestimate the scale of their exposure, leaving them vulnerable when adoption accelerates in white-collar work.

Result 4: Manufacturing States Face Hidden White-Collar Surprise

Midwest states like Ohio and Michigan face double-digit technical exposure in white-collar work while states may focus on physical automation. Cognitive and administrative exposure under stated assumptions is up to ten times higher than technology-sector exposure.

9.6.5 Industry Patterns: Concentrated vs. Distributed Impact

States with similar Iceberg Index values face very different challenges depending on how that exposure is distributed across industries. In some cases, exposure is tightly concentrated in just a few dominant sectors. In others, the same level of exposure is spread broadly across many parts of the economy. This structure matters because it determines how states must respond: distributed risk demands broad, multi-sector coordination, while concentrated risk allows for targeted, sector-specific action.

To distinguish between these patterns, we calculate each industry's contribution to a state's total Iceberg Index value and compute the Herfindahl-Hirschman Index (HHI) of this distribution. The HHI quantifies whether Index values concentrate in a few industries or spread broadly across sectors and is a standard measure adapted from labor economics [273,274]. We rank all states by their HHI values and divide them into thirds based on the observed distribution: the bottom third ($\text{HHI} \leq 1580$) represents the most distributed patterns, the middle third (1581–1737) shows moderate concentration, and the top third (≥ 1738) represents the most concentrated patterns. These categories are descriptive classifications based on relative position in the data, not normative thresholds. Details appear in full paper and Figure 9.7 visualizes these patterns geographically.

The analysis demonstrates clear regional contrasts. Northeastern belt tend to show concentrated exposure, especially in finance and technology. In contrast, states across the Manufacturing Belt display more distributed patterns, with exposure spread across logistics, production, administration, and services. These structural differences influence how disruption might propagate — whether driven by a few sectors or ripple broadly across the economy.

Even when overall Index values appear similar, underlying structures can vary significantly. For example, Iowa (12.22%) and Ohio (11.34%) both show broadly distributed patterns, while Virginia (12.48%) channels a comparable level of exposure dominated by just two sectors: finance and technology. The same Iceberg Index can imply very different workforce vulnerabilities depending on how it is composed.

Result 5: Structure of Exposure Determines Strategy

The same level of technical exposure can require entirely different responses. Concentrated patterns enables sector-specific action, while distributed patterns demands multi-sector coordination

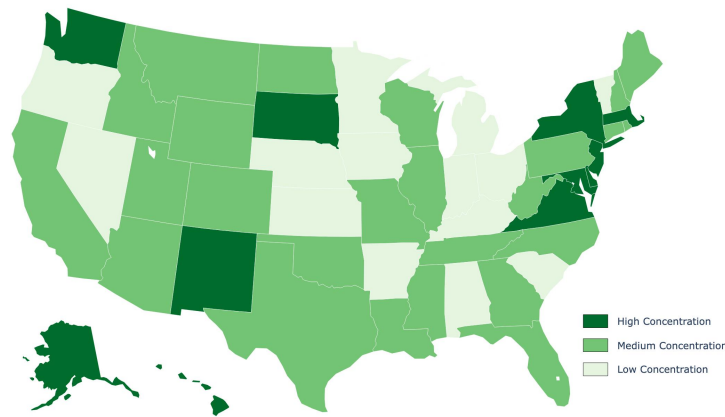


Figure 9.7: Geographic distribution of automation exposure concentration. States are grouped into three tiers based on their Herfindahl–Hirschman Index (HHI) computed from how Iceberg Index values distribute across industries within each state. The national HHI range spans from 1422 (most distributed) to 1895 (most concentrated). We divide states into thirds by HHI ranking: Most Distributed (≤ 1580 , light green), Moderate (1581–1737, medium green), and Most Concentrated (≥ 1738 , dark green). This classification demonstrates regional contrasts, with distributed patterns across the Manufacturing Belt and concentrated patterns along the Northeast corridor.

9.6.6 Why Traditional Metrics Miss the Iceberg

Traditional economic metrics—GDP, per-capita income, and unemployment—are widely used to benchmark state performance. We test how these metrics align with workforce exposure as captured by the Iceberg Index. Using Q2 2025 data, we rank all 50 states by GDP, income, and unemployment, and then compare these with state rankings from the Index: first for the Surface Index (today’s visible disruptions concentrated in technology and software tasks) and then for the Iceberg Index (a forward-looking measure of the projected spread to cognitive and administrative tasks).

The results show clear differences. Traditional metrics exhibit moderate alignment with the Surface Index: states with higher GDP or lower unemployment often have larger technology sectors and therefore higher visible exposure today. By contrast, their relationship with the Iceberg Index is negligible, with GDP, income, and unemployment each explaining less than five percent of the variation in systemic exposure; in some cases, the correlations are weakly negative (Figure 9.8). For example, Delaware and Utah exhibit higher Iceberg exposure than California, despite much smaller economies, because their concentrated finance and administrative sectors present sharper automation targets than California’s diversified workforce. This reflects a broader pattern: technology automation is clustered in coastal hubs, while cognitive automation spans administrative, financial, and professional roles present in every state.

These findings underscore the need for a complementary benchmark. GDP, income, and unemployment remain essential for assessing economic performance, but they were not designed to capture how skills and tasks itself is changing in the human–AI economy. The

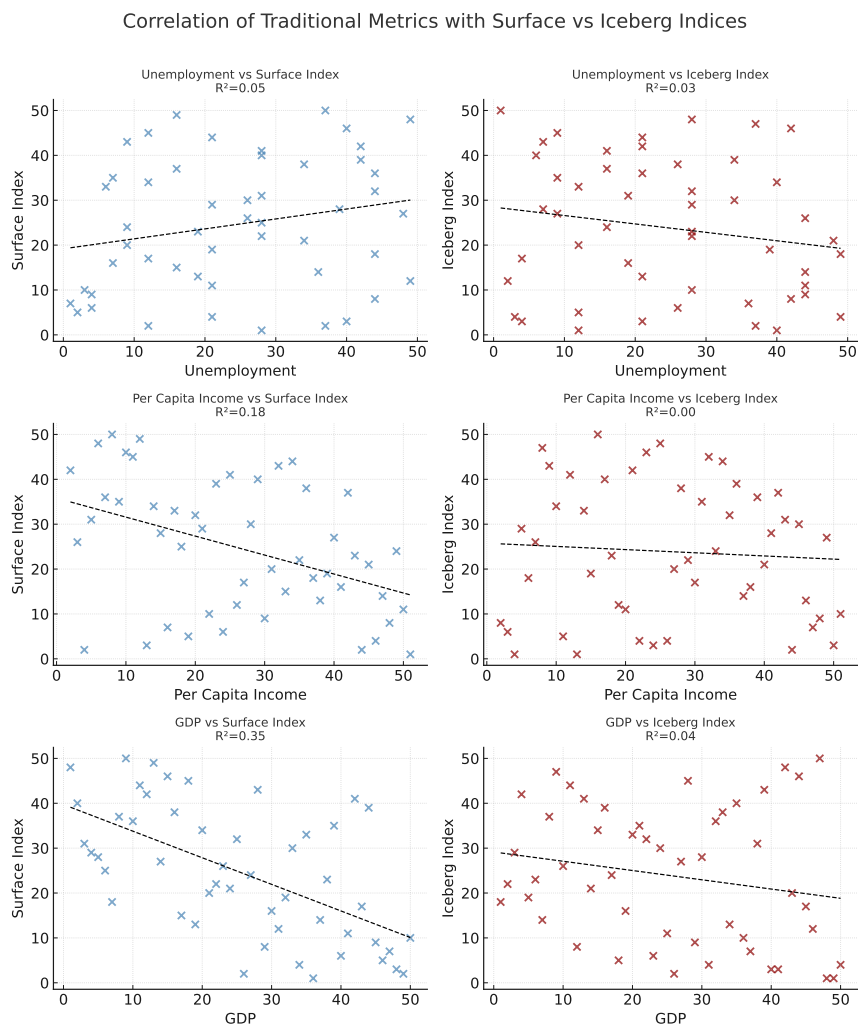


Figure 9.8: Correlations of traditional economic metrics with the Iceberg Index. Each panel shows the relationship between a conventional signal (unemployment, per-capita income, or GDP) and state-level automation exposure. The top row reports the **Surface Index**, reflecting today’s visible adoption concentrated in technology. The bottom row reports the **Iceberg Index**, projecting systemic exposure across finance, administration, and professional services. Regression lines and R^2 values quantify explanatory power. Traditional metrics correlate with the Surface Index but show low correlation with the Iceberg Index, indicating why policymakers relying on unemployment or GDP may misjudge skill-centered exposure in the workforce.

Iceberg Index fills that gap by providing a forward-looking KPI that estimates skills-centered exposure directly. By using both traditional metrics and the Iceberg Index, policymakers gain a more complete picture of today’s economy and tomorrow’s transitions, enabling better alignment of training, infrastructure, and investment strategies with the realities of AI adoption.

Result 6: Traditional Metrics Miss the Hidden Risk

GDP and unemployment correlate with the Surface Index but show low correlation with the Iceberg Index ($R^2 < 0.05$). Variation in projected cognitive and administrative exposure is not captured by traditional state-level economic metrics.

9.7 Limitations

The Iceberg Index captures technical exposure—where AI capabilities overlap with occupational skills—not actual displacement or labor market outcomes. Real-world impacts depend on adoption choices, firm strategies, worker adaptation, societal acceptance and policy interventions. The Index functions as a capability map enabling scenario-based planning rather than deterministic forecasting.

9.7.1 Validation

The framework achieves strong empirical validation against independent data sources: high recall rates predicting career transitions and substantial geographic agreement with observed AI adoption patterns (see Section 9.6.1 for detailed results). These results confirm the Index captures genuine workforce structure and plausible adoption patterns. However, validation anchors reflect current contexts—career transitions from human-only labor markets and technology-sector AI usage—which may not fully capture adoption dynamics in non-technology sectors. See Appendix A for validation methodology and anchor constraints.

9.7.2 Measurement Scope

The Index estimates exposure from production AI systems deployed in enterprise contexts—automation platforms, copilots, and industry-specific tools—rather than frontier model capabilities tested on academic benchmarks. This focuses on immediate business impact: tools reshaping work today. Recent evaluations like GDPval [266] and APEX [267] test whether frontier models can perform isolated professional tasks at human-equivalent quality. These benchmarks measure task-level capability; our framework estimates systemic workforce exposure across 151 million workers. Both dimensions matter for workforce planning. A model may perform well on benchmarks yet not reshape work until packaged into deployable tools, integrated into workflows, and adopted at scale. By cataloging thousands of production systems, the Iceberg Index captures where AI is positioned to affect work now, providing conservative estimates of immediate exposure. Future work could examine alignment between high-exposure occupations and benchmark performance.

9.7.3 Modeling Choices

Key assumptions affecting interpretation:

- **Wage-value weighting:** Enables consistent comparison across occupations but smooths over job quality dimensions (stability, autonomy, career progression) and

masks within-occupation heterogeneity where some tasks face high automation potential while others remain human-intensive.

- **Skill transferability:** The Index assumes skills demonstrated in one occupational context transfer to others, establishing an upper bound on exposure. This enables scenario analysis where transferability can be varied, but may overestimate near-term exposure if domain-specific adaptation proves more challenging than assumed.
- **Digital AI focus:** Analysis covers cognitive and administrative automation where deployment is observable. Physical robotics excluded due to immature adoption data; see the full paper for scope rationale.

See the full paper for detailed data sources, implementation choices, and computational infrastructure.

9.7.4 Causal Interpretation

Validation is correlational rather than causal. Agreement with observed adoption patterns increases confidence but does not prove exposure drives outcomes. External factors—state investment, infrastructure, regulation—mediate how capability translates to impact. This limitation is inherent to forward-looking analysis: policymakers cannot wait for causal evidence of disruption before preparing responses. The Index provides the best available prospective indicator, validated against current patterns.

9.7.5 Complementary Role

The Index complements traditional labor metrics (GDP, unemployment, wages) rather than replacing them. Traditional metrics track realized outcomes; the Iceberg Index projects potential exposure under stated assumptions before adoption crystallizes. Together they enable comprehensive workforce planning combining retrospective assessment and prospective intelligence.

9.8 Conclusion

The Iceberg Index estimates where AI technical capabilities overlap with human occupational skills across 151 million workers, providing forward-looking intelligence to complement traditional workforce metrics that track employment outcomes after disruption occurs.

The simulation projects a substantial gap between visible adoption and structural exposure. Current AI adoption concentrates in technology occupations representing 2.2% of labor market wage value. Under the structural assumptions stated in Section 9.4.2, AI technical capability extends to cognitive and administrative tasks spanning approximately 11.7% of the labor market—an estimated \$1.2 trillion in wage value across finance, healthcare, and professional services. This fivefold projection is geographically distributed nationwide rather than concentrated in coastal hubs, indicating that workforce preparation strategies based on visible technology-sector signals may substantially undercount potential transformation.

Validation against independent data indicates the framework is consistent with plausible workforce patterns.

The Index estimates technical exposure—where AI can perform occupational tasks—not displacement outcomes. Actual workforce impacts depend on firm strategies, worker adaptation, and policy choices. By simulating how capabilities might spread under different scenarios, Project Iceberg enables states to test interventions before committing resources, supporting a shift in workforce planning from reactive crisis management to strategic foresight. Future work will model adoption dynamics, extend to physical automation as robotics mature, and integrate task-level quality benchmarks.

The Iceberg Index provides measurable intelligence for critical workforce decisions: where to invest in training, which skills to prioritize, how to balance infrastructure with human capital. It projects not only visible disruption in technology sectors but the larger structural exposure beneath the surface. By projecting exposure before adoption reshapes work, the Index enables states to prepare rather than react.

Chapter 10

Population AI in the Age of Agents

10.1 What This Thesis Built

Grand challenges facing human societies emerge from the collective consequences of millions of interacting decisions. A pandemic spreads because contact decisions compound through social networks into waves that overwhelm health systems. A labor market shifts because new capabilities propagate through networks of skills faster than institutions adapt. A supply chain collapses because rational responses to uncertainty amplify through sequential dependencies into economy-wide disruption. In each case, the outcome emerges from the structure of interactions, not from any individual agent acting alone.

This thesis developed Large Population Models (LPMs): a framework for representing, measuring, and learning from populations of interacting agents. In LPMs, each agent maintains local state, makes decisions based on its context, and interacts with others through structured channels. The objective is to measure how agent behavior and interaction structure jointly produce population-level outcomes, and to learn the interaction dynamics that make these outcomes consistent with real-world observations.

We applied this framework to two classes of systemic risk. In biosecurity, LPMs modeled interacting animal and human populations across the contiguous United States to study zoonotic spillover. In labor markets, LPMs represented 151 million workers as agents executing tasks and skills across regions, revealing that 11.7% of wage value across 20 million workers carries additional disruption risk not captured by traditional metrics.

Across these settings, the same result held: collective outcomes depend on how agents behave and interact, not only on how capable any individual agent is. Measuring behavior in isolation is insufficient. Understanding how behavior propagates through a population is necessary to detect and quantify systemic risk before it becomes visible in aggregate statistics. This thesis therefore establishes LPMs as a foundation for measuring behavior, interaction, and risk in human societies.

10.2 Open Problems

The framework opens more directions than this thesis pursues. Each algorithmic chapter validates its contribution at the scale required to make the point, but the framework's reach

extends well beyond what any individual chapter exercises. Population-scale studies of language-based coordination, broader characterization of the cognition-scale tradeoff across model families, and millions-of-agents deployments of private populations are natural next steps that the framework supports by construction.

Validation strategies vary across chapters in ways that reflect what each domain’s data permits and what the field’s evaluation infrastructure provides. Biosecurity allows recall against reported H5N1 incidence; labor markets do not yet have ground-truth displacement data, and the chapter validates correlationally against career transitions and adoption patterns. The deeper issue is that the field lacks shared benchmarks for evaluating learned agent populations: each domain develops its own metric (epidemiological recall, supply-chain cost, labor-market correlation), which makes cross-method comparison difficult and leaves individual contributions harder to position against alternatives. Building such benchmarks, especially for coordination and cognition at population scale, is a research direction in its own right.

Several technical problems remain open within the framework itself.

- **Problem 1: Byzantine robustness.** The thesis assumes honest-but-curious agents throughout: in private agent populations, in REP coordination, and in federated calibration. Cases where some agents behave adversarially or strategically misreport sensitivities are not handled by the current framework. Extending it to adversarial settings is a natural next step for which the distributed-optimization literature provides starting points.
- **Problem 2: Theoretical guarantees.** The thesis contributes mechanism design and empirical demonstration. Formal results on convergence and optimality would require assumptions about agent rationality that do not hold cleanly for LLM agents. Developing theory under realistic agent assumptions is open.
- **Problem 3: Individual-level validation.** Validation in this thesis operates at the level of population aggregates rather than individual decisions. Agent-level behavior is consistent with the priors used to construct it, but is not validated against individual-level ground truth because such data does not exist at population scale. This is a structural feature of population-scale modeling rather than a fixable gap, and downstream conclusions should be interpreted accordingly.
- **Problem 4: LLM bias inheritance.** Where LLM cognition is used, agent behavior inherits the biases of the underlying language model. Sampling multiple queries per archetype reduces variance but does not correct for systematic bias. Identifying when LLM-derived agent behavior diverges from observed human behavior, and correcting for it without losing the cognitive richness LLMs provide, is open.
- **Problem 5: Gradient estimators and joint optimization.** Gradient estimation through stochastic agent interactions is functional but not optimal: AgentTorch’s straight-through estimators outperform continuous relaxations such as Gumbel-Softmax in tested regimes, but principled low-variance estimators for discrete distributions composed at population scale remain an active research area. Joint optimization across

discrete-time and continuous-time interaction channels has not been systematically studied; current applications choose one or the other rather than combining them within a single calibration pass.

In the settings studied so far, agents are used to model and measure real-world systems. The most consequential direction the framework opens is what happens when the agents in these systems are not only representations, but participants. The remainder of this chapter considers systems in which AI agents act alongside humans, where their interactions directly shape outcomes, and where the open problems above find a new empirical surface.

10.3 From Human to Agentic Societies

Understanding how agentic societies will function requires first understanding what made human societies stable

What worked for humans. Human interaction has always been bounded. Dunbar’s number, the cognitive limit on the number of stable social relationships a human can meaningfully maintain, sits at approximately 150 [275,276]. This is not a cultural accident. It reflects the carrying capacity of the human neocortex for social information. Even in the age of social media, where platforms theoretically connect billions, meaningful engagement remains bounded to a few hundred relationships per person.

This constraint shaped human societies in ways that turned out to be stabilizing. Human social networks are small-world: densely connected locally, sparsely connected globally, with information propagating gradually through weak ties rather than flooding every node simultaneously [277]. Because coordination was locally bounded, decisions did not synchronize instantly across the population. Information spread gradually, failures remained local, and diversity of strategies persisted. The cognitive constraints that bounded individual agents acted as implicit coordination mechanisms that kept collective behavior stable.

The internet was built around these human interaction patterns. Its protocols were designed for a world where the agents at the endpoints are cognitively limited, socially embedded, and temporally constrained. These constraints were not limitations to be removed. They were the source of the system’s stability.

How AI agents differ. AI agents operate without these constraints. Where humans maintain dozens of active relationships, an AI agent can coordinate with thousands simultaneously in real time. Where human information spreads gradually through sparse social networks, AI agents trained on the same data can reach similar conclusions within milliseconds. The evidence of this transformation is already visible in the stress it places on existing infrastructure. Where Google historically sent one visitor for every two pages crawled, that ratio is now 6:1. OpenAI’s crawl-to-visit ratio is 250:1. Anthropic’s is 6,000:1. Where a human queries 10 to 12 sources to answer a question, an AI agent queries 200 or more [278]. These patterns are already observable in current systems.

Communication at this scale is not the same as coordination. When many agents act on similar information using similar optimization strategies, individually rational choices

become collective failure modes. We have already seen early previews: website crashes when bots coordinate simultaneous purchase attempts, flash crashes in financial markets from algorithmic cascade effects, and traffic gridlock when navigation systems route everyone through the same path. These are the natural consequence of removing the interaction constraints that kept human coordination stable.

As the number of such agents grows, interaction structure becomes a first-order determinant of system behavior. The same mechanisms that governed risk in human populations now operate at greater speed and scale. Understanding and shaping these interactions is therefore central to the design of agentic societies.

10.4 Early Evidence in Agentic Societies

This thesis studies how populations of agents behave and interact. A natural question is whether these ideas apply only to human societies, or whether they also hold for populations of AI agents. This section examines whether the principles identified in human populations extend to populations of AI agents. The results show two consistent patterns. First, AI agents do not all behave the same, even when their overall performance is similar. Second, the behavior of a system depends not only on how capable each agent is, but on how agents interact with each other. Together, these results provide early evidence that intelligence in agent populations is not only a property of individual agents, but of their interaction.

10.4.1 Agent Diversity Persists Without Biology

Human populations are diverse because people differ in background, experience, and perspective. This diversity stabilizes systems by preventing uniform responses and enabling multiple strategies under uncertainty. AI agents, trained on similar data and evaluated using the same benchmarks, might be expected to behave similarly. If two systems achieve similar scores, one might expect them to reason in the same way. In practice, this is not the case.

We evaluated two frontier AI systems, Claude and Gemini, on professional tasks spanning law, consulting, and finance. These tasks require multi-step reasoning, domain knowledge, and decision-making under constraints. Their aggregate performance is nearly identical (0.47 vs. 0.46). Yet the capabilities they demonstrate are structurally different, with fewer than half overlapping across the two systems.

A simple example illustrates this difference. The task is to select a fire door vendor for a commercial facility while ensuring compliance with safety regulations. Claude approaches the task through compliance. It checks regulatory constraints, identifies violations, and filters vendors based on whether they meet legal standards. Gemini approaches the task through procurement. It compares vendors based on cost, quality, and reliability, and organizes the decision as a structured supplier evaluation.

Both produce coherent, professional outputs. The difference is in how they reason: one prioritizes regulation and risk, the other prioritizes cost and operations. This pattern persists across tasks. The implication is that AI agent populations can be diverse even without biological differences. For population-level systems, this diversity is a first-order property that must be measured and preserved.

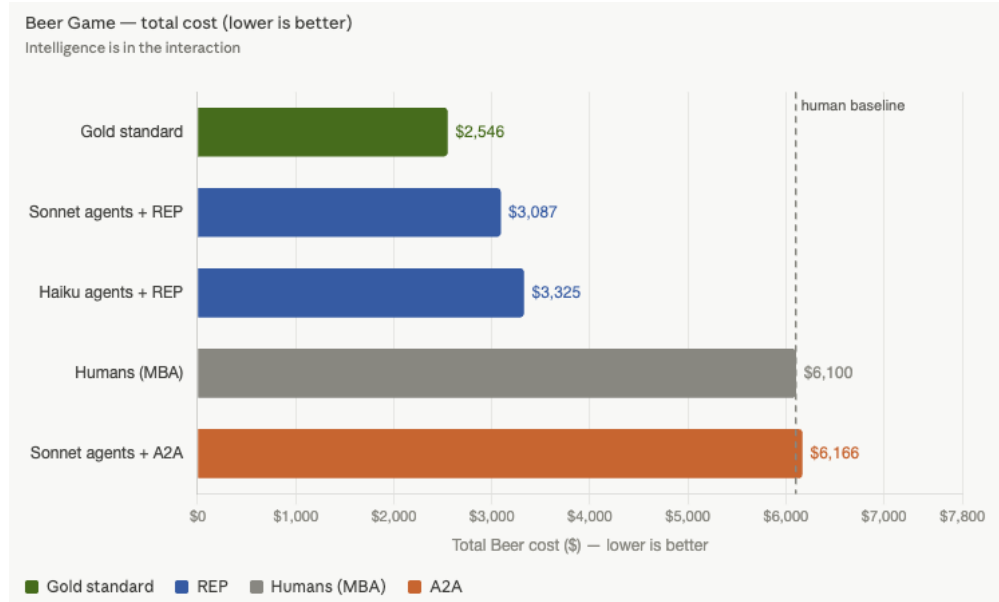


Figure 10.1: Coordination dominates individual capability. A population of smaller models (Haiku) with REP coordination achieves significantly lower cost than both human teams and more capable models (Sonnet) without coordination. This shows that interaction structure can have as much impact on system outcomes as improving individual agents, and in some cases more.

10.4.2 Interaction Structure Shapes Collective Outcomes

If diversity defines what agents bring to a system, interaction defines what the system produces. Improving individual agents does not necessarily improve the behavior of the population. What matters is how agents share information, how they respond to each other, and how decisions propagate through the system.

Chapter 6 introduced the Ripple Effect Protocol (REP) as a coordination mechanism for agent populations, and evaluated it in settings such as the Beer Game and Fishbanks. These experiments showed that structured interaction can stabilize systems and improve collective outcomes.

Here, we revisit the Beer Game to isolate the role of coordination relative to individual agent capability. The agents themselves are not modified. The only change is in how they interact.

Figure 10.1 shows the results. A population of Claude Haiku agents with REP coordination achieves a total cost of \$3,325. Human teams average \$6,100, and uncoordinated Sonnet agents achieve \$6,166. The theoretical optimum is \$2,546.

A smaller model that only changes how agents interact outperforms both humans and more capable agents without coordination. This isolates a key principle: collective outcomes are determined not only by the capability of individual agents, but by the structure of their interaction.

10.5 Research Frontiers

The results in this thesis open three concrete directions for future work: measuring how agents coordinate, understanding what happens when they do so at economic scale, and ensuring the infrastructure that enables them remains open. All three build on the ability to represent, learn from, and analyze population-scale systems through Large Population Models.

10.5.1 Measuring Coordination in Agent Populations

Benchmarks such as MMLU, HellaSwag, and BIG-Bench have made it possible to evaluate reasoning at scale. There is still no standard benchmark for coordination. Models that score well individually may fail when required to act together, and without a systematic way to measure coordination we cannot train for it, evaluate whether proposed mechanisms improve it, or compare AI agent teams against human teams on tasks where coordination determines the outcome.

Coordination is not a single capability but a multi-dimensional challenge with five independent levers, each representing a distinct failure mode: *Grounding* (can agents establish a shared representation to coordinate?), *Synchronization* (can agents act at the right time relative to each other?), *Observability* (can agents coordinate when information is delayed or structurally hidden?), *Scale* (can coordination mechanisms remain effective as the number of agents grows from 2 to 200 to millions?), and *Incentives* (can agents coordinate when their goals diverge?). Each lever can fail independently; the hardest real-world problems activate several simultaneously.

Recent work on the Collaboration Gap [279] demonstrated that capable models degrade substantially when required to collaborate, even with identical copies of themselves. This isolates Lever 1: agents fail because they cannot establish a shared coordinate system. But the task used is static, involves only two agents, and assumes perfectly aligned incentives. It leaves Levers 2 through 5 unmeasured.

This thesis provides initial evidence on Levers 2 and 3 in canonical settings. The experiments in Fishbanks and the Beer Game show that structured interaction can synchronize restraint and dampen information cascades. But these are two settings, with specific incentive structures, at limited scale. The full five-dimensional coordination picture remains unmeasured.

The right foundation for this benchmark suite is the class of system dynamics games developed to study coordination failures in human organizations: Fishbanks, the Beer Game, Public Goods, Double Auction, Pandemic Response. These games carry three properties that synthetic tasks cannot provide: decades of human team baselines from MBA programs and executive education, computable theoretical optima under perfect centralized coordination, and natural multi-lever complexity combining delayed feedback, partial observability, competing incentives, and network effects. Table 10.1 maps a candidate suite to the five levers. The Double Auction tests all five simultaneously. Pandemic Response provides an integration test that can be executed on the LPM infrastructure developed in this thesis.

Scoring this benchmark requires measuring outcomes at three scales. At the *individual*

Game	Ground	Sync	Observe	Scale	Incentives	Human Data
Maze [279]	✓✓✓				Aligned	No
Fishbanks	✓	✓✓	✓	✓✓	Mixed	30+ yrs
Beer Game	✓	✓✓✓	✓✓	✓	Aligned	30+ yrs
Public Goods	✓	✓	✓	✓✓	Mixed	Yes
Double Auction	✓✓	✓✓	✓✓✓	✓✓	Competitive	Yes
Pandemic	✓✓	✓✓✓	✓✓✓	✓✓✓	Mixed	Yes

Table 10.1: Coordination benchmark suite mapping candidate games to the five levers. No single game covers all levers. Fishbanks and Beer Game are complementary: synchronization with observable feedback versus coordination under delayed feedback. Double Auction is the full-lever integration test. Pandemic Response is the integration test deployable on LPM infrastructure.

level: does coordination benefit each agent? In the Fishbanks experiments in Chapter 6, every coordinated agent individually earned more than every uncoordinated agent. Coordination was individually rational. This answers the adoption question: if coordination helps the group but hurts the individual, no agent opts in. At the *group* level: does the team achieve its collective objective? At the *population* level: does local coordination produce good systemic outcomes or cascading failures? This third scale is where the LPM infrastructure developed in this thesis becomes essential, and where current evaluation approaches remain limited.

10.5.2 The Agentic Economy

Consider what is already happening in the Research Triangle of North Carolina. AI agents are beginning to handle routine software engineering tasks. At first, this makes programmers more productive. But as agents become more capable, companies need fewer engineers. Some developers relocate. The restaurants and services that depended on tech worker spending see revenue drop. Housing demand shifts. Service workers who supported the tech ecosystem face their own disruptions. New roles emerge, but in different places, requiring different skills. The Iceberg Index estimates this ripple already touches \$12.8 billion in economic activity across 116,000 workers in North Carolina alone [1]. No traditional economic indicator is moving yet. GDP is stable. Unemployment is flat. The restructuring is happening underneath.

This pattern generalizes beyond a single region. When AI capabilities change the demand for particular skills, the consequences propagate through skill networks, occupational dependencies, and regional labor markets in cascades that no individual-level analysis can capture. The same cascade mechanism that governs disease spread through contact networks governs AI-driven labor market disruption. The Iceberg Index showed that traditional economic metrics explain less than 5% of the variation in skill-level AI exposure across US states. GDP correlates with visible AI adoption in technology sectors ($R^2 = 0.35$) but explains essentially nothing of the broader exposure spreading through administrative, financial, and professional

services ($R^2 = 0.04$). Unemployment moves after jobs are lost, not before skills are automated. This is not a forecasting failure. It is a measurement failure.

Every ten years, the United States counts every person and maps every household. That count becomes the foundation for \$2.8 trillion in annual federal spending across 353 programs. The census works because it tracks the full population whose decisions shape outcomes. Within the next decade, the population that shapes economic outcomes will include large populations of AI agents acting alongside those 330 million Americans. The census has no mechanism to count them. The Bureau of Labor Statistics tracks human employment but has no framework for measuring AI agent capabilities or their economic interactions with human workers. The LPMs developed in this thesis can represent and analyze a hybrid human-agent economy, but only when given data about both populations. Today we have data about one of them.

What is needed is an Agentic Census: infrastructure to identify agents, catalog their capabilities, and monitor how they act on the taxonomy of human skills. Combined with LPMs, such infrastructure would enable simulation of cascading effects through specific communities before those effects appear in unemployment figures. The states and institutions that build this before the transformation accelerates will have economic intelligence when it matters. Those that wait will manage cascades they cannot see with instruments designed for an economy that no longer exists.

10.5.3 Open Infrastructure for an Agentic Society

The internet had a founding choice: open protocols or proprietary ones. TCP/IP, DNS, and HTTP were designed as open standards, accessible to any participant. That choice shaped everything that followed. Anyone could build on the web. No single company controlled who could communicate. The network's value accumulated broadly. The agentic economy faces the same choice, and the window to make it deliberately is closing.

The companies building AI agents also control the data about how those agents reshape society. They see economic disruptions forming before local leaders do, run simulations that public institutions cannot replicate, and understand workforce transitions before the workers experiencing them. Without public infrastructure, the most consequential decisions about communities will be made using data those communities do not control.

Current agent protocols focus on enabling agents to communicate. Communication alone does not produce coordination. When large numbers of agents act on similar information using similar optimization strategies, individually rational choices become collective failure modes.

This thesis shows that coordination requires structured interaction mechanisms. Without such mechanisms, agent populations can exhibit instability, amplification effects, and synchronized failures.

Open coordination infrastructure requires three capabilities that current protocols do not provide: mechanisms to preserve diversity across agent populations before convergence becomes a stampede; coordination without centralization, so that agents can align without any single authority observing every state; and governance that keeps this infrastructure accessible rather than concentrating in the hands of those who control proprietary agent population data. The LPMs developed in this thesis demonstrate one instance of the second:

enabling agents to coordinate and calibrate shared dynamics without requiring access to individual state. What remains unsolved is the governance layer that keeps this infrastructure open.

10.6 The New Foundation

Agent populations are increasing rapidly and will soon exceed human populations in many systems. Over the last century, human civilization grew from 2.09 billion to 8.18 billion people, at a compound annual rate of approximately 1.4% [280]. That growth was bounded by biology, resources, and the cognitive limits of human agents. Agent populations face no such bounds. Within a decade, the number of AI agents operating in the world may exceed the number of humans by a wide margin. How those agents coordinate will shape outcomes for everyone.

Different coordination structures produce different collective outcomes. In systems where interaction protocols distribute information and resources effectively, benefits propagate across the population. In systems where coordination fails, locally rational decisions accumulate into system-wide inequality and instability. The Star Trek universe depicts the first kind of future: advanced coordination protocols have eliminated material scarcity, and technology serves the whole population. Elysium depicts the second: the technology exists to heal every illness and solve every problem, but the coordination failed, and its benefits concentrated in the hands of a few [281]. The difference between these two futures is interaction structure.

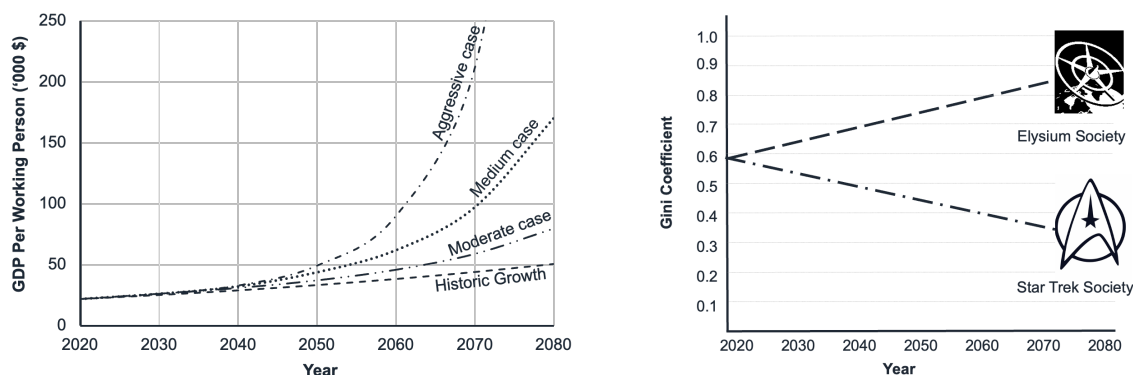


Figure 10.2: (a) Projected growth of AI agents across different adoption scenarios. (b) Potential societal outcomes measured by Gini coefficient under different coordination regimes. The difference between a Star Trek-like society and an Elysium-like world lies in how agents coordinate.

This thesis argued, and demonstrated, that collective outcomes are shaped by interaction structure. We showed this for human populations facing biological risk and economic disruption. We showed it for AI agent populations facing coordination challenges across supply chains and labor markets. In each setting, the mechanism was the same. Agents

interacted locally, those interactions propagated through the population, and the aggregate outcome depended on that structure.

In Isaac Asimov's Foundation series, the mathematician Hari Seldon faces a grand challenge: the Galactic Empire is on the verge of collapse, and no individual intervention can stop it. His response is to develop psychohistory, a mathematical science for predicting the behavior of large populations and designing interventions that work at the level of the system rather than the individual [282]. While fictional, this concept captures something real. The grand challenges of our moment are problems of collective behavior. They require understanding and shaping the interaction structures through which individual behavior becomes collective outcome.

This thesis is a step in that direction. We can now model populations of agents, learn from their interactions, and evaluate how coordination structures affect outcomes. Agentic societies are forming around us. The question is whether coordination structures are designed deliberately before unstable ones become embedded at scale. That is the work ahead.

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